

Atlas of Human Health

A Unified Taxonomy of Biological, Psychological, Social, Environmental, Functional, and Care Domains

History, medical education, diagnostic systems, research ontologies, pharmacology, psychology, professional silos, and a proposed multi-axial health knowledge architecture

Purpose

This packet provides a comprehensive map of human health as a knowledge domain. It is designed to support future technical reviews, research programs, measurement systems, educational curricula, clinical decision support, and device-development roadmaps without mistaking any one classification system for the whole of health.

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Scope, use, and safety

This document is a **systems-level atlas**, not a personal diagnosis or treatment plan. It explains how health can be divided, linked, measured, researched, and acted upon. It is intentionally broader than a textbook organized by organ system and more operational than a philosophical definition of health. The central problem is that the word *health* is used for several different things at once:

- the absence, presence, severity, or probability of disease;
- biological integrity and physiological regulation;
- symptoms, distress, cognition, emotion, and behavior;
- functional capacity and participation in ordinary life;
- subjective well-being, meaning, connection, and flourishing;
- resilience and the ability to adapt to changing demands;
- social, economic, environmental, occupational, and political conditions;
- access to safe, effective, equitable, coordinated care;
- population distributions of risk, burden, opportunity, and outcome.

No single tree can represent all of these without loss. A disease classification answers a different question from a disability classification, a pharmacologic taxonomy, a medical-school curriculum, a research ontology, a patient-reported outcome system, or a public-health surveillance framework. The design presented here therefore has two layers:

1. a **master navigation tree** that makes the knowledge space learnable; and
2. a **faceted coordinate system** that lets one concept occupy multiple biological, psychological, social, functional, temporal, environmental, evidentiary, and intervention dimensions.

The review draws on historical scholarship, the WHO Family of International Classifications, ICD-11, ICF, ICHI, DSM-5-TR, ICD-11 mental-disorder guidance, RDoC, HiTOP, medical-education standards, professional competency frameworks, clinical terminologies, research ontologies, public-health frameworks, pharmacology resources, burden-of-disease methods, and major critiques of reductionism, multimorbidity, overmedicalization, data fragmentation, and inequity. Current classification releases and institutional materials were checked through **13 August 2026**. (1–6)

The packet is comprehensive in structure and selective in citation. It does not attempt to cite every paper ever published in every health domain. Rather, it identifies the organizing concepts, landmark developments, representative evidence, current standards, known limitations, and a reproducible architecture for expanding each branch into its own domain packet.

Important limitation

A taxonomy can clarify reality, but it can also conceal it. Labels are models built for purposes. They should not be treated as natural boundaries when the underlying biology, behavior, experience, or social process is continuous, overlapping, uncertain, or changing.

Executive synthesis

0.1. The central conclusion

Health is best represented not as a list of medical specialties and not as a single hierarchy of diseases, but as a **multi-scale, life-course, person-in-context system**. The most useful architecture combines fifteen major branches with nine cross-cutting axes. The branches answer, “What part of the health landscape are we navigating?” The axes answer, “At what scale, through which mechanism, over what time course, under which determinants, with what functional consequence, measured how, and changed by which intervention?”

The proposed fifteen branches are:

1. **Whole-person health states and outcomes**
2. **Biological foundations and regulation**
3. **Organ-system health**
4. **Cross-system pathological processes**
5. **Brain, cognition, emotion, and behavior**
6. **Psychosocial, relational, cultural, and existential health**
7. **Development, reproduction, aging, and the life course**
8. **Infection, immunity, host-microbe biology, and biosecurity**
9. **Environmental, occupational, One Health, and planetary health**
10. **Function, disability, rehabilitation, and participation**
11. **Population health, prevention, and public health**
12. **Health care, professions, and delivery systems**
13. **Measurement, evidence, data, and informatics**
14. **Interventions and therapeutics**
15. **Ethics, law, rights, equity, and governance**

The nine cross-cutting axes are:

1. **Domain or target** — what aspect of health is under consideration;
2. **Scale or level** — molecule to ecosystem;
3. **Mechanism or process** — infection, inflammation, ischemia, degeneration, learning, discrimination, and so forth;
4. **Time and life course** — acute, chronic, episodic, developmental, reproductive, aging, intergenerational;
5. **Determinants and exposures** — genes, nutrition, behavior, relationships, work, housing, pollution, care, policy;
6. **Function and lived outcome** — symptoms, capacity, activities, participation, quality of life, survival;
7. **Context and population** — individual, family, community, culture, geography, institution, population;
8. **Measurement and evidence** — observation, device, laboratory, imaging, report, trial, causal inference, uncertainty;
9. **Intervention and governance** — prevention, treatment, rehabilitation, accommodation, policy, rights, resource allocation.

Key synthesis

A usable health record, research dataset, curriculum, or device program should be able to locate any health concept on all nine axes. “Hypertension,” for example, is not merely a cardiovascular diagnosis. It is a continuously distributed vascular state, a measurement problem, a causal risk factor, a treatment target, a population-health indicator, a source of disparities, and a longitudinal process whose meaning changes with age, pregnancy, comorbidity, setting, and measurement method.

0.2. What existing systems do well

ICD-11 supports internationally comparable morbidity and mortality classification and modern digital coding. **ICF** represents body functions and structures, activities, participation, and contextual factors. **ICHI** classifies interventions. Together they form the conceptual core of the WHO Family of International Classifications, but adoption and interoperability remain uneven. (1,7–9)

DSM-5-TR and the ICD-11 mental-disorder framework provide clinically actionable descriptive diagnoses. They improve communication, eligibility decisions, service organization, and research sampling, but categorical thresholds can reify heterogeneous syndromes. RDoC and HiTOP respond by emphasizing dimensions, functional systems, and empirically derived spectra; neither is a universal substitute for clinical classification. (10–13)

Medical education increasingly integrates foundational sciences with organ systems, clinical experiences, professional competencies, social context, patient safety, and interprofessional practice. Yet the hidden organizing structure remains heavily departmental and specialty-based. Learners often encounter the same person as separate cardiology, psychiatry, neurology, pharmacology, epidemiology, and social-medicine problems rather than as one dynamic system. (14–17)

Clinical terminologies and data standards are complementary: SNOMED CT expresses clinical meaning; LOINC identifies observations; RxNorm normalizes medications; FHIR exchanges structured resources; OMOP harmonizes analytic data; UMLS connects vocabularies; HPO, Mondo, Gene Ontology, Disease Ontology, ChEMBL, and IUPHAR/BPS support phenotype, disease, molecular-function, compound, and target research. The difficulty is not that no standards exist; it is that too many standards exist for different purposes, at different granularities, with incomplete semantic alignment. (5,6,18–26)

0.3. Where the present organization fails

Modern health systems are exceptionally capable at resolving narrowly defined acute problems, but they often perform poorly when the main challenge is **interaction**: multimorbidity, chronic pain, frailty, long-duration symptoms, mental-physical comorbidity, disability, polypharmacy, adverse social conditions, environmental exposure, or conflicting goals. Single-disease guidelines can become internally inconsistent when applied simultaneously. (27–29)

The recurring failure modes are:

- **organ and specialty silos**, which divide causal pathways that cross tissues and professions;
- **categorical reification**, which turns useful thresholds into assumed natural kinds;
- **mental-physical dualism**, which misclassifies reciprocal brain-body processes;
- **underrepresentation of function and lived experience**, especially when biomarkers improve without meaningful daily-life improvement;
- **underweighting of social and environmental determinants**, despite their causal relevance;
- **billing and administrative distortion**, where what can be coded becomes what is seen;

- **semantic fragmentation**, preventing longitudinal and cross-institutional synthesis;
- **inequitable reference standards and algorithms**, which can encode structural disadvantage as biology;
- **reactive care**, with prevention and upstream action structurally separated from clinical care;
- **evidence mismatch**, where average treatment effects are applied to people unlike study populations;
- **overdiagnosis and medicalization alongside underdiagnosis**, sometimes occurring in different groups at the same time.

0.4. The proposed repair

This atlas recommends a **federated, multi-axial architecture**, not replacement of all existing systems. Existing codes remain attached for reimbursement, surveillance, safety, and interoperability. A new integrative layer stores:

- the health concept and its standard identifiers;
- biological and psychosocial mechanisms with confidence levels;
- temporal trajectory and life-stage context;
- determinant and exposure history;
- symptoms, function, participation, and personally valued outcomes;
- measurement provenance and uncertainty;
- interventions, mechanisms, burden, benefit, adverse effects, and feasibility;
- ethical, cultural, legal, and equity considerations;
- relationships among conditions, rather than merely a flat problem list.

This makes it possible to move from a collection of labels toward a **computable health profile**: a versioned model of a person or population that remains explicit about uncertainty, purpose, and values.

How to use the atlas

The document is organized in five layers.

Part I — Foundations and history explains why societies have repeatedly reorganized health knowledge as new instruments, causal theories, institutions, and moral priorities emerged.

Part II — The unified taxonomy defines the fifteen branches and nine axes. It is the principal reference for deciding where a new topic belongs.

Part III — Existing classification ecosystems explains how medicine, psychiatry, public health, pharmacology, research, informatics, and professional education currently divide the world, and what each system is designed to accomplish.

Part IV — Failure modes analyzes why present systems often fail to integrate the whole person and why classification can generate blind spots.

Part V — A practical architecture provides a record structure, worked examples, profile design, research schema, device-program implications, curriculum redesign, and an expansion roadmap.

The appendices provide a deep tree, crosswalk matrices, maps of professions, measurements, interventions, and a glossary.

Design principle

When starting a new domain review, first locate the topic within the master tree. Then assign all nine axes. Only after that should the team decide which medical specialty, research ontology, diagnosis code, measurement suite, and intervention framework to use. This order prevents the available tool or department from silently defining the problem.

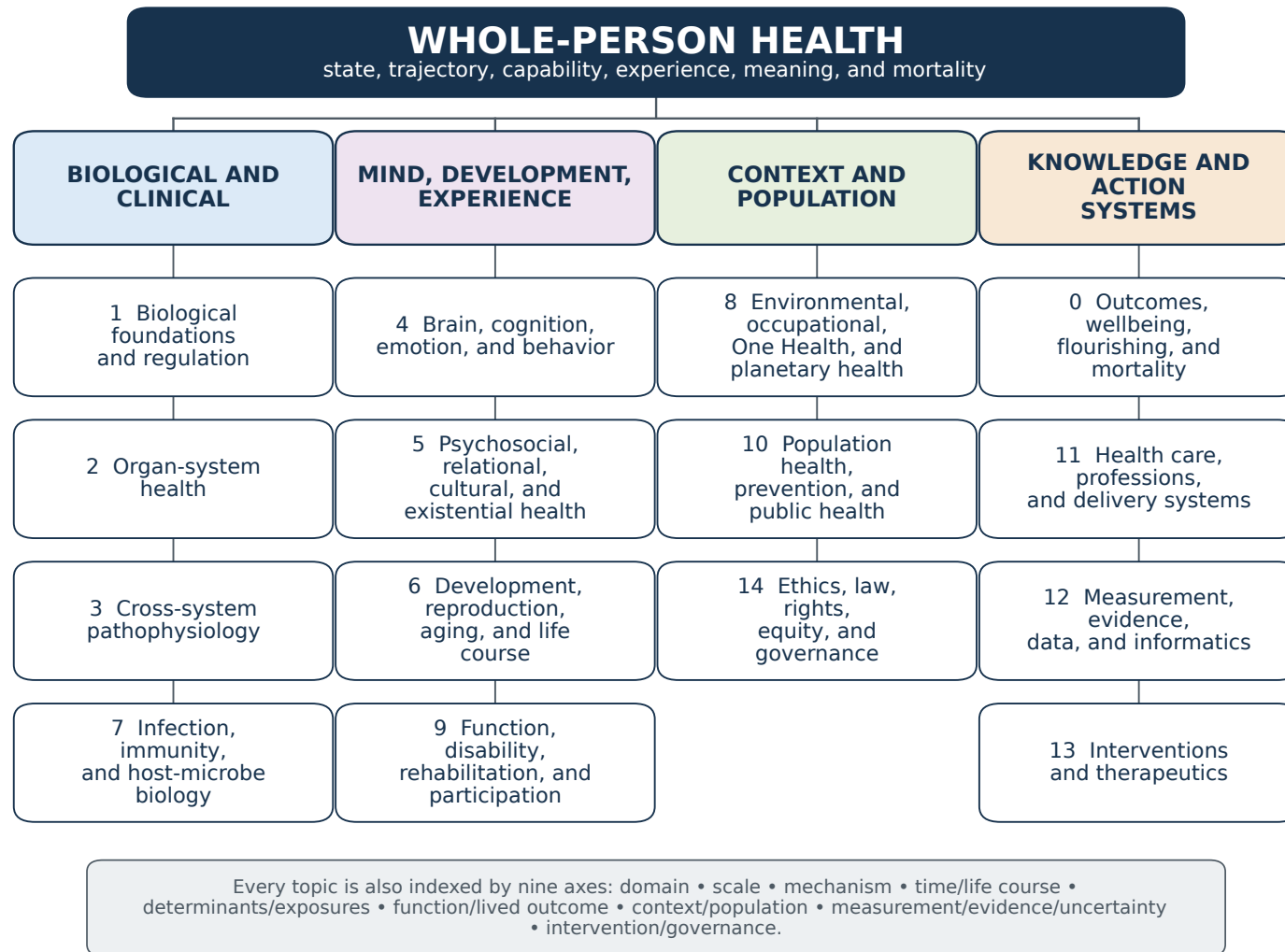


Figure 1.: The master atlas. Fifteen navigational branches cover health states, mechanisms, contexts, measurements, interventions, institutions, and governance. The tree is intentionally complemented by cross-cutting axes because many health phenomena belong to several branches simultaneously.

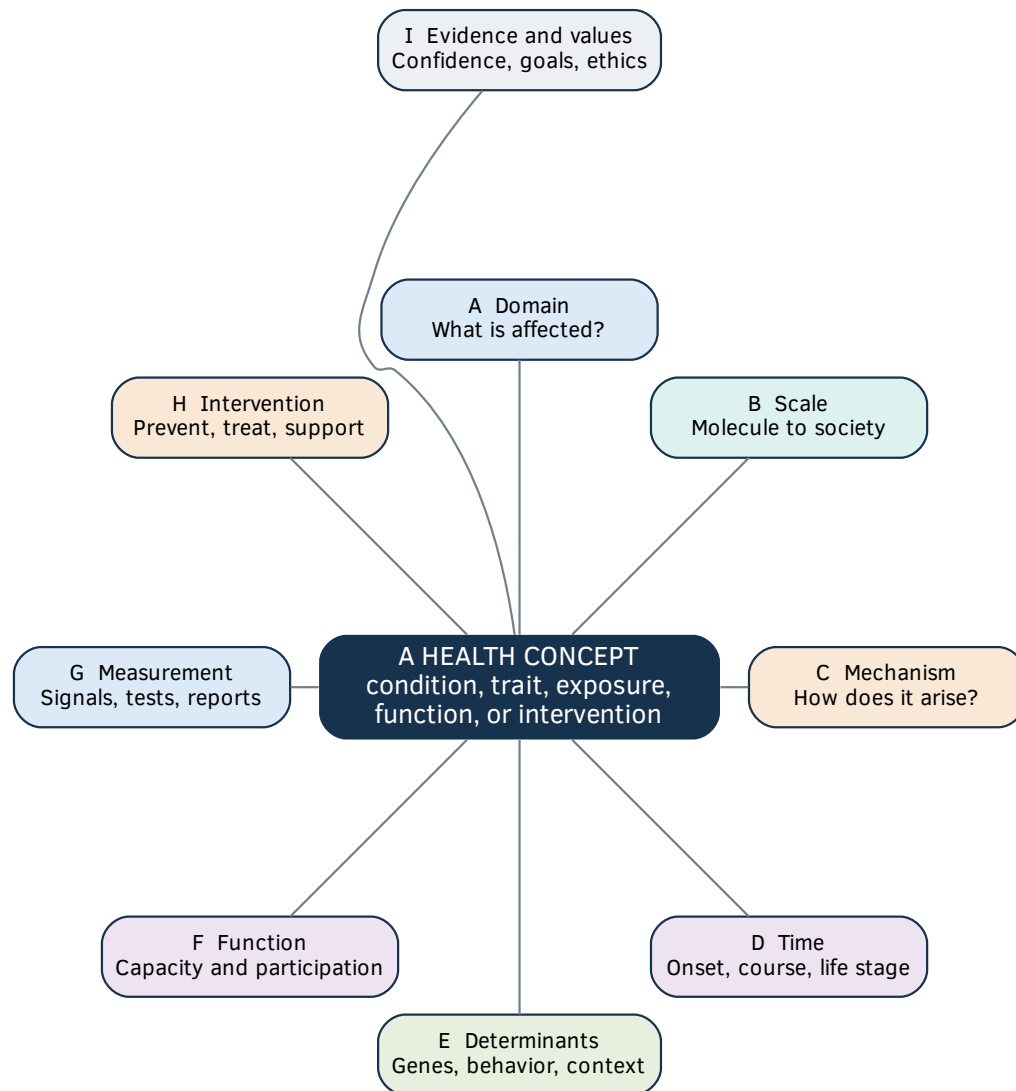


Figure 2.: *The faceted health coordinate. A concept is represented by a tuple rather than a single box. This permits one phenomenon to be viewed simultaneously as biology, experience, function, context, measurement, evidence, intervention, and governance.*

1. Part I — Foundations: What Health Is and How Its Organization Evolved

2. What “health” can mean

2.1. Health is an overloaded construct

In ordinary conversation, *health* may mean that no disease has been diagnosed. In physiology it may mean regulated variables remaining within viable ranges. In psychology it may mean adaptive cognition, emotion, behavior, and relationships. In rehabilitation it may mean capacity and participation despite permanent impairment. In public health it may mean the distribution of survival, disability, exposure, opportunity, and care across a population. In ethics it may mean capability, agency, dignity, and freedom from avoidable harm. These meanings overlap, but they are not interchangeable.

A useful taxonomy begins by separating at least eight concept families.

Concept family	Central question	Typical indicators	Main blind spot when used alone
Disease status	What pathological entity is present?	diagnosis, lesion, pathogen, mutation	may miss function, distress, context, and risk before disease
Physiological state	How is the organism regulating itself?	pressure, flow, rhythm, chemistry, reserve	“normal” values may coexist with suffering or low participation
Risk and susceptibility	What future events are more probable?	genetics, exposures, biomarkers, scores	probability is not destiny and can be population-specific
Symptoms and experience	What does the person feel or notice?	pain, fatigue, dyspnea, mood, cognition	subjective measures can be discounted or confounded
Function	What can the person do?	mobility, self-care, communication, work	capacity depends on environment and support
Well-being and flourishing	Is life going well by the person’s values?	affect, purpose, connection, security	can obscure serious pathology or structural injustice
Resilience and adaptation	Can the system respond to disturbance?	recovery, reserve, flexibility, coping	adaptation to harmful conditions is not the same as justice
Population health	How are health and opportunity distributed?	mortality, DALYs, incidence, inequity	population averages can conceal individual variation

The WHO Constitution famously described health as physical, mental, and social well-being rather than merely the absence of disease. That formulation was historically important because it widened the legitimate concern of health institutions. Its “complete well-being” language has also been criticized as unattainable, static, and potentially medicalizing ordinary imperfection. Later proposals emphasize the ability to adapt and self-manage, while critics note that adaptation alone can normalize deprivation or shift responsibility from institutions to individuals. (30–32)

The correct response is not to choose one sentence as the final definition. It is to make the **purpose of the definition explicit**. A public-health constitution needs an aspirational scope. A clinical trial needs measurable endpoints. A disability-rights framework needs to separate impairment from environmental barriers. A device needs an operational signal model. A person deciding whether life is going well needs room for values and meaning.

2.2. Disease, illness, sickness, disorder, injury, and disability

These terms carry different perspectives:

- **Disease** usually denotes a biological or behavioral pathological process recognized by a professional knowledge system.
- **Illness** emphasizes the person’s lived experience of symptoms, distress, limitation, or altered identity.
- **Sickness** emphasizes the social position, expectations, legitimacy, and institutions surrounding ill health.
- **Disorder** is often used when a syndrome entails clinically significant disturbance or dysfunction but its boundaries or mechanisms are heterogeneous.
- **Injury** emphasizes damage from an external or internal event, including mechanical, thermal, chemical, electrical, radiological, or psychological trauma.
- **Impairment** concerns a body function or structure; **activity limitation** concerns execution of tasks; **participation restriction** concerns involvement in life situations. The ICF explicitly connects these to personal and environmental context. (8,33)

A person may have disease without illness, illness without an established disease label, impairment without poor quality of life, or disability that appears only in an inaccessible environment. Conversely, a person may feel well while carrying a high-risk process such as severe hypertension or an early malignancy. A mature taxonomy must preserve all of these possibilities.

2.3. Health as state, capacity, trajectory, and relationship

Health is not only a state at one moment. It can be represented as:

1. **State** — the current configuration of symptoms, physiology, function, and context;
2. **Capacity** — reserve and capability under challenge;
3. **Trajectory** — direction, rate of change, variability, and reversibility;
4. **Relationship** — fit between the person, goals, social world, built environment, and available support.

This distinction matters technically. Resting heart rate is a state. Chronotropic response to exercise is a capacity. Progressive decline in exercise tolerance is a trajectory. Whether limited walking prevents employment depends on the environment and accommodations. The same principle applies to cognition, immunity, mood, sleep, respiratory function, and social participation.

Definition

****Operational definition used in this atlas:**** Health is the dynamic configuration of biological integrity and regulation, subjective experience, cognition and behavior, functional capability and participation, social and environmental conditions, and future risk, interpreted across the life course and relative to the person’s values and population context.

3. Why classifications exist

3.1. Classification is an instrument, not a mirror

A classification groups observations according to a purpose. Its categories can support communication, counting, causal research, prediction, treatment selection, reimbursement, eligibility, regulation, teaching, or advocacy. These purposes impose different design requirements.

A mortality classification needs mutually exclusive rules for an underlying cause of death. A clinician needs a problem list that permits several interacting diagnoses. A molecular biologist needs relationships among genes, pathways, phenotypes, and compounds. A psychologist may need continuous dimensions. An insurer needs billable units. A person may need a narrative that explains suffering without reducing identity to a code.

The same phenomenon can therefore receive several valid representations. “Major depressive disorder” may be:

- a DSM or ICD diagnosis;
- a dimensional profile of mood, reward, threat, sleep, cognition, and function;
- a set of symptom-network relationships;
- a longitudinal state affected by trauma, inflammation, medication, social isolation, and sleep;
- a cause of disability in burden-of-disease analysis;
- a reason for a psychotherapy or pharmacotherapy intervention;
- an eligibility criterion for a study;
- a personally meaningful but incomplete explanation.

Confusion begins when a representation optimized for one purpose is treated as if it were the ontology of nature.

3.2. Fundamental design choices

Every health classification makes choices along several continua.

Design choice	Pole A	Pole B	Consequence
Category structure	discrete classes	continuous dimensions	reliability and actionability versus fidelity to graded variation
Hierarchy	one parent per concept	polyhierarchy/network	simplicity versus biological and contextual realism
Basis	anatomy/phenotype	mechanism/etiology	observable stability versus causal depth
Time	cross-sectional	longitudinal	snapshot convenience versus trajectory understanding
Unit	disease	person/system	analytic focus versus multimorbidity integration
Perspective	professional observation	lived experience	standardization versus meaning and relevance

Design choice	Pole A	Pole B	Consequence
Context	context-free	environment-dependent	portability versus ecological validity
Uncertainty	forced assignment	probabilistic membership	decisiveness versus epistemic honesty
Governance	centrally curated	federated/community-involved	consistency versus pluralism and responsiveness

A robust architecture does not demand one answer everywhere. It records the choices and allows transformations between representations.

3.3. Taxonomy, ontology, terminology, nosology, and data model

These words are often used loosely.

- A **taxonomy** arranges entities into classes, usually hierarchical.
- An **ontology** formally specifies entities, properties, and relations, often allowing multiple inheritance and machine reasoning.
- A **terminology** standardizes names and codes used to express concepts.
- A **nosology** classifies diseases or disorders.
- A **phenomenology** describes what is observed or experienced without necessarily specifying cause.
- A **data model** defines how observations, entities, events, and relationships are stored.
- A **knowledge graph** represents entities and typed relations in a queryable network.

ICD is principally a statistical classification with clinical extensions. SNOMED CT is a compositional clinical terminology. LOINC identifies observations and documents. FHIR is an exchange specification. OMOP is an analytic common data model. UMLS is a metathesaurus and integration resource. These are complementary, not competitors in a single contest. (5–7,18,20,21)

Important limitation

A coding hierarchy can appear scientifically authoritative because it is precise, but precision of syntax is not the same as certainty of mechanism. Administrative categories frequently contain heterogeneous biology; molecular categories may have uncertain clinical significance; and patient-important outcomes may not map neatly to either.

4. A global history of health concepts

4.1. Before modern biomedicine

Human societies have always distinguished health from illness, but explanations and practices varied with cosmology, ecology, available technologies, social order, and accumulated observation. Healing traditions commonly combined material interventions, diet, plants, surgery, ritual, relationship, and moral interpretation. It is historically inaccurate to treat all premodern medicine as one undifferentiated supernatural stage or to portray modern Western medicine as the sole linear endpoint.

Ancient Egyptian and Mesopotamian records describe symptoms, prognoses, wounds, remedies, and ritual causation. Greek traditions associated with Hippocrates sought naturalistic patterns in environment, regimen, and bodily balance; later humoral models organized illness around blood, phlegm, yellow bile, and black bile. Galenic synthesis shaped Mediterranean and European medicine for centuries. Ayurveda organized health through constitutional patterns, diet, behavior, environment, and balances often described through *doshas*. Classical Chinese medicine developed dynamic relational models involving *yin-yang*, *qi*, channels, organ networks, season, and environment. Indigenous traditions worldwide maintained place-based knowledge, communal care, spiritual meaning, and ecological relationships. Islamic medicine preserved, criticized, and extended Greek and regional knowledge; Ibn Sina's *Canon* systematized diagnosis, pharmacology, hygiene, and therapeutics. (34,35)

These systems are not directly interchangeable with molecular physiology. Some historical claims are unsupported by modern evidence, while some practices yielded effective observations or compounds. Their continuing conceptual contribution is that health was often viewed as a **relationship among person, behavior, community, and environment**, not solely as a lesion in an isolated organ.

4.2. Anatomy, circulation, microscopy, and pathological localization

Renaissance dissection challenged inherited anatomical authority. Vesalius's work strengthened direct observation of structure. Harvey's account of blood circulation reframed the cardiovascular system as a mechanical circuit with directional flow, valves, and a pump. Microscopy opened cellular and microbial scales. Morgagni related symptoms to organ lesions; Bichat emphasized tissues; Virchow emphasized cellular pathology. The stethoscope, thermometer, sphygmomanometer, microscopy, laboratory chemistry, radiography, electrocardiography, and later imaging progressively made invisible processes measurable.

This created a powerful organizing rule: **localize the abnormality**. Medicine became arranged by anatomy, tissue, pathology, and instrument. Localization remains indispensable. It also encouraged a recurrent error: assuming that the most visible lesion is the complete explanation of the person's condition.

4.3. Population counting, sanitation, and social medicine

The development of vital statistics and cause-of-death classification made population patterns visible. John Graunt's analysis of mortality bills, later statistical work, and epidemiologic comparisons supported the idea that disease could be counted and related to conditions. Sanitation, clean water, sewage systems, housing reform, workplace regulation, nutrition, and vaccination demonstrated that population health could change without individualized bedside treatment.

Virchow's work linked epidemics and deprivation to social arrangements and framed medicine as inseparable from politics and social reform. The later history of public health repeatedly alternated between upstream structural action and narrower technical programs. (36–38)

4.4. Germ theory and the causal revolution

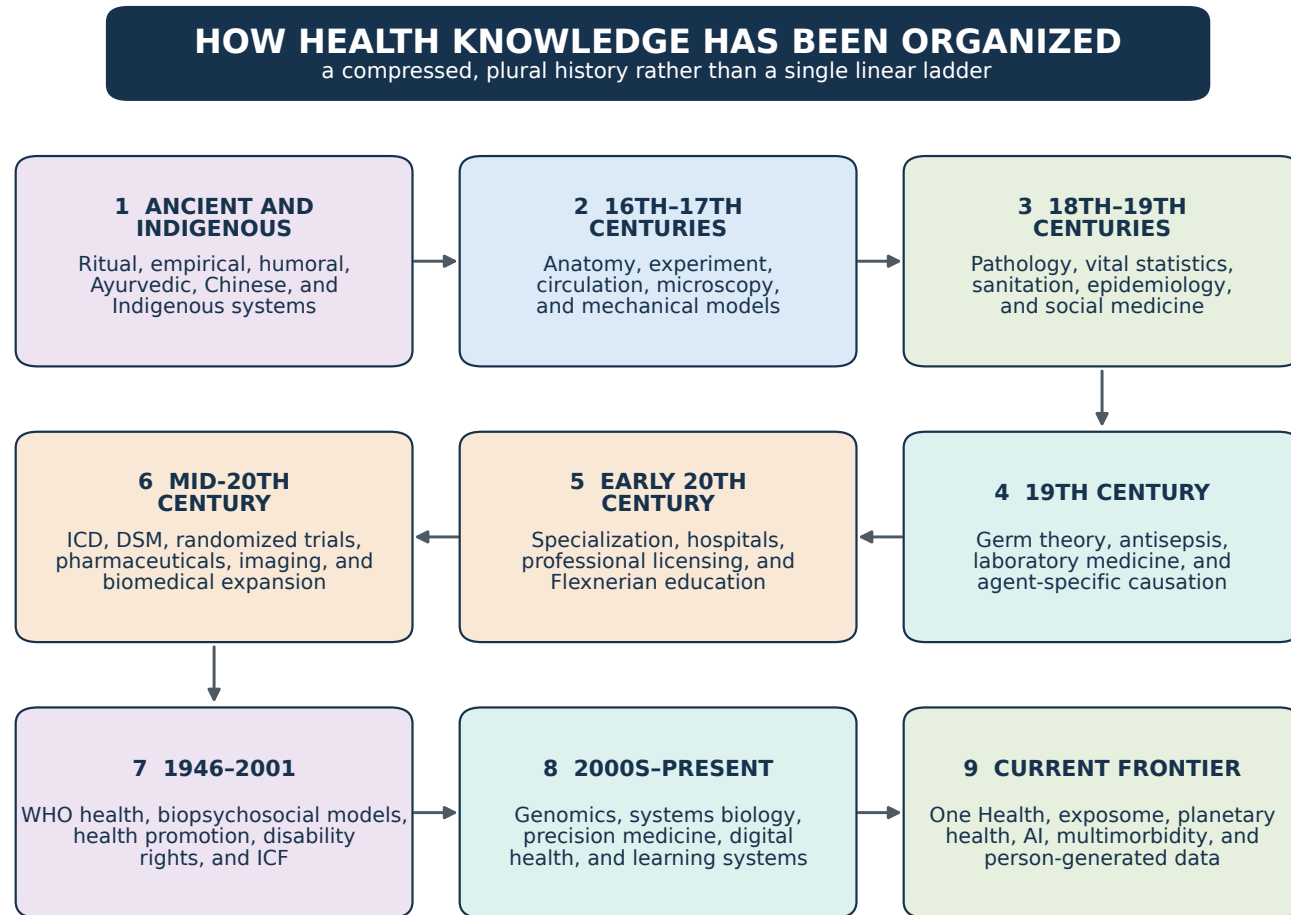
The nineteenth-century convergence of microscopy, experimental methods, Pasteur's work, Koch's postulates, antiseptics, and laboratory culture transformed infectious-disease explanation. Specific agents could be linked to specific diseases; transmission could be interrupted; vaccines and antimicrobials could be developed. Germ theory did not eliminate environmental and social causation: crowding, nutrition, sanitation, labor conditions, animal reservoirs, and host immunity still shape infection. Yet the agent-centered model was so successful that it became a template for seeking a single specific cause elsewhere. (39,40)

Modern infectious science now uses a richer triangle: **host–agent–environment**, expanded by genomics, immune phenotypes, microbiome ecology, evolutionary dynamics, and global mobility. The lesson is not that germ theory was too reductionist; it is that a precise causal agent can be embedded in a multi-level causal system.

4.5. Psychiatry, neurology, and changing boundaries of mind

Historical concepts of madness, melancholy, possession, moral failing, nervous disease, and psychiatric disorder changed with institutions, philosophy, neurology, psychoanalysis, psychometrics, pharmacology, and social movements. The boundary between neurology and psychiatry was never biologically clean. Epilepsy, dementia, delirium, movement disorders, functional neurological symptoms, substance effects, trauma, and developmental conditions repeatedly crossed it.

Twentieth-century psychiatric classification moved from theory-laden formulations toward more operational symptom criteria, especially with DSM-III in 1980. This increased standardization but did not establish that each diagnostic category was a distinct biological entity. Later dimensional and process-based frameworks attempt to map cognition, emotion, motivation, arousal, and social processes across conventional diagnoses. (12,41–43)



Traditions overlap, exchange knowledge, and persist; instruments, institutions, causal theories, politics, and social movements repeatedly reorganize the map.

Figure 3.: A compressed history of health organization. The sequence is not a single Western ladder: multiple traditions coexist, exchange knowledge, and are repeatedly reorganized by new instruments, institutions, causal theories, political priorities, and social movements.

5. The rise of scientific medicine and specialization

5.1. Laboratory medicine and the Flexner era

By the late nineteenth and early twentieth centuries, medical education varied widely in quality. The Flexner Report promoted university affiliation, scientific laboratories, standardized prerequisites, and structured clinical training. These reforms helped institutionalize rigorous biomedical education. They also contributed to the closure of many schools, including institutions serving Black communities and women, and reinforced a model centered on research universities and hospital-based specialties. (44–46)

The enduring educational architecture became:

1. foundational sciences such as anatomy, physiology, pathology, microbiology, pharmacology, and biochemistry;
2. clinical clerkships organized by specialty;
3. postgraduate training in increasingly differentiated disciplines;
4. board certification and subspecialization.

This architecture maps well to bodies with identifiable organ systems and to professions with defined scopes. It maps less well to chronic multi-system conditions, social determinants, prevention, rehabilitation, and coordination.

5.2. Why specialties multiplied

Specialization is a rational response to knowledge volume, technical complexity, procedural skill, and risk. Cardiothoracic surgery, electrophysiology, interventional radiology, transplant medicine, neonatal intensive care, and molecular pathology require concentrated expertise. New instruments often create new specialties or subspecialties because interpretation and intervention become inseparable from technology.

But specialization has transaction costs:

- each clinician sees a partial state;
- goals and treatment plans may conflict;
- no specialty owns the interactions;
- responsibility can diffuse across referrals;
- patients become coordinators of their own fragmented care;
- reimbursement may reward procedures more than integration;
- educational identity can become tied to organs rather than persons.

The challenge is therefore not to abolish specialties, but to add structures that preserve **system coherence**.

5.3. The randomized trial and evidence-based medicine

Random allocation, control groups, blinding, prespecified outcomes, and statistical inference transformed evaluation of interventions. Evidence-based medicine later emphasized integrating the best available

research with clinical expertise and patient values. Reporting standards such as CONSORT, PRISMA, and STROBE improved transparency for trials, reviews, and observational studies; GRADE provides a structured approach to certainty and recommendation strength. (47–51)

This movement corrected reliance on authority and uncontrolled experience. It also created new abstractions: the average treatment effect, the eligibility-defined study population, the prespecified endpoint, and the evidence hierarchy. These are essential tools but not complete representations of health. External validity, heterogeneity, competing outcomes, multimorbidity, long time horizons, implementation conditions, and values remain difficult. Excessive worship of hierarchy can underweight mechanism, context, qualitative evidence, natural experiments, and patient experience. (52,53)

6. From biomedical to biopsychosocial and functional models

6.1. The biomedical model: strength and boundary

The biomedical model is sometimes caricatured as believing that only molecules matter. In practice, its strongest form is a methodological commitment: identify observable mechanisms, lesions, pathogens, physiological disturbances, and interventions that alter them. This has enabled anesthesia, surgery, antibiotics, vaccination, intensive care, imaging, transplantation, and targeted therapy.

Its limitation arises when the analytic strategy becomes an ontology: when health is assumed to be nothing more than proximal biological abnormality. Behavior, expectation, learning, relationship, labor, poverty, culture, and policy all have causal paths through nervous, endocrine, immune, cardiovascular, metabolic, and developmental processes. Their biological embodiment does not make them “merely biological,” because the upstream level remains necessary for explanation and intervention.

6.2. Engel’s biopsychosocial challenge

Engel argued in 1977 that biomedicine required a model spanning biological, psychological, and social levels. The model’s enduring value is its rejection of one-level explanation and its insistence that the clinical encounter is itself part of the system. Later formulations connect reciprocal causal processes across levels and emphasize collaborative, person-centered practice. (54–57)

The model’s weakness is operational ambiguity. “Consider biological, psychological, and social factors” can become an unweighted checklist in which every factor is mentioned but no causal structure is specified. Critics also note that it can be used to psychologize poorly understood physical symptoms or to place social problems within individual treatment rather than institutional change. (58)

A stronger version requires:

- specified mechanisms and feedback loops;
- temporal ordering;
- evidence for causal versus correlational claims;
- explicit scales and contexts;
- measurable outcomes;
- intervention points at the level where change is possible;
- protection against dismissing unexplained disease as “psychosocial.”

6.3. Functioning and the ICF

The ICF offers a different integration. It classifies:

- body functions and structures;
- activities;
- participation;
- environmental factors;
- personal factors as relevant context, though personal factors are not formally classified in the same way.

Disability emerges from interaction rather than residing entirely inside a body. A spinal-cord injury can produce different participation outcomes depending on mobility technology, accessible transport, housing, work design, income, attitudes, and legal protection. This model aligns with rehabilitation, occupational therapy, disability studies, and rights-based policy. (8,59)

ICF does not replace diagnosis. It answers a different question: what is the person's functioning in context? An integrated record needs both.

6.4. Salutogenesis, resilience, and flourishing

Pathogenesis asks why disease occurs. Salutogenesis asks what supports movement toward health. Antonovsky's concept of a "sense of coherence" emphasized comprehensibility, manageability, and meaningfulness. Contemporary resilience research examines adaptive capacity across biology, psychology, family, community, and institutions. Flourishing frameworks add domains such as happiness, mental and physical health, meaning, character, relationships, and financial stability. (60,61)

These perspectives correct a deficit-only view. They must not be used to imply that people can think their way out of unsafe work, discrimination, pollution, violence, or inadequate care. Resilience can be a property of individuals and communities, but **resilience requirements are often produced by structural failure.**

7. Life course, systems, and ecology

7.1. Developmental and life-course health

Health trajectories begin before birth and are shaped by development, timing, accumulation, critical periods, and intergenerational processes. Barker's work on fetal and early-life conditions helped stimulate developmental-origins research. Life-course health development integrates biology, behavior, family, community, and policy across changing developmental tasks. (62,63)

A life-course model distinguishes:

- **critical or sensitive periods**, when exposure has disproportionate effects;
- **accumulation**, in which repeated exposures build burden;
- **pathway effects**, in which one event changes later opportunities;
- **chains of risk**, where linked events amplify probability;
- **social mobility and turning points**, which can alter trajectory;
- **intergenerational transmission**, biological and social;
- **reserve and compensation**, which can delay visible impairment.

This explains why a cross-sectional diagnosis is often an endpoint of a much longer process.

7.2. Allostasis and stress physiology

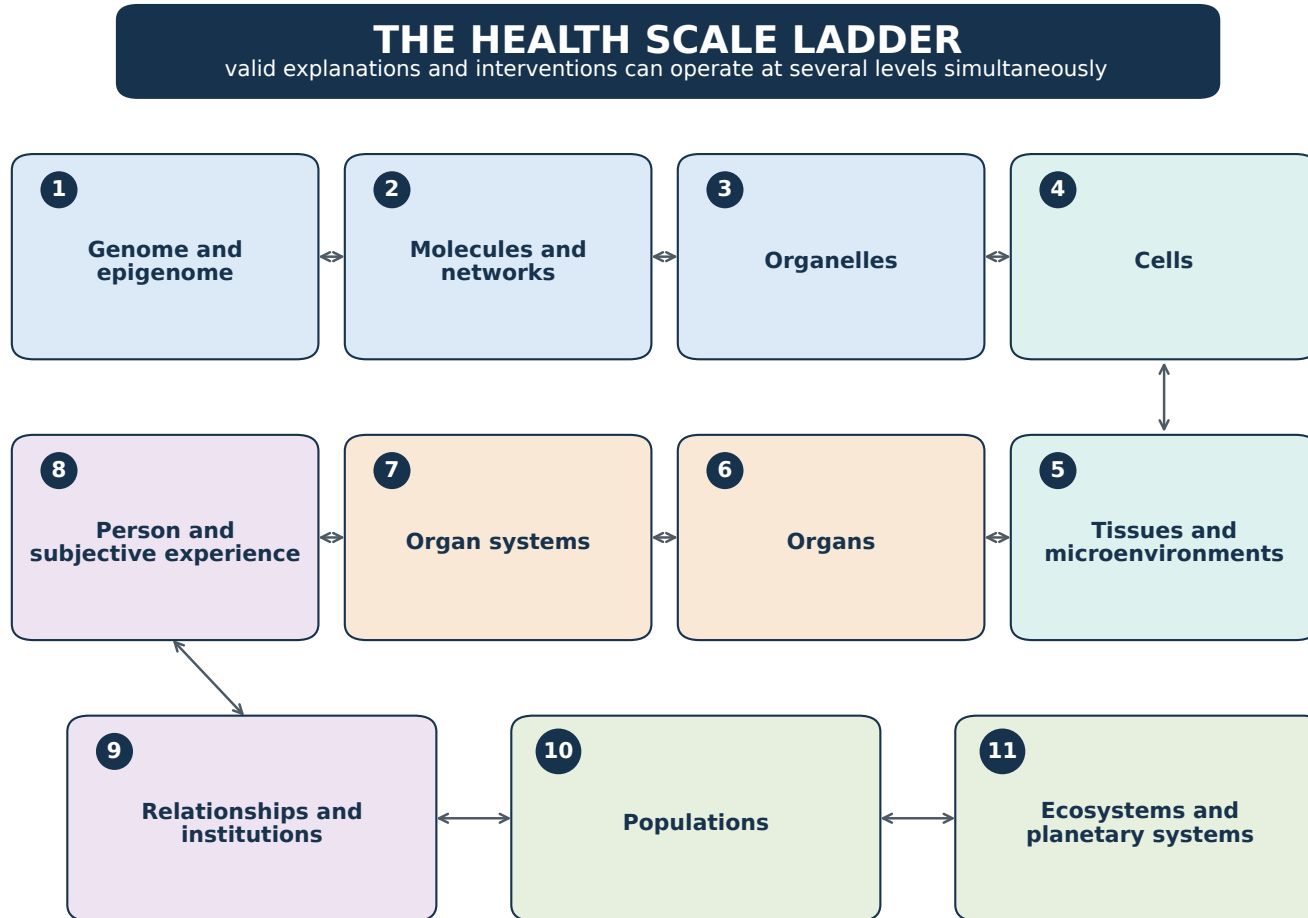
Homeostasis emphasizes regulated stability. Allostasis emphasizes achieving stability through change—altering autonomic, endocrine, immune, metabolic, and behavioral responses to meet demand. Repeated or poorly regulated activation can create allostatic load. Stress is not a single scalar; its health effects depend on intensity, duration, controllability, predictability, meaning, recovery, developmental timing, and resources. (64,65)

The stress concept is useful when it specifies pathway. It becomes unhelpful when used as a residual explanation for any unexplained symptom. A good model asks: what demand, appraised how, through which neural and endocrine circuits, causing what behavioral adaptation, over what duration, interacting with which material conditions and biological susceptibilities?

7.3. Systems biology and network medicine

Systems biology represents interacting genes, proteins, cells, organs, behaviors, and environments as dynamic networks. Disease can arise from network perturbation, loss of robustness, maladaptive feedback, or failure of coordination rather than one defective component. Network medicine maps diseases to modules and shared pathways, helping explain comorbidity and pleiotropy.

This perspective supports multi-omics and computational modeling but does not automatically solve integration. A molecular network may still omit subjective experience, institutional behavior, or social causation. A truly multi-level system must permit causal explanations at several scales without assuming that the lowest scale is always the most informative.



Influence is bidirectional: molecular changes can alter experience, while policy, relationships, behavior, and environments can reshape biology.

Figure 4.: The health scale ladder. Explanations can be valid at several levels simultaneously. Reduction to molecules may clarify mechanism; higher-level variables may remain the most efficient targets for prediction or intervention.

7.4. Precision medicine, exposome, and digital phenotyping

Precision medicine seeks more accurate prevention and treatment by incorporating variability in genes, environment, lifestyle, biomarkers, and response. Its strongest applications identify mechanistically distinct subgroups or predictive biomarkers. Its risks include overfitting, inequitable datasets, inflated interpretation of weak signals, and neglect of population-wide interventions.

The **exposome** was proposed to represent the totality of environmental exposures across life, complementing the genome. It includes chemical, physical, biological, social, behavioral, and internal exposures and their dynamic interactions. Measurement remains difficult because exposure is temporally variable, correlated, spatially patterned, and partly mediated through behavior and social structure. (66,67)

Digital phenotyping and wearable sensing can capture continuous behavior, physiology, location patterns, communication proxies, sleep, and activity. These technologies may reveal trajectories missed by clinic visits. They also introduce surveillance, consent, data ownership, algorithmic bias, context collapse, and false-positive risks. More data does not automatically create a valid health model; provenance and purpose remain essential.

7.5. One Health and planetary health

One Health recognizes that human, animal, plant, and ecosystem health are interdependent, particularly for zoonotic infection, antimicrobial resistance, food systems, environmental contamination, and vector ecology. International agencies have developed joint plans linking surveillance, prevention, preparedness, and environmental action. (68,69)

Planetary health examines the health of human civilization and the natural systems on which it depends. Climate change, biodiversity loss, land-use change, freshwater stress, pollution, and food-system transformation alter heat exposure, air quality, infectious ecology, nutrition, displacement, conflict risk, and mental health. (70)

These frameworks expand the unit of analysis beyond the patient and even beyond the human population. They do not imply that every clinical decision requires a planetary model; they establish that some health determinants cannot be governed within clinical institutions alone.

8. Philosophical and technical problems in drawing boundaries

8.1. Categories versus continua

Many clinical variables are continuous: blood pressure, glucose, bone density, mood symptoms, attention, pulmonary function, and pain. Categories are imposed because action requires decisions. Thresholds can improve reliability and standardize access, but they often represent risk gradients, resource constraints, or consensus rather than natural discontinuities.

The practical solution is to store both:

- the **continuous measurement and its uncertainty**; and
- the **decision category and the rule used to assign it**.

This preserves clinical action without erasing underlying variation.

8.2. Natural kinds, practical kinds, and prototypes

Some conditions approach natural-kind structure when a specific pathogen, genetic variant, toxic exposure, or lesion reliably produces a characteristic process. Even then, host response and phenotype vary. Other categories are **practical kinds**: groupings stable enough to guide action despite heterogeneous causes. Still others function as prototypes or syndromes with fuzzy boundaries.

A taxonomy should therefore annotate the basis of a class:

- etiologic;
- mechanistic;
- anatomical;
- histopathologic;
- physiological;
- symptomatic;
- behavioral;
- functional;
- prognostic;
- treatment-response;
- administrative;
- self-identified or community-defined.

8.3. Normality is not one thing

“Normal” can mean statistically common, within a reference interval, biologically functional, clinically low risk, socially expected, personally preferred, or morally approved. These meanings should never be collapsed.

A laboratory reference interval usually describes a central portion of values in a selected reference population; it is not automatically the optimal range or a diagnostic boundary. A rare trait can be healthy. A common exposure can be harmful. A socially disfavored identity is not a disorder. An adaptation can be statistically unusual but functionally valuable.

8.4. Causation across levels

The claim that a social exposure “gets under the skin” should not be interpreted as making the social exposure unreal once a molecular mediator is found. Discrimination can alter opportunity, stress physiology, health behavior, environmental exposure, and care access. Molecular measurements describe pathways; they do not replace the higher-level cause. Conversely, social explanation should not be used to deny biological investigation.

A multi-level causal model asks four separate questions:

1. What variables predict the outcome?
2. Which variables are causal under a specified intervention?
3. Through which mediators does the effect operate?
4. At which level is intervention effective, feasible, equitable, and acceptable?

8.5. Values inside classification

Classification is partly empirical and partly normative. Decisions about impairment, distress, dangerousness, capacity, acceptable risk, resource priority, and treatment need involve values. This does not make them arbitrary. It means governance must be transparent, evidence-informed, rights-respecting, and open to affected communities.

Critical appraisal

The deepest mistake in health organization is not reductionism alone. It is ****unmarked reduction****: using a narrow representation without stating what it leaves out, then allowing that representation to control access, identity, research funding, and policy as though no alternative perspective existed.

9. Part II — A Unified Taxonomy of Human Health

10. Architecture of the atlas

10.1. Why a tree is still useful

A tree is cognitively efficient. It gives learners and project teams a stable place to begin, supports modular documents, and prevents major areas from disappearing. The master tree therefore answers a navigation question: **Which broad family of health knowledge is this?** It is not intended to assert that each entity has one true parent.

The root is *human health in context*. Its fifteen branches cover outcomes, biological foundations, organ systems, cross-system mechanisms, mind and behavior, psychosocial life, life course, host–microbe ecology, environment, function, population health, care systems, measurement, intervention, and governance.

10.2. Why the tree must be faceted

Long COVID may involve infection, immunity, autonomic regulation, vascular biology, respiration, cognition, fatigue, post-exertional symptom exacerbation, disability, work participation, social legitimacy, measurement uncertainty, and health-system coordination. Placing it only under infectious disease or respiratory medicine would destroy information.

The faceted coordinate for any concept is:

$$H = (D, L, M, T, X, F, C, E, I)$$

where:

- *D*: domain or target;
- *L*: scale or level;
- *M*: mechanism or process;
- *T*: time and life course;
- *X*: determinants and exposures;
- *F*: function and lived outcome;
- *C*: context and population;
- *E*: evidence, measurement, and uncertainty;
- *I*: intervention and governance.

This is not a numerical score. Each coordinate is a structured set of terms with provenance, dates, relationships, and confidence levels.

10.3. The nine axes in detail

10.3.1. Axis 1 — Domain or target

The domain identifies what is being described: an organism-level state, molecular pathway, organ system, cognitive function, social relationship, functional activity, population distribution, care process, measurement, intervention, or ethical concern. Several domain tags are expected.

10.3.2. Axis 2 — Scale or level

The level locates the relevant unit:

1. molecule and chemical;
2. gene and epigenetic regulation;
3. organelle and cell;
4. tissue and local microenvironment;
5. organ;
6. integrated physiological system;
7. whole person;
8. relationship and household;
9. institution and organization;
10. community and built environment;
11. population and political economy;
12. ecosystem and planetary system.

A mechanism can move across levels. Lead exposure is environmental and political in origin, chemical at entry, molecular and cellular in action, neurodevelopmental in consequence, educational in manifestation, and intergenerational in distribution.

10.3.3. Axis 3 — Mechanism or process

Mechanisms include genetic variation, development, infection, immune activation, inflammation, ischemia, thrombosis, degeneration, neoplasia, toxic injury, endocrine and metabolic dysregulation, neural plasticity, conditioning, reinforcement, social exclusion, resource deprivation, and organizational failure. The same endpoint can arise through different mechanisms, and one mechanism can produce several endpoints.

10.3.4. Axis 4 — Time and life course

Time labels include onset, duration, frequency, periodicity, latency, recurrence, progression, recovery, remission, seasonality, developmental timing, pregnancy, aging, and intergenerational effects. Every measurement should have a timestamp; every interpretation should state the relevant time window.

10.3.5. Axis 5 — Determinants and exposures

Determinants include inherited and de novo variation, prenatal conditions, nutrition, activity, sleep, substance exposure, relationships, trauma, work, income, education, housing, discrimination, geography, pathogens, pollutants, health care, and policy. “Lifestyle” should not be used as a catch-all that obscures constraints, marketing, infrastructure, addiction, or unequal opportunity.

10.3.6. Axis 6 — Function and lived outcome

This axis records symptoms, capacity, activities, participation, role performance, autonomy, quality of life, meaning, satisfaction, safety, caregiver burden, and survival. It is where laboratory change is tested against what matters in life.

10.3.7. Axis 7 — Context and population

Context includes age, sex-related biology, gender, pregnancy, ancestry, language, culture, disability, occupation, socioeconomic position, geography, family structure, care setting, legal status, and historical conditions. These variables can modify exposure, measurement validity, mechanism, treatment effect, and access.

10.3.8. Axis 8 — Measurement and evidence

Every observation should identify modality, device or instrument, protocol, calibration, reference standard, uncertainty, missingness, operator, setting, preprocessing, and interpretation rule. Evidence labels should distinguish direct observation, self-report, proxy report, clinician judgment, algorithmic inference, causal evidence, and expert consensus.

10.3.9. Axis 9 — Intervention and governance

This axis includes primordial prevention, primary prevention, screening, diagnosis, treatment, rehabilitation, palliation, accommodation, social support, environmental modification, public policy, emergency response, regulation, and rights protection. It also records who authorizes, pays, delivers, monitors, and is accountable.

Design principle

The nine-axis representation resolves a common dispute: whether a condition is “biological,” “psychological,” or “social.” Most clinically important conditions are biological **and** psychological **and** social at different causal and descriptive levels. The scientific task is to specify the relationships, not to choose a team.

11. Branch 1 — Whole-person health states and outcomes

This branch contains the high-level variables used to summarize how a person or population is doing. It is deliberately not a residual “general health” category; it is the place where lower-level findings are integrated and tested for relevance.

11.1. Core subdomains

11.1.1. Vital status and longevity

- survival, mortality, cause-specific mortality, premature mortality;
- life expectancy and healthy life expectancy;
- years of life lost;
- recovery, remission, recurrence, and event-free survival.

11.1.2. Morbidity and burden

- incidence, prevalence, severity, duration;
- symptom burden and treatment burden;
- years lived with disability and disability-adjusted life years;
- multimorbidity, complexity, and frailty. ([71–73](#))

11.1.3. Subjective health and well-being

- self-rated health;
- positive and negative affect;
- life satisfaction;
- meaning, purpose, and coherence;
- flourishing and suffering;
- security and hope.

11.1.4. Resilience and reserve

- physiological reserve;
- cognitive reserve;
- coping flexibility;
- social and community resilience;
- recovery time after challenge;
- ability to adapt without unacceptable cost.

11.1.5. Safety and freedom from avoidable harm

- exposure to violence, neglect, unsafe care, occupational hazards, environmental toxins, and preventable infection;
- risk of self-harm, harm from others, and iatrogenic harm;
- medication safety and diagnostic safety.

11.1.6. Person-defined goals

Health cannot be integrated without goals. One person may prioritize survival, another alertness, fertility, pain relief, mobility, independence, caregiving, creative work, or time at home. Goal-concordant care is not an optional “soft” outcome; it determines whether an intervention is beneficial for this person.

11.2. Summary-state dashboards

Whole-person dashboards should avoid one composite score that hides tradeoffs. A better design displays a vector:

$$\mathbf{W} = (S, B, Y, F, P, Q, R, G)$$

where *S* is survival/risk, *B* biological burden, *Y* symptoms, *F* function, *P* participation, *Q* quality of life, *R* resilience/reserve, and *G* goal attainment. Each component has uncertainty and trend.

Worked example

A cancer therapy may reduce tumor burden and improve survival probability while worsening fatigue, cognition, fertility, neuropathy, work participation, and financial security. The whole-person branch holds all outcomes together so that “response” is not reduced to one imaging measure.

12. Branch 2 — Biological foundations and regulation

This branch captures the mechanisms that make all organ systems possible. It is the biological substrate beneath anatomy-based medicine.

12.1. Molecular and chemical foundations

- atoms, ions, water, pH, redox state, gases, metabolites, lipids, carbohydrates, proteins, and nucleic acids;
- binding, catalysis, diffusion, osmosis, electrochemical gradients, and phase behavior;
- pharmacologic and toxicologic interactions;
- energy capture, transfer, storage, and dissipation.

12.2. Genetics, genomics, and epigenetics

- inherited variants, de novo variants, structural variation, mitochondrial genetics;
- gene expression and regulation;
- penetrance, expressivity, pleiotropy, mosaicism, and gene–environment interaction;
- epigenetic marks and chromatin state;
- polygenic architecture and ancestry-dependent performance;
- somatic evolution in cancer and aging.

Genetics is not a separate destiny branch. Genetic effects depend on developmental stage, tissue, exposure, behavior, and social context. A variant may be highly penetrant in one condition and only weakly predictive in another. Population stratification and unequal reference datasets can produce misleading inferences.

12.3. Cells, tissues, and extracellular organization

- membranes, organelles, trafficking, cytoskeleton, adhesion, and polarity;
- stem cells, differentiation, senescence, apoptosis, necrosis, and autophagy;
- extracellular matrix, fibrosis, remodeling, and mechanotransduction;
- tissue niches, barriers, vascular supply, and local immune ecology.

12.4. Information and signaling

- electrical excitability;
- neurotransmission and synaptic plasticity;
- endocrine, paracrine, autocrine, and juxtacrine signaling;
- receptor systems, second messengers, and transcriptional responses;
- circadian timing and oscillatory coordination.

12.5. Homeostasis, allostasis, and control

The body is a network of coupled control systems. Relevant concepts include set points, defended ranges, feedforward control, negative and positive feedback, delays, hysteresis, adaptation, redundancy, reserve, and tradeoffs. Examples include thermoregulation, glucose control, fluid balance, oxygen delivery, acid-base regulation, and immune tolerance.

A simple feedback representation is:

$$e(t) = r(t) - y(t), \quad u(t) = K[e(t), \dot{e}(t), \int e(t) dt]$$

where r is a regulated target, y the sensed state, and u the corrective response. Biological systems differ from engineered controllers because targets can change, sensors are noisy and distributed, actuators share resources, and the controller itself learns and ages.

12.6. Metabolism and energy

- nutrient sensing and substrate use;
- mitochondrial function;
- anabolic and catabolic pathways;
- oxygen delivery and utilization;
- energy expenditure, storage, and thermogenesis;
- metabolic flexibility and reserve.

12.7. Repair, regeneration, and aging

- wound healing and tissue regeneration;
- inflammation resolution;
- proteostasis and DNA repair;
- cellular senescence and stem-cell exhaustion;
- aging hallmarks, frailty, and reduced reserve;
- compensatory remodeling that is initially adaptive and later harmful.

13. Branch 3 — Organ-system health

Organ systems remain the most practical backbone for anatomy, physiology, clinical examination, and many devices. The branch should be treated as a set of interacting subsystems rather than sealed compartments.

13.1. Cardiovascular and circulatory health

Scope includes cardiac anatomy, electrophysiology, excitation–contraction coupling, valves, coronary circulation, arterial and venous systems, microcirculation, lymphatic return, pressure–flow relations, oxygen transport, rhythm, vascular biology, hemostasis, and exercise response. Major condition families include atherosclerotic disease, hypertension, arrhythmia, heart failure, cardiomyopathy, valve disease, congenital heart disease, aortic disease, peripheral vascular disease, venous thromboembolism, and shock.

The prior cardiovascular packet is the dedicated deep dive for this branch. In the atlas, cardiovascular health is also linked to renal sodium handling, endocrine regulation, inflammation, respiration, autonomic control, metabolism, behavior, sleep, pregnancy, environment, and social conditions.

13.2. Respiratory health

Scope includes airway structure, ventilation, diffusion, gas exchange, pulmonary circulation, respiratory mechanics, control of breathing, mucociliary clearance, and defense. Major families include obstructive disease, restrictive and interstitial disease, infection, pulmonary vascular disease, sleep-related breathing disorders, respiratory failure, pleural disease, occupational lung disease, and malignancy.

Respiratory function is tightly coupled to cardiovascular output, hemoglobin, acid–base balance, neuromuscular control, sleep, exercise, air quality, occupation, and infection ecology.

13.3. Nervous-system health

Scope includes central and peripheral nervous systems, autonomic function, sensory and motor systems, neural development, plasticity, and cerebrovascular supply. Families include stroke, epilepsy, headache, neurodegeneration, demyelination, infection, trauma, tumors, movement disorders, neuropathy, neuromuscular disease, and functional neurological disorders.

The nervous-system branch is distinct from the mind/cognition branch for navigation, but the boundary is permeable. Cognition and emotion are embodied functions of brain–body systems; neurological disease has psychological and social consequences; mental disorders have neural and physiological correlates.

13.4. Endocrine and metabolic health

Scope includes hypothalamic–pituitary axes, thyroid, adrenal, pancreatic endocrine function, gonadal hormones, growth, calcium and bone metabolism, appetite, energy balance, and adipose signaling. Families include diabetes, thyroid disorders, adrenal disorders, pituitary disease, dyslipidemia, inborn errors of metabolism, metabolic syndrome, and obesity-related complications.

“Metabolic health” should not be reduced to body size. It includes glucose regulation, lipoprotein biology, liver and muscle metabolism, adipose distribution and function, inflammation, fitness, sleep, medication effects, and social determinants of food and activity.

13.5. Gastrointestinal, hepatobiliary, and pancreatic health

Scope includes ingestion, motility, digestion, absorption, barrier function, enteric nervous control, liver metabolism and detoxification, bile production, pancreatic exocrine function, and gut microbial ecology. Families include reflux, motility disorders, inflammatory bowel disease, celiac disease, infection, liver disease, pancreatitis, malabsorption, functional gastrointestinal disorders, and cancer.

13.6. Renal and urinary health

Scope includes filtration, tubular transport, fluid and electrolyte balance, acid-base control, endocrine functions, blood-pressure regulation, urinary storage and elimination, and reproductive/urinary anatomical overlap. Families include acute kidney injury, chronic kidney disease, glomerular and tubular disease, stones, infection, obstruction, incontinence, and malignancy.

Renal status modifies drug handling, cardiovascular risk, anemia, mineral-bone health, cognition, and pregnancy. Kidney disease exemplifies why organ-system and cross-system representations must coexist.

13.7. Hematologic health

Scope includes hematopoiesis, erythrocytes, leukocytes, platelets, coagulation, fibrinolysis, blood groups, marrow, and oxygen carrying. Families include anemia, hemoglobinopathy, cytopenia, myeloproliferative disease, leukemia, lymphoma overlap, bleeding disorders, and thrombophilia.

13.8. Immune and lymphatic health

Scope includes innate and adaptive immunity, tolerance, inflammation, lymphatic transport, lymphoid organs, barrier immunity, vaccination, and immune surveillance. Families include immunodeficiency, allergy, autoimmunity, autoinflammation, transplant rejection, and immune-mediated injury. Infection and immunity receive their own integrative branch because they cross every organ.

13.9. Musculoskeletal and connective-tissue health

Scope includes bone, cartilage, skeletal muscle, tendon, ligament, fascia, joints, biomechanics, movement, and connective tissue. Families include fracture, osteoporosis, osteoarthritis, inflammatory arthritis, myopathy, tendon injury, spine disorders, congenital and genetic connective-tissue disorders, and pain syndromes.

Musculoskeletal health depends on neural control, cardiovascular and respiratory capacity, nutrition, endocrine state, occupation, built environment, injury prevention, rehabilitation, and social participation.

13.10. Integumentary health

Scope includes skin, hair, nails, glands, barrier function, thermoregulation, sensation, immune defense, repair, appearance, and social meaning. Families include dermatitis, infection, wound, burn, ulcer, pigment disorder, autoimmune disease, and skin cancer.

13.11. Sensory health

This subbranch covers vision, hearing, vestibular function, smell, taste, touch, proprioception, nociception, and sensory integration. Sensory loss affects communication, mobility, cognition, safety, education, employment, and social connection. Assistive technologies and environmental design are therefore part of sensory health, not merely downstream accommodations.

13.12. Oral, dental, and craniofacial health

Scope includes teeth, periodontium, oral mucosa, salivary function, jaw, mastication, swallowing interfaces, speech, craniofacial development, pain, infection, and cancer. Oral health is often administratively separated from medicine despite links to nutrition, diabetes, pregnancy, cardiovascular risk markers, infection, sleep, and self-esteem.

13.13. Reproductive and sexual health

Scope includes reproductive anatomy and physiology, fertility, contraception, sexually transmitted infection, sexual function, gender-affirming care, reproductive endocrinology, and reproductive rights. Sexual health includes safety, consent, pleasure, function, identity, relationship context, and freedom from coercion.

13.14. Maternal, fetal, and perinatal health

Pregnancy is a dynamic multi-organ state, not a diagnosis isolated within obstetrics. It changes cardiovascular load, coagulation, immune regulation, renal filtration, respiration, metabolism, pharmacokinetics, and mental health. Fetal, placental, maternal, and family health are linked but not identical, creating real ethical and clinical tradeoffs.

13.15. Integrated homeostatic systems

Some systems are defined by coordination rather than anatomy:

- autonomic regulation;
- sleep and circadian organization;
- thermoregulation;
- fluid, electrolyte, and acid-base balance;
- exercise and cardiorespiratory fitness;
- pain and interoception;
- stress response;
- energy balance;
- sexual and reproductive coordination.

These are frequent “homeless” domains in specialty trees. The atlas gives them explicit cross-system status.

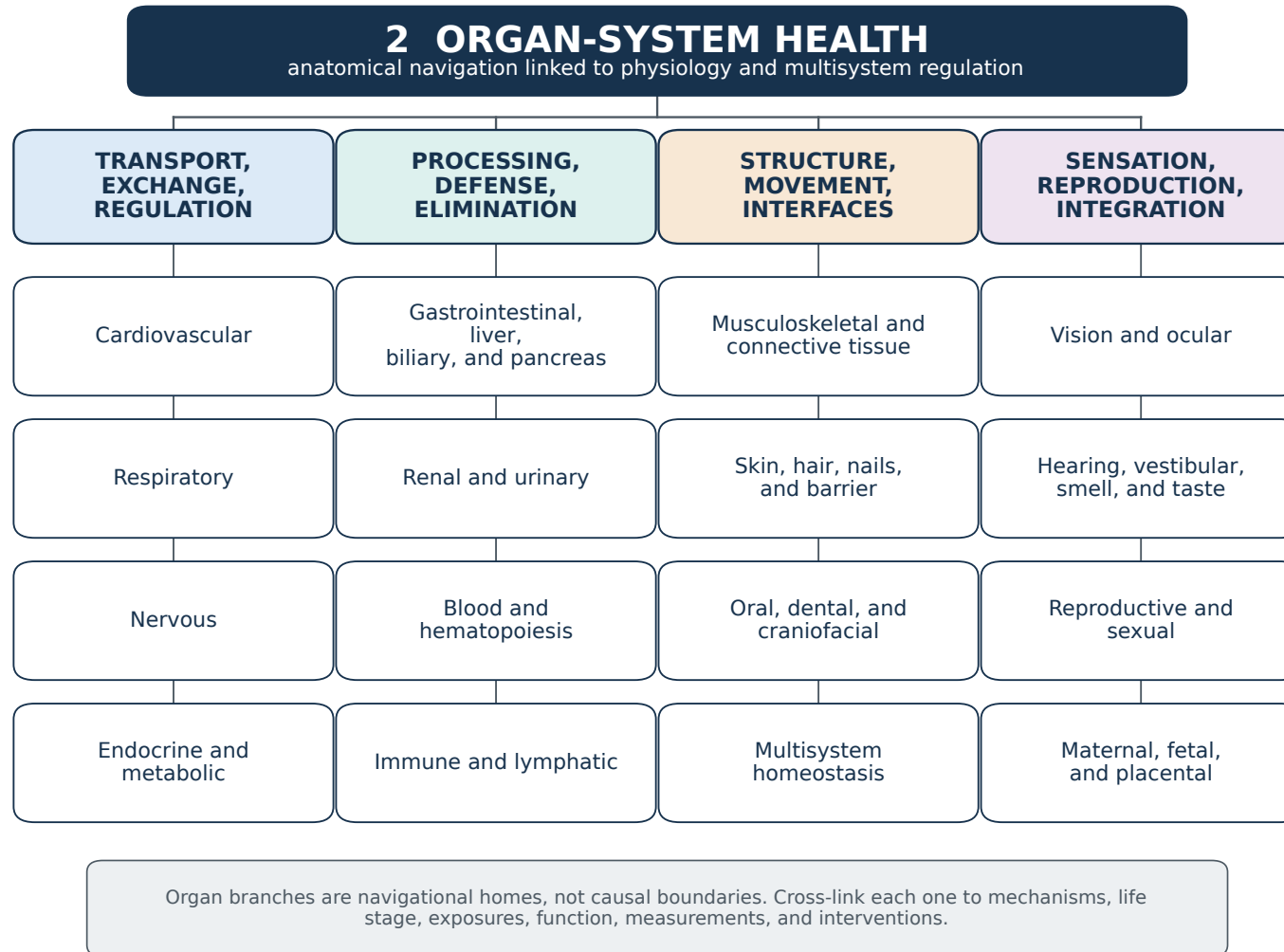


Figure 5.: Organ-system navigation. The lower band contains integrative systems that cannot be assigned to one organ alone. Every organ branch should be cross-linked to mechanisms, life course, function, determinants, measurements, and interventions.

14. Branch 4 — Cross-system pathological processes

This branch organizes disease by *how it happens*, regardless of where it occurs. Mechanism-centered organization is essential for understanding shared causal pathways, comorbidity, and therapies that act across organs.

14.1. Genetic, genomic, and congenital processes

- monogenic, chromosomal, mitochondrial, and polygenic mechanisms;
- developmental malformations;
- imprinting, mosaicism, and epigenetic dysregulation;
- gene–environment and gene–treatment interaction;
- inherited susceptibility versus deterministic disease.

14.2. Infectious processes

- invasion, colonization, toxigenesis, immune evasion, persistence, latency, and transmission;
- bacterial, viral, fungal, parasitic, prion, and polymicrobial disease;
- local, systemic, congenital, vector-borne, foodborne, airborne, contact, and healthcare-associated infection;
- post-infectious and chronic sequelae.

14.3. Inflammation and immune dysregulation

- acute and chronic inflammation;
- allergy and hypersensitivity;
- autoimmunity and autoinflammation;
- immunodeficiency;
- cytokine dysregulation;
- immune tolerance failure;
- transplant and graft reactions;
- immune exhaustion and immunosenescence.

14.4. Neoplasia

- initiation, clonal evolution, tumor microenvironment, invasion, metastasis, angiogenesis, immune escape, and treatment resistance;
- benign, premalignant, malignant, hematologic, and solid neoplasms;
- hereditary susceptibility, exposure, infection-associated cancer, and stochastic somatic change.

14.5. Vascular, ischemic, thrombotic, and hemorrhagic processes

- endothelial dysfunction;

- atherosclerosis and arteriosclerosis;
- thrombosis and embolism;
- ischemia and reperfusion injury;
- aneurysm and dissection;
- microvascular dysfunction;
- hemorrhage and hemostatic failure.

These processes connect heart disease, stroke, kidney disease, retinal disease, pregnancy complications, pulmonary embolism, and peripheral vascular disease.

14.6. Endocrine and metabolic dysregulation

- hormone excess, deficiency, resistance, and timing disorders;
- substrate excess or deficiency;
- toxic metabolite accumulation;
- metabolic inflexibility;
- dyslipidemia;
- protein-energy malnutrition;
- vitamin and mineral imbalance.

14.7. Degeneration and loss of repair

- proteotoxicity, mitochondrial dysfunction, oxidative injury, senescence, stem-cell exhaustion, fibrosis, and network failure;
- neurodegeneration, osteoarthritis, organ fibrosis, and age-related decline;
- compensation that masks early damage until reserve is exhausted.

14.8. Trauma and injury

- mechanical, blast, thermal, cold, electrical, radiological, chemical, and psychological trauma;
- acute damage, secondary injury cascades, repair, chronic pain, disability, and post-traumatic adaptation;
- unintentional injury, violence, self-harm, occupational injury, and disaster.

14.9. Toxicologic and exposure-related processes

- dose, route, duration, latency, mixture effects, bioaccumulation, susceptibility, and endocrine disruption;
- medication toxicity, substance-related injury, pollutants, heavy metals, pesticides, solvents, radiation, and occupational agents;
- interaction with nutrition, genetics, age, pregnancy, and kidney/liver function.

14.10. Mechanical and structural dysfunction

- obstruction, compression, instability, rupture, stenosis, insufficiency, overload, malalignment, and altered compliance;

- pressure-volume mismatch;
- repetitive strain and biomechanical adaptation;
- device–tissue interactions.

14.11. Pain, interoceptive, and functional syndromes

Pain can be nociceptive, neuropathic, nociplastic, inflammatory, ischemic, visceral, or mixed. Symptoms can persist after tissue healing through altered processing, prediction, learning, autonomic changes, sleep disruption, and social reinforcement—without being imaginary. Functional syndromes require rigorous positive characterization, exclusion of dangerous alternatives, and avoidance of false mind–body dichotomies.

14.12. Iatrogenic and care-related harm

- adverse drug events and interactions;
- procedural complications;
- healthcare-associated infection;
- radiation exposure;
- overdiagnosis and overtreatment;
- delayed, missed, or incorrect diagnosis;
- fragmentation, communication failure, and access delay;
- withdrawal, rebound, dependence, and treatment burden.

14.13. Aging, frailty, and loss of reserve

Aging is not one disease. It changes risk through accumulated damage, altered repair, immunosenescence, endocrine change, body-composition change, vascular stiffness, sensory loss, and reduced reserve. Frailty is a system-level state of vulnerability to stressors. It predicts that a small disturbance may cause disproportionate decline.

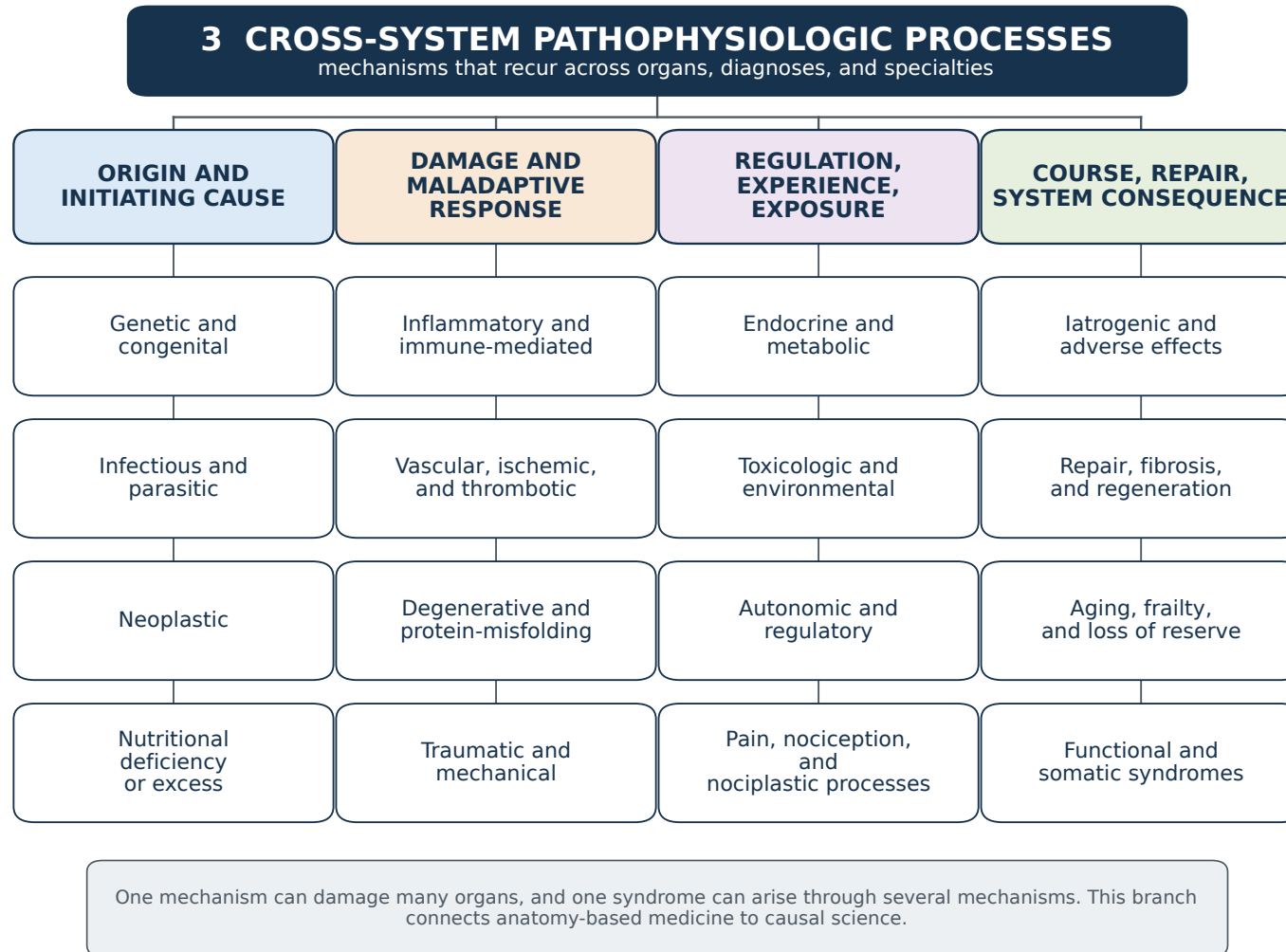


Figure 6.: Cross-system mechanisms. A single mechanism can affect several organs; one clinical syndrome can be produced by several mechanisms. This branch is the principal bridge between anatomy-based medicine and causal science.

15. Branch 5 — Brain, cognition, emotion, and behavior

This branch describes functions that emerge from nervous-system activity interacting with body, development, learning, relationship, culture, and environment. It includes mental health but is broader than psychiatric disorder.

15.1. Consciousness, arousal, and awareness

- wakefulness, alertness, sleep–wake transitions, anesthesia, delirium, coma, and altered states;
- awareness of self, body, environment, and internal state;
- attentional engagement and vigilance;
- disorders of consciousness and fluctuating cognition.

Consciousness is both a neurological function and a clinical condition affecting consent, safety, communication, and prognosis.

15.2. Perception and interoception

- visual, auditory, somatosensory, vestibular, olfactory, and gustatory perception;
- body ownership, pain, temperature, hunger, thirst, dyspnea, heartbeat perception, and visceral sensation;
- sensory integration, prediction, salience, and perceptual learning;
- hallucination, illusion, neglect, agnosia, and altered sensory weighting.

15.3. Attention and cognitive control

- selective, sustained, divided, and spatial attention;
- working memory;
- inhibition, task switching, planning, error monitoring, and decision-making;
- cognitive effort, fatigue, and executive reserve.

15.4. Learning and memory

- encoding, consolidation, storage, retrieval, and reconsolidation;
- declarative, procedural, emotional, spatial, and autobiographical memory;
- reinforcement learning, habit, extinction, and generalization;
- developmental learning differences and acquired amnesia.

15.5. Language and communication

- speech production, comprehension, reading, writing, pragmatics, gesture, and augmentative communication;
- aphasia, dysarthria, apraxia of speech, language-development differences, and social communication;

- communication access, health literacy, and interpreter needs.

15.6. Emotion, threat, and regulation

- positive and negative affect;
- threat detection and fear learning;
- anger, sadness, joy, shame, guilt, disgust, and attachment-related emotion;
- regulation through attention, appraisal, behavior, relationship, physiology, and environment;
- emotional granularity and flexibility.

15.7. Motivation, reward, and goal-directed behavior

- wanting, liking, reinforcement, incentive salience, effort, reward prediction, and habit;
- anhedonia, apathy, impulsivity, compulsivity, and addiction;
- goal formation, persistence, conflict, and self-efficacy.

15.8. Social cognition and attachment

- recognition of emotion and intention;
- empathy, perspective-taking, trust, cooperation, competition, and moral judgment;
- attachment, belonging, rejection sensitivity, and social learning;
- cultural scripts and group identity.

15.9. Personality, identity, and self

- temperament and trait dimensions;
- self-concept, narrative identity, body image, gender identity, values, and agency;
- stability and change across life;
- personality pathology as maladaptive patterns rather than a total description of a person.

15.10. Behavior and health behavior

Behavior includes movement, eating, sleep routines, medication use, substance use, sexual behavior, risk behavior, caregiving, help-seeking, and adherence. A behavior is rarely explained by knowledge alone. It is shaped by reward, stress, habit, social norms, marketing, cost, opportunity, environment, addiction, executive function, and trust.

15.11. Sleep and circadian health

Sleep is a neurobehavioral state with respiratory, cardiovascular, immune, metabolic, cognitive, and emotional consequences. Subdomains include duration, continuity, architecture, timing, regularity, circadian alignment, sleepiness, parasomnias, insomnia, sleep-disordered breathing, and environmental conditions.

15.12. Neurodevelopment

- sensory, motor, language, cognitive, emotional, and social development;
- attention-deficit/hyperactivity, autism-spectrum, intellectual-developmental, learning, communication, and motor conditions;
- variability, support needs, adaptive function, strengths, and environmental fit;
- early adversity and developmental plasticity.

15.13. Psychiatric and behavioral disorder families

This atlas follows neither one DSM-like list nor one dimensional model exclusively. For navigation, major families include:

- neurodevelopmental conditions;
- psychotic-spectrum phenomena;
- bipolar and depressive syndromes;
- anxiety, obsessive-compulsive, and trauma-related syndromes;
- dissociative and somatic-symptom-related syndromes;
- feeding and eating disorders;
- elimination and sleep–wake disorders;
- sexual dysfunction and gender-related clinical needs where applicable;
- disruptive, impulse-control, and conduct syndromes;
- substance-related and addictive disorders;
- neurocognitive disorders;
- personality dysfunction;
- paraphilic disorders as clinically and ethically defined;
- medication-induced and medical-condition-related syndromes.

Categorical diagnoses should be supplemented with dimensions of distress, impairment, risk, cognition, emotion, behavior, sleep, substance exposure, trauma, and social context. ([4](#),[10](#),[11](#),[43](#))

15.14. Cognitive aging and dementia

Cognitive aging includes normal variation, reserve, sensory effects, medication effects, mood, sleep, vascular burden, neurodegeneration, and social engagement. Dementia syndromes should be represented by cognitive profile, functional impact, suspected etiology, biomarkers where valid, trajectory, behavioral symptoms, caregiver context, safety, and goals.

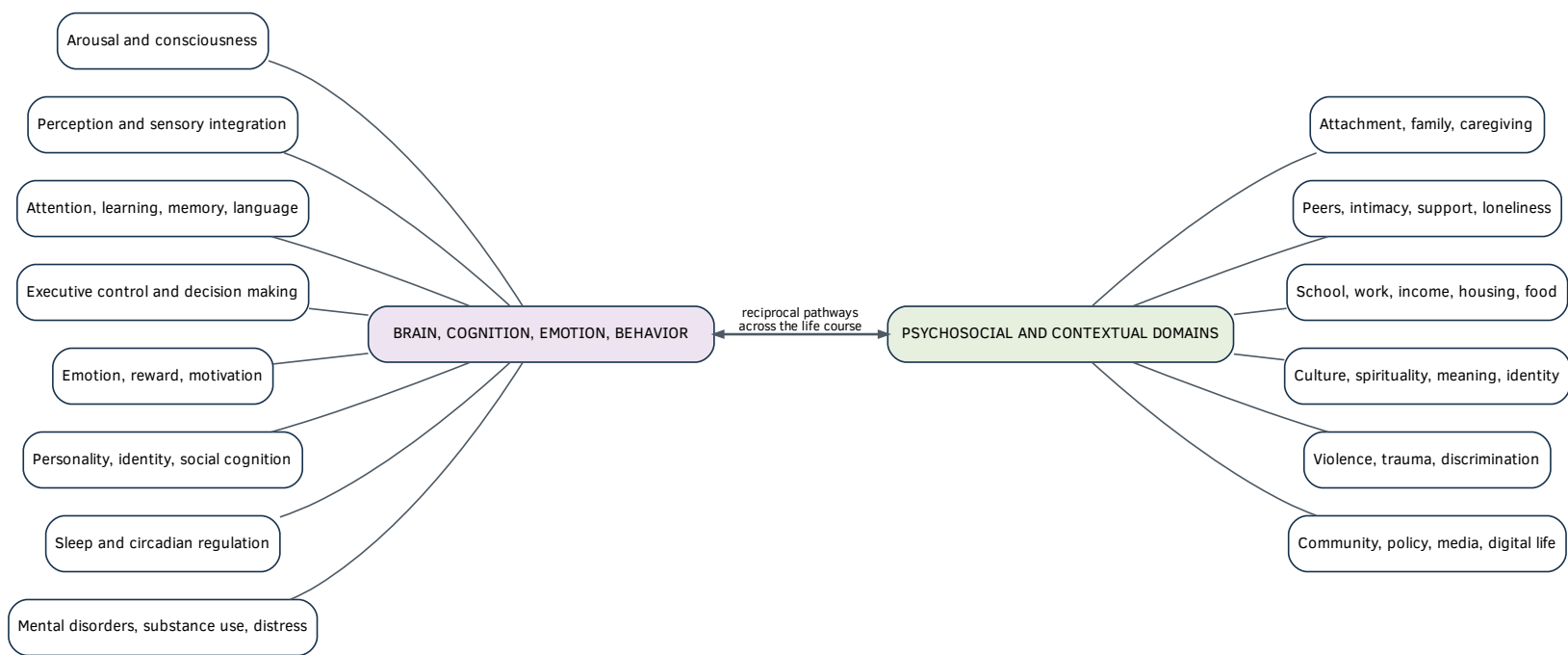


Figure 7.: Brain, mind, behavior, and psychosocial life are analytically distinguishable but causally reciprocal. The bridge is bidirectional: social and cultural conditions shape neural development and behavior, while cognition, emotion, and behavior reshape relationships and environments.

16. Branch 6 — Psychosocial, relational, cultural, and existential health

This branch contains the interpersonal and meaning-bearing conditions of life. It should not be treated merely as a collection of “risk factors” attached to biological disease.

16.1. Attachment, family, and intimate relationships

- attachment security and caregiving;
- family structure and function;
- parenting, child development, co-parenting, and intergenerational relationships;
- intimate partnership, sexual relationship, conflict, coercion, and support;
- separation, bereavement, adoption, foster care, and family reunification.

Relationships can buffer stress, support treatment, and create meaning; they can also be sources of violence, coercion, neglect, or chronic strain.

16.2. Social connection and loneliness

Social health includes network size, relationship quality, perceived support, reciprocity, belonging, loneliness, isolation, and community integration. Meta-analytic evidence links social relationships with mortality risk, though mechanisms and causal pathways include behavior, stress regulation, inflammation, material support, and care access. (74)

A person can be socially isolated without feeling lonely or lonely in a crowd. Both objective and subjective dimensions matter.

16.3. Violence, abuse, trauma, and safety

- child maltreatment, intimate-partner violence, sexual violence, elder abuse, bullying, community violence, war, trafficking, and institutional abuse;
- direct exposure, witnessing, threat, coercive control, and historical trauma;
- acute injury, post-traumatic symptoms, developmental effects, chronic disease risk, substance use, and social consequences.

Adverse childhood experiences research demonstrated graded associations with later health, but a cumulative count is not a deterministic diagnosis. Type, severity, timing, duration, protective relationships, and structural context matter. (65)

16.4. Work, school, and role health

- workload, control, demand, reward, insecurity, schedule, shift work, exposure, injury, and organizational justice;
- learning environment, inclusion, bullying, academic demand, support, and special education;
- role identity, unemployment, retirement, and meaningful contribution;
- work–family conflict and caregiving burden.

16.5. Material conditions

- income and wealth;
- food security and nutrition access;
- housing quality, stability, crowding, and homelessness;
- transport, utilities, digital access, and neighborhood resources;
- debt, financial toxicity, and legal vulnerability.

These are causal determinants and intervention targets, not merely “background.”

16.6. Culture, language, religion, and spirituality

Culture shapes symptom expression, explanatory models, trust, healing practices, family roles, identity, stigma, and treatment preferences. Language access is a safety issue. Religion and spirituality may provide meaning, community, ritual, coping, moral conflict, or trauma. Cultural competence should not reduce people to group stereotypes; it requires humility, inquiry, and attention to power.

16.7. Discrimination and structural inequity

- racism, sexism, ableism, ageism, classism, homophobia, transphobia, xenophobia, and religious discrimination;
- interpersonal, institutional, structural, historical, and algorithmic forms;
- segregation, environmental injustice, unequal schooling, labor exclusion, legal discrimination, and differential care.

Race is a social and political classification with ancestry-correlated biological patterns in some contexts; it should not be used as a crude biological proxy when the actual causal variable can be measured. Race-based clinical corrections can entrench inequity when they delay care or misattribute socially produced differences to innate biology. (75,76)

16.8. Migration, legal status, and displacement

Health may be affected by migration stress, family separation, trauma, detention, language barriers, occupational exploitation, legal exclusion, interrupted care, climate displacement, and cultural transition. Migrant status is heterogeneous; it should not be treated as a universal risk label.

16.9. Digital and media environments

- social media, gaming, online communities, misinformation, cyberbullying, harassment, parasocial relationships, and digital work;
- health information access and telehealth;
- attention capture, sleep disruption, surveillance, and algorithmic shaping;
- assistive technology and connection for isolated or disabled people.

Effects depend on content, pattern, purpose, person, and context; screen-time totals alone are generally weak explanations.

16.10. Meaning, values, morality, and existential health

- purpose, coherence, identity, dignity, hope, grief, mortality awareness, spiritual struggle, and moral injury;
- autonomy and authenticity;
- reconciliation, forgiveness, and legacy;
- suffering that is not reducible to symptom counts.

Existential distress can occur with or without psychiatric disorder. Palliative care, psychotherapy, chaplaincy, peer support, community, and creative practice may address different aspects.

17. Branch 7 — Development, reproduction, aging, and the life course

This branch organizes health by time, transitions, and developmental tasks rather than by organ.

17.1. Preconception and intergenerational health

- parental health, genetics, fertility, nutrition, infection, toxic exposure, medication, and social conditions;
- gamete and epigenetic mechanisms;
- reproductive autonomy and access;
- intergenerational transmission through biology, resources, behavior, and institutions.

17.2. Prenatal, fetal, and placental development

- implantation, placentation, organogenesis, growth, and maturation;
- maternal–fetal exchange and immune tolerance;
- congenital anomalies, growth restriction, preterm birth, and exposure effects;
- pregnancy complications as predictors of later maternal and offspring health.

17.3. Infancy and early childhood

- attachment, feeding, sleep, sensory and motor development, language, immunization, and safety;
- growth and developmental surveillance;
- early intervention;
- caregiver mental health and family resources;
- exposure to adversity and protective relationships.

17.4. Middle childhood and adolescence

- learning, peer relationships, identity, puberty, sleep timing, risk behavior, autonomy, and school context;
- chronic illness self-management;
- mental-health emergence;
- sexual health and consent;
- transition from pediatric to adult services.

17.5. Adulthood

- work, partnership, parenting, caregiving, reproductive decisions, chronic-disease prevention, substance use, and injury;
- life transitions, migration, unemployment, and social support;
- cumulative exposure and early manifestations of long-latency disease.

17.6. Reproductive transitions

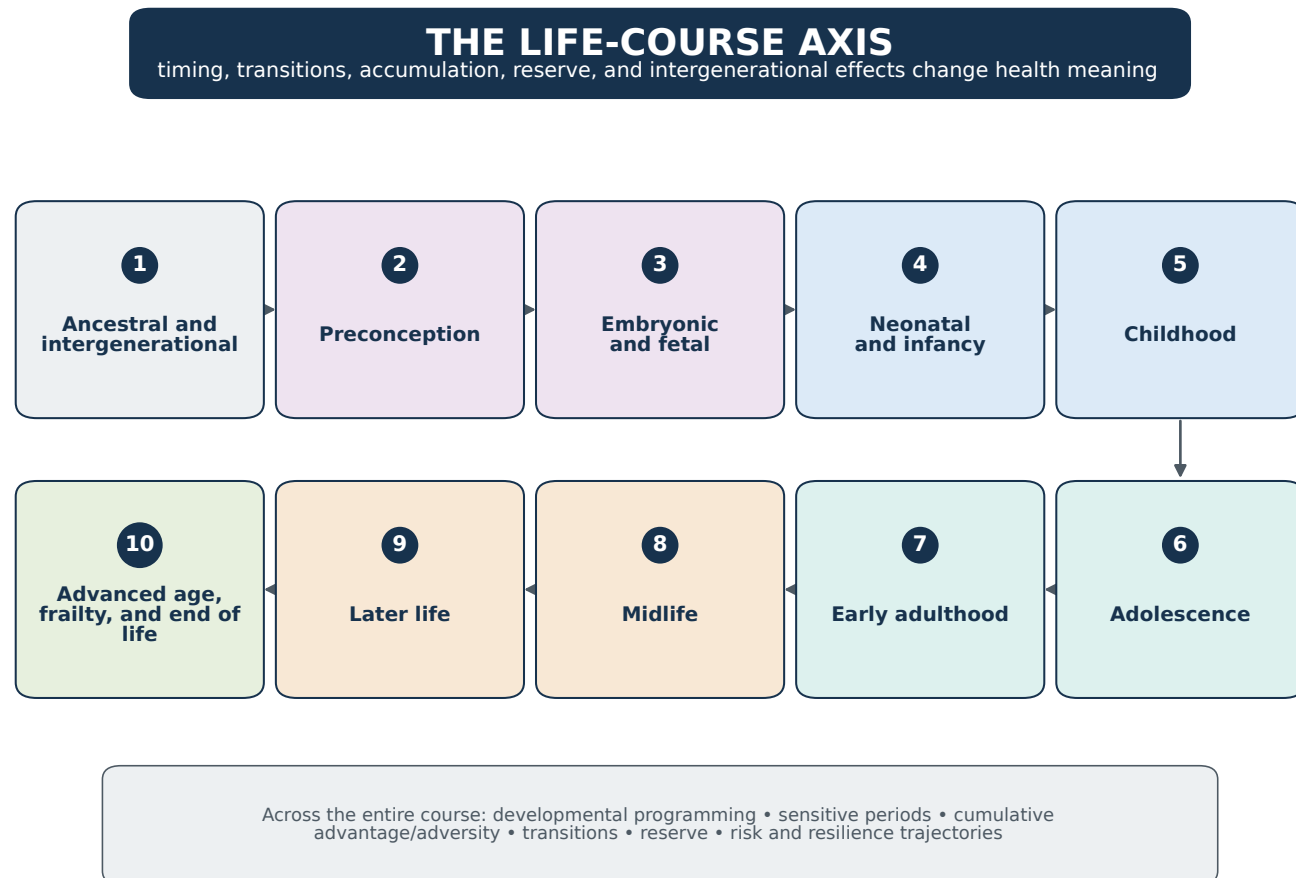
- puberty, menstruation, fertility, pregnancy, postpartum, lactation, contraception, infertility, menopause, and andropause-related changes;
- sexual function and identity across life;
- endocrine, cardiovascular, bone, mental-health, and social consequences.

17.7. Aging and later life

- sensory, cognitive, musculoskeletal, cardiovascular, renal, immune, and metabolic change;
- multimorbidity, frailty, falls, polypharmacy, and social isolation;
- retirement, caregiving, bereavement, housing, transport, and ageism;
- autonomy, supported decision-making, advance care planning, and end-of-life goals.

17.8. Death, dying, grief, and bereavement

End-of-life health includes symptom control, communication, dignity, caregiver support, cultural practice, spiritual needs, and goal-concordant decisions. Death is both a biological event and a social process. Bereavement can involve normal grief, prolonged grief disorder, depression, trauma, practical disruption, and community ritual.



The same exposure, biomarker, diagnosis, or intervention can have different implications at different ages and transition points.

Figure 8.: *The life-course axis. Timing, transition, accumulation, and reserve alter the meaning of the same exposure, measurement, diagnosis, or intervention.*

18. Branch 8 — Infection, immunity, host–microbe biology, and biosecurity

This branch integrates processes that are usually divided among infectious disease, microbiology, immunology, epidemiology, veterinary medicine, and public health.

18.1. Host defense architecture

- physical and chemical barriers;
- innate sensing and effector systems;
- adaptive immunity, memory, and tolerance;
- mucosal immunity;
- immunologic development and aging;
- nutritional and metabolic support;
- genetic susceptibility and acquired immunodeficiency.

18.2. Microbial ecology

Humans live within microbial ecosystems. The microbiome includes bacteria, viruses, fungi, archaea, and their genes and metabolites across body sites. Associations between microbial composition and disease are abundant; causal inference is harder because diet, medication, geography, age, disease, and measurement methods alter composition. The Human Microbiome Project established reference approaches, while subsequent work emphasizes function and ecology over a single “healthy microbiome.” (77,78)

18.3. Infection process

- exposure and dose;
- route of entry;
- colonization and invasion;
- virulence and toxin production;
- host response and immunopathology;
- local versus systemic spread;
- latency, persistence, reactivation, and shedding;
- clearance, immunity, chronic sequelae, and post-infectious syndromes.

18.4. Transmission and population dynamics

- contact structure, superspreading, seasonality, vectors, reservoirs, mobility, and behavior;
- reproduction number, generation interval, serial interval, and attack rate;
- asymptomatic and presymptomatic transmission;
- surveillance bias and case definition;
- vaccine coverage, effectiveness, waning, and immune escape.

18.5. Vaccination and immune prevention

Vaccination is simultaneously an immunologic intervention, individual preventive service, population program, manufacturing system, communication challenge, and governance issue. Evaluation includes safety, immunogenicity, efficacy, effectiveness, durability, transmission effect, equity, and public trust.

18.6. Antimicrobial resistance

Resistance emerges through mutation, gene transfer, selection, prescribing, agriculture, sanitation, manufacturing discharge, global travel, and inadequate infection prevention. A One Health approach is necessary because resistant organisms and genes move among people, animals, food, water, and environments.

18.7. Outbreaks, epidemics, pandemics, and biosecurity

- detection and laboratory confirmation;
- case definitions and contact tracing;
- genomic surveillance;
- risk communication;
- infection prevention and control;
- healthcare capacity;
- countermeasure development;
- supply chains and workforce protection;
- biosafety, biosecurity, and dual-use governance.

Biosecurity includes natural, accidental, and deliberate biological risks. Preparedness must avoid stigmatization, excessive surveillance, and inequitable restrictions while preserving rapid response.

18.8. Immune-mediated disease

The branch also includes allergy, autoimmunity, autoinflammation, immunodeficiency, transplant immunology, tumor immunity, maternal–fetal immune interactions, and chronic inflammatory consequences. These are cross-linked to organ systems and therapeutics.

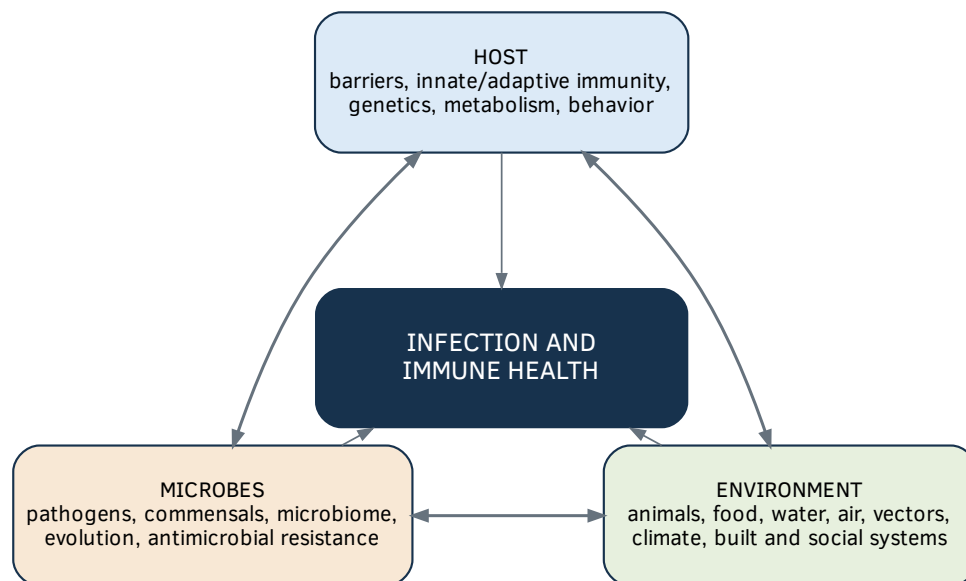


Figure 9.: *Host–microbe–environment integration. Infection and immune health cannot be understood from the pathogen alone; susceptibility, ecology, behavior, infrastructure, and governance shape exposure and outcome.*

19. Branch 9 — Environmental, occupational, One Health, and planetary health

19.1. Physical environment

- heat, cold, humidity, altitude, noise, vibration, light, radiation, and weather extremes;
- air, water, soil, and indoor environmental quality;
- housing, ventilation, crowding, dampness, mold, and combustion;
- built-environment design, transport, walkability, green space, and accessibility.

19.2. Chemical environment

- ambient and household air pollutants;
- metals, solvents, pesticides, persistent organic pollutants, endocrine-active compounds, and microplastics;
- drinking-water contaminants;
- occupational and consumer-product exposures;
- mixtures, cumulative exposure, latency, and vulnerable windows.

19.3. Biological environment

- pathogens, vectors, allergens, molds, harmful algal blooms, animal contact, and biodiversity;
- food and water microbiology;
- ecosystem disruption and zoonotic spillover.

19.4. Occupational health

Occupational health includes injury, ergonomics, noise, radiation, chemical exposure, infection, heat, shift work, psychosocial demand, job control, harassment, precarity, and organizational safety culture. Work can provide income, identity, structure, and community; it can also create exposure and exclusion.

19.5. Climate and planetary change

Climate-related health pathways include:

- heat illness and cardiovascular/renal strain;
- wildfire smoke and air pollution;
- changing vector and infectious ecology;
- food and water insecurity;
- injury from extreme events;
- displacement and conflict;
- mental-health effects, grief, and anxiety;
- health-system disruption;
- unequal burden on communities with fewer resources.

Planetary health includes climate but extends to biodiversity, land use, oceans, freshwater, biogeochemical cycles, and the sustainability of food, energy, transport, and healthcare systems. (70)

19.6. Environmental justice

Exposure and protection are not distributed randomly. Zoning, segregation, labor markets, infrastructure investment, colonial history, and political power shape who lives near hazards and who benefits from resources. Environmental measurement should therefore be linked to demography, occupation, housing, and governance—not interpreted as an individual lifestyle variable.

19.7. One Health operations

One Health connects human clinical data, veterinary surveillance, wildlife ecology, agriculture, food systems, environmental sampling, and antimicrobial use. It is especially relevant to zoonoses, antimicrobial resistance, food safety, toxic exposure, and ecosystem change. (68,69)

20. Branch 10 — Function, disability, rehabilitation, and participation

20.1. From impairment to participation

The ICF's core insight is that functioning is multi-level and context-dependent. The taxonomy for this branch includes:

1. **body functions** — physiological and psychological functions;
2. **body structures** — anatomical parts;
3. **activities** — execution of tasks;
4. **participation** — involvement in life situations;
5. **environmental facilitators and barriers**;
6. **personal context and goals**.

A diagnosis does not determine function. Two people with the same lesion may differ because of severity, reserve, pain, cognition, assistive technology, home design, employment, support, and opportunity.

20.2. Functional domains

- mobility, transfers, balance, endurance, and dexterity;
- self-care and domestic life;
- communication;
- learning and applying knowledge;
- cognition and executive function;
- interpersonal interaction;
- education and work;
- community, civic, and recreational participation;
- sexual function and parenting;
- sleep, energy, and symptom interference.

20.3. Rehabilitation

Rehabilitation includes restoration, compensation, adaptation, skill learning, assistive technology, environmental modification, caregiver training, and participation support. Major professions include physical therapy, occupational therapy, speech-language pathology, rehabilitation medicine, psychology, nursing, prosthetics/orthotics, audiology, vision rehabilitation, social work, and vocational services.

20.4. Disability models

- **medical model:** disability as impairment requiring cure or management;
- **social model:** disability produced by environmental and social barriers;
- **biopsychosocial/ICF model:** interaction among health condition, function, activity, participation, and context;
- **rights model:** disabled people as rights holders entitled to equality, accessibility, autonomy, and participation.

Each contributes. The social and rights models correct the assumption that the person is the sole site of intervention. The UN Convention on the Rights of Persons with Disabilities provides a governance framework for participation and non-discrimination. (59)

20.5. Assistive and adaptive technology

- mobility aids, prostheses, orthoses, exoskeletons;
- hearing aids, cochlear implants, visual aids;
- augmentative and alternative communication;
- environmental controls and smart-home systems;
- cognitive supports and reminders;
- accessibility software;
- workplace and educational accommodation.

Technology success should be evaluated by real-world use, fit, maintenance, burden, stigma, affordability, and participation—not only laboratory performance.

21. Branch 11 — Population health, prevention, and public health

21.1. Units and distributions

Population health asks not only the average level of health but its distribution. Core descriptors include incidence, prevalence, mortality, survival, burden, inequality, exposure distribution, service coverage, and preventable fraction. Stratification should be scientifically justified and ethically interpreted.

21.2. Prevention continuum

- **primordial prevention:** prevent emergence of risk-producing social and environmental conditions;
- **primary prevention:** prevent disease before onset;
- **secondary prevention:** detect early disease or risk states and intervene;
- **tertiary prevention:** reduce complications and disability;
- **quaternary prevention:** protect against unnecessary medical activity and harm. (79,80)

21.3. Screening

Screening evaluates apparently asymptomatic populations. Valid programs require an important condition, an understood natural history, a suitable test, effective early intervention, acceptable harms, feasible follow-up, and favorable balance over the full pathway—not merely high analytic accuracy. (81,82)

Harms include false positives, false negatives, overdiagnosis, incidental findings, anxiety, procedure complications, inequitable follow-up, and opportunity cost.

21.4. Health promotion

The Ottawa Charter framed health promotion through healthy policy, supportive environments, community action, personal skills, and reoriented health services. It emphasized that health is created in the settings of everyday life, not only in clinics. (83)

21.5. Public-health functions

Public-health systems assess population health, investigate hazards, communicate, mobilize communities, develop policy, enforce protection, assure access, build workforce and infrastructure, evaluate, and innovate. The updated U.S. Essential Public Health Services explicitly center equity. (84)

21.6. Social determinants of health

The WHO defines social determinants as the conditions in which people are born, grow, work, live, and age and the wider forces shaping them. Domains include economic policy, education, labor, housing, food systems, social protection, discrimination, and political power. (85–87)

21.7. Commercial determinants

Commercial actors shape health through products, marketing, pricing, lobbying, labor conditions, data practices, and influence on science and policy. The category includes harmful products but also broader corporate practices and political economy. (88)

21.8. Global health and health security

Global health examines transnational determinants, inequity, infectious threats, noncommunicable disease, maternal and child health, injury, migration, humanitarian crises, and health systems. Health security focuses on capacities to prevent, detect, and respond to threats; it should be balanced with universal health needs and rights.

21.9. Burden of disease

The Global Burden of Disease framework estimates mortality, years of life lost, years lived with disability, DALYs, and attributable risk across places and time. It enables comparison but depends on modeling assumptions, disability weights, data quality, causal attribution, and uncertainty. Rankings should not substitute for local priorities or lived experience. (73,89,90)

22. Branch 12 — Health care, professions, and delivery systems

22.1. Care continuum

- self-care and family care;
- community and public-health services;
- primary care;
- emergency and acute care;
- specialty and subspecialty care;
- surgical and procedural care;
- mental-health and substance-use care;
- rehabilitation;
- long-term services and supports;
- palliative and hospice care;
- laboratory, imaging, pharmacy, and diagnostic services;
- remote, home-based, and digital care.

22.2. Professional domains

Health work is distributed among physicians, nurses, pharmacists, dentists, psychologists, social workers, therapists, public-health practitioners, laboratorians, engineers, informaticians, community health workers, peer specialists, caregivers, administrators, and many others. Scope varies by jurisdiction.

The atlas organizes professions by **functions**, not prestige:

1. assessment and diagnosis;
2. monitoring and surveillance;
3. prevention and health promotion;
4. intervention and treatment;
5. rehabilitation and adaptation;
6. care coordination and navigation;
7. communication, counseling, and education;
8. community and structural intervention;
9. research, quality, and safety;
10. governance, logistics, and stewardship.

22.3. Delivery-system properties

- access, timeliness, continuity, coordination, comprehensiveness, and cultural/linguistic responsiveness;
- safety, effectiveness, person-centeredness, equity, and efficiency;
- workforce capacity, training, and well-being;
- financing, incentives, procurement, and supply chains;
- data interoperability and learning systems;
- emergency preparedness and resilience.

The *Crossing the Quality Chasm* framework articulated six aims—safe, effective, patient-centered, timely, efficient, and equitable care—that remain useful as system properties. (91)

22.4. Primary care and integration

Primary care is structurally important because it maintains longitudinal context, first contact, comprehensiveness, and coordination. Yet primary care cannot integrate the system without adequate time, information, multidisciplinary support, specialty collaboration, and payment.

22.5. Person-centered and integrated care

WHO's integrated people-centered approach emphasizes coordinated services organized around life needs rather than diseases and institutions. Integration includes clinical, professional, organizational, informational, and policy levels. (92)

23. Branch 13 — Measurement, evidence, data, and informatics

23.1. Measurement families

- history and narrative;
- observation and physical examination;
- self-report and proxy report;
- laboratory chemistry, hematology, microbiology, genomics, and pathology;
- electrical, optical, acoustic, pressure, flow, motion, temperature, and chemical sensors;
- imaging;
- physiological challenge and functional testing;
- ecological and environmental monitoring;
- administrative and service data;
- qualitative research and ethnography;
- digital traces and passive sensing.

23.2. Measurement quality

Every measure should be evaluated for:

- construct validity;
- content and face validity;
- criterion validity;
- analytic accuracy and precision;
- reliability and repeatability;
- sensitivity to change;
- calibration and traceability;
- reference population;
- measurement invariance across groups;
- burden, usability, and acceptability;
- clinical or decision utility.

23.3. Evidence generation

- mechanistic experiments;
- observational epidemiology;
- randomized trials;
- diagnostic-accuracy studies;
- natural experiments and quasi-experiments;
- implementation research;
- comparative effectiveness;
- qualitative and mixed methods;
- evidence synthesis and guidelines;
- post-market and real-world surveillance.

No design is universally highest. The question determines the design.

23.4. Data and knowledge representation

- identifiers and terminology;
- event and time-series models;
- provenance and versioning;
- uncertainty and missingness;
- privacy, consent, access, and security;
- interoperability and semantic mapping;
- knowledge graphs and causal models;
- algorithm validation and monitoring;
- patient-contributed and community-governed data.

24. Branch 14 — Interventions and therapeutics

24.1. Intervention targets

Interventions can act on:

- molecules and receptors;
- cells, tissues, and organs;
- physiological control systems;
- behavior and learning;
- symptoms and coping;
- function and environment;
- relationships and caregiving;
- organizations and care pathways;
- communities, markets, infrastructure, and policy;
- ecosystems and planetary drivers.

24.2. Major intervention families

24.2.1. Biological and pharmaceutical

- small molecules, biologics, vaccines, cell and gene therapies;
- fluids, nutrition, blood products, oxygen, and replacement therapies;
- antimicrobials, immunomodulation, endocrine therapy, analgesia, psychopharmacology;
- drug delivery and monitoring.

24.2.2. Procedural and surgical

- open, minimally invasive, endoscopic, catheter-based, image-guided, ablative, reconstructive, transplant, and regenerative procedures;
- anesthesia, perioperative care, and recovery pathways.

24.2.3. Device-based

- diagnostic, monitoring, assistive, prosthetic, implantable, wearable, therapeutic, and life-support devices;
- electrical stimulation, pacing, defibrillation, dialysis, ventilation, infusion, and closed-loop control.

24.2.4. Psychological and behavioral

- cognitive, behavioral, exposure-based, interpersonal, psychodynamic, family, acceptance-based, mindfulness-based, motivational, contingency, and trauma-focused approaches;
- habit and environmental redesign;
- sleep, exercise, nutrition, and substance-use interventions.

24.2.5. Rehabilitation and accommodation

- therapy, training, compensatory strategies, assistive technology, environmental adaptation, vocational support, and caregiver training.

24.2.6. Social and structural

- income support, housing, food programs, education, labor protection, transport, childcare, violence prevention, legal aid, and anti-discrimination enforcement.

24.2.7. Public-health and environmental

- vaccination, sanitation, clean air, water safety, screening programs, product regulation, taxation, urban design, vector control, and climate mitigation/adaptation.

24.2.8. Palliative and supportive

- symptom relief, communication, advance planning, spiritual care, caregiver support, grief care, and goal-concordant decision-making.

24.3. Benefit, harm, burden, and feasibility

An intervention profile should include:

$$U = B - H - T - O$$

where B is expected benefit, H expected harm, T treatment burden, and O opportunity cost. These quantities are multidimensional and person-specific; the expression is conceptual, not a universal scalar calculation.

24.4. Pharmacology as a cross-cutting map

Drugs can be classified by anatomy/therapeutic use, chemical structure, molecular target, mechanism of action, physiological effect, indication, route, formulation, pharmacokinetics, pharmacodynamics, toxicity, and interaction. No one classification captures all. ATC provides a hierarchical use-oriented system; RxNorm normalizes clinical drug concepts; FDA pharmacologic classes combine mechanism, physiologic effect, and structure; ChEMBL and IUPHAR/BPS link compounds to targets and bioactivity. (19,25,26,93,94)

25. Branch 15 — Ethics, law, rights, equity, and governance

25.1. Foundational ethical dimensions

- respect for autonomy and supported decision-making;
- beneficence and nonmaleficence;
- justice and fair distribution;
- dignity, privacy, confidentiality, and identity;
- truthfulness, transparency, and trust;
- solidarity and reciprocity;
- stewardship and sustainability.

25.2. Consent and capacity

Consent requires information, understanding, voluntariness, and decision-making capacity for the specific decision. Capacity can fluctuate and is not identical to diagnosis. Pediatric assent, surrogate decision-making, emergency exceptions, research consent, and data reuse require distinct rules.

25.3. Rights and non-discrimination

Health governance includes rights to access, accessibility, privacy, bodily integrity, reproductive autonomy, freedom from abuse, and participation. Disability rights require reasonable accommodation and removal of barriers. Public-health powers require necessity, proportionality, evidence, least-restrictive means, and procedural safeguards.

25.4. Equity

Equity asks whether avoidable, unfair differences in exposure, opportunity, care, and outcome are being reduced. Equality of identical treatment may be inequitable when needs or barriers differ. Measurement should separate biological variation from socially patterned exposure and institutional bias.

25.5. Data and algorithm governance

- consent and secondary use;
- data minimization and purpose limitation;
- security and breach response;
- representativeness and bias evaluation;
- explainability appropriate to risk;
- human oversight and contestability;
- performance drift and post-deployment monitoring;
- ownership, benefit sharing, and community governance.

25.6. Resource allocation

Allocation decisions occur at bedside, institution, insurer, government, and global levels. Criteria may include need, benefit, urgency, cost-effectiveness, equality, priority to the worse off, reciprocity, and social value. No technical metric eliminates normative tradeoffs.

25.7. Research governance and dual use

Research ethics includes scientific validity, favorable risk–benefit balance, fair selection, independent review, informed consent, privacy, monitoring, and dissemination. Bioscience, AI, and sensor systems may have dual-use risks requiring proportional oversight without suppressing beneficial science.

Key synthesis

Ethics is not a final checkpoint after the “real” technical work. Values enter when categories are defined, thresholds selected, populations sampled, outcomes chosen, devices calibrated, algorithms deployed, and resources allocated. Governance is therefore a full branch of health architecture.

26. Part III — How Major Institutions Organize Health Today

27. The classification ecosystem: purpose before comparison

Existing systems should be judged against the task they were built to perform. A recurring error is to ask whether ICD, DSM, SNOMED CT, RDoC, or a molecular ontology is “the best classification.” They occupy different layers.

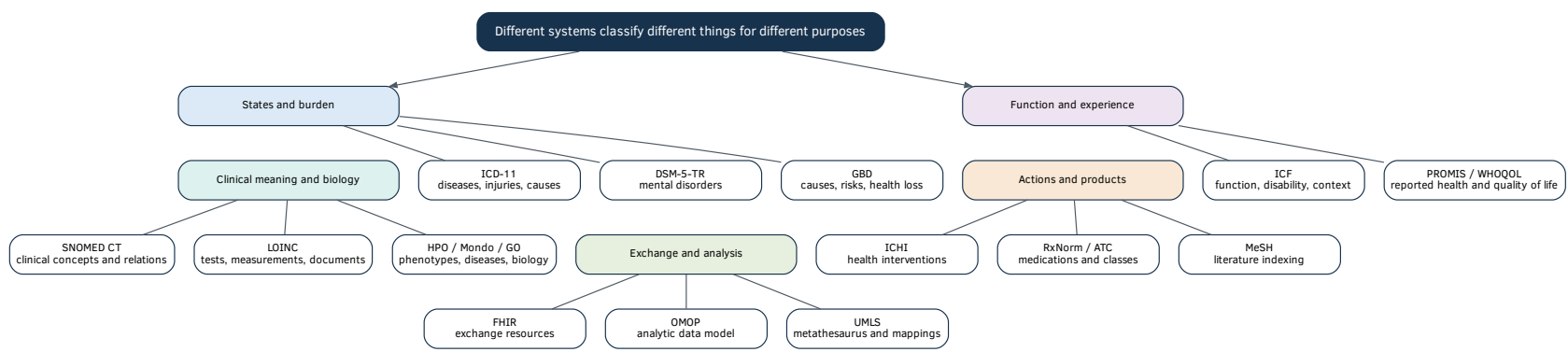


Figure 10.: The classification ecosystem. Systems for statistics, diagnosis, functioning, interventions, terminology, exchange, analytics, research, pharmacology, and outcomes overlap but are not interchangeable.

System family	Primary purpose	Main unit	Strength	Limitation if treated as total model
ICD	morbidity/mortality classification	disease, disorder, injury, cause	global comparability and reporting	limited whole-person and mechanistic detail
ICF	functioning and disability	function, activity, participation, context	links body and environment	not a disease nosology
ICHI	health interventions	intervention target, action, means	intervention comparability	implementation and outcome detail still separate
DSM/ICD mental disorders	clinical psychiatric diagnosis	syndrome/category with criteria	communication and clinical action	heterogeneity and threshold effects
RDoC/HiTOP	research dimensions	process, construct, spectrum	cross-diagnostic science	not a complete care or reimbursement system
SNOMED CT	clinical terminology	compositional concept	rich clinical expression	complexity and variable implementation
LOINC	observations/documents	test or observation identity	normalizes what was measured	does not itself store result meaning or quality
RxNorm/ATC	medicines	clinical drug / therapeutic class	medication normalization	indication, patient response, and causal context separate
FHIR	exchange	resource	modern interoperability	syntactic exchange does not ensure semantic equivalence
OMOP	observational analytics	standardized analytic tables/concepts	multi-database research	source detail can be lost in harmonization
UMLS	vocabulary integration	concept mapping	bridges terminologies	mappings are not always exact equivalence
HPO/Mondo/GO/DO	biomedical research ontology	phenotype, disease, function	computable relations	partial clinical/context representation
GBD	population burden	outcome and risk estimates	comparable global trends	model dependence and limited individual meaning
PROMIS/WHOQOL	lived outcomes	reported function/well-being	person-centered measurement	response and cultural comparability challenges

28. ICD-11: diseases, disorders, injuries, and causes of death

28.1. What ICD is for

The International Classification of Diseases is the global standard for recording and reporting diseases, disorders, injuries, causes of death, and related health information. ICD supports mortality statistics, morbidity reporting, epidemiology, service planning, quality measurement, reimbursement adaptations, and international comparison. ICD-11 came into effect for reporting in 2022 and continues to receive maintained digital releases; WHO announced a 2026 release in February 2026. (2,7,95)

ICD-11 was designed as a digital classification with a **foundation component** and linearizations for specific uses. Concepts can have multiple parents in the foundation while a statistical linearization imposes mutually exclusive structure where required. Postcoordination can add detail to a stem code through extension codes, improving representation of anatomy, severity, temporality, laterality, causation, and other attributes.

28.2. Strengths

- international governance and comparability;
- broad coverage across physical and mental health;
- coding tools, API-oriented implementation, and multilingual support;
- explicit maintenance rather than decade-scale frozen editions;
- capacity for greater detail through postcoordination;
- linkage to mortality rules and health-statistical systems.

28.3. Limitations

- categories often combine etiology, anatomy, syndrome, and administrative tradition;
- coding practice varies by country, institution, training, and incentive;
- one underlying cause of death can oversimplify causal chains;
- detailed postcoordination may be difficult to implement consistently;
- lived experience, function, social context, and uncertainty are not the primary representation;
- reimbursement adaptations can distort the statistical ontology.

28.4. How ICD should be used in the atlas

ICD codes should remain attached to health concepts for reporting and interoperability. The atlas adds dimensions not meant to be carried by ICD alone: mechanisms, continuous measurements, function, determinants, uncertainty, longitudinal trajectories, and relationships among conditions.

29. ICF, ICHI, and the WHO Family of International Classifications

29.1. ICF: functioning in context

The ICF shifted disability classification from a linear consequence of disease toward interaction among health condition, body functions and structures, activities, participation, and contextual factors. It can describe functioning for people with or without a conventional diagnosis. Qualifiers indicate extent of impairment or difficulty, and environmental factors may be barriers or facilitators. (8,33,96)

Its conceptual power is greater than its routine clinical adoption. Implementation barriers include coding burden, ambiguous boundaries between activity and participation, incomplete classification of personal factors, limited integration into electronic records, and weak reimbursement incentives.

29.2. ICHI: what was done

The International Classification of Health Interventions organizes interventions using axes such as target, action, and means. It is intended to cover clinical, public-health, rehabilitative, and traditional-medicine interventions. (9,97)

This is important because procedure codes are often national, payer-specific, and focused on billable clinical acts. A global intervention classification can support comparison, but it still needs linkage to indication, actor, dose/intensity, setting, implementation fidelity, cost, outcome, and adverse effect.

29.3. WHO-FIC as a conceptual triad

A practical interpretation is:

- **ICD:** what disease, disorder, injury, or health problem is present;
- **ICF:** how the person functions in context;
- **ICHI:** what intervention occurred.

This triad is much closer to a whole-health architecture than ICD alone. The atlas adds determinants, physiology, measurement provenance, evidence, goals, and governance as additional layers. (1)

30. DSM-5-TR and ICD-11 mental-disorder classification

30.1. The role of descriptive diagnosis

Psychiatric diagnoses group patterns of cognition, emotion, behavior, physiology, distress, impairment, course, and exclusion criteria. DSM-5-TR, published in 2022 and maintained through subsequent text updates, is widely used in the United States and internationally for clinical communication, research criteria, training, service eligibility, and reimbursement. ICD-11 provides the global statistical classification and WHO clinical descriptions and diagnostic requirements. (3,10,11)

The systems are not identical, but harmonization efforts reduce unnecessary divergence. Both use syndromal categories because clinicians, institutions, and patients need actionable labels even when mechanisms remain uncertain.

30.2. What categorical diagnosis accomplishes

- provides a shared language;
- signals expected symptom clusters and differential diagnoses;
- supports care pathways and evidence retrieval;
- enables eligibility and epidemiology;
- can validate that suffering is recognized;
- permits safety and service planning.

30.3. Scientific limitations

30.3.1. Heterogeneity

Two people can meet criteria for the same diagnosis through different symptom combinations and causal pathways. Comorbidity is common because categories overlap and underlying dimensions cross boundaries.

30.3.2. Threshold dependence

A person just below a symptom-count or duration threshold may not be meaningfully different from one just above it. Thresholds are often necessary for action but should not be confused with biological discontinuities.

30.3.3. Reliability and validity

Operational criteria can improve reliability, yet field-trial reliability varies by diagnosis. Reliability is not the same as validity: clinicians can agree on a category that still contains several mechanisms. (98)

30.3.4. Context and culture

Distress, normality, impairment, and symptom expression are culturally mediated. Diagnostic systems include cultural guidance, but routine implementation may remain shallow. Structural adversity can be individualized as disorder, while stigma can also prevent recognition of treatable illness.

30.4. Best practice in the atlas

Use a DSM/ICD diagnosis as one layer, supplemented by:

- dimensional symptom and function profiles;
- course and episode structure;
- medical, substance, sleep, and medication contributors;
- developmental and trauma history;
- social and cultural context;
- mechanism hypotheses with confidence;
- risk and protective factors;
- person-defined goals;
- response and adverse-effect trajectories.

31. RDoC, HiTOP, and dimensional alternatives

31.1. Research Domain Criteria

NIMH's RDoC is a research framework organized around functional domains and constructs studied across units of analysis—from genes and molecules through circuits, physiology, behavior, self-report, and paradigms. Current domains include negative valence, positive valence, cognitive systems, social processes, arousal/regulatory systems, and sensorimotor systems. It was designed to stimulate research across conventional diagnoses, not to replace clinical diagnosis in routine care. (4,99,100)

Strengths: mechanism-oriented, dimensional, cross-diagnostic, multi-unit, compatible with neuroscience and behavior.

Limitations: constructs and measures continue to evolve; social structure and lived meaning can be underrepresented; translation into clinical decisions is incomplete; biology-centered research can reproduce reductionism if higher levels are treated as peripheral.

31.2. HiTOP

The Hierarchical Taxonomy of Psychopathology uses patterns of symptom covariation to organize psychopathology dimensionally and hierarchically—from specific components and syndromes to subfactors, spectra, and broad superspectra. It addresses arbitrary thresholds, within-diagnosis heterogeneity, and high comorbidity. (13,43,101)

Strengths: empirical quantitative basis, graded severity, hierarchical resolution, explicit representation of shared liability.

Limitations: factor structure depends on measures and samples; dimensions do not automatically reveal mechanism; clinical thresholds and treatment pathways still require decisions; cultural generalizability must be tested.

31.3. Complementary use

DSM/ICD answer, “Which recognized clinical syndrome best organizes care?” HiTOP asks, “What dimensional structure best summarizes covariation?” RDoC asks, “Which functional systems and units of analysis should research examine?” A health architecture can retain all three.

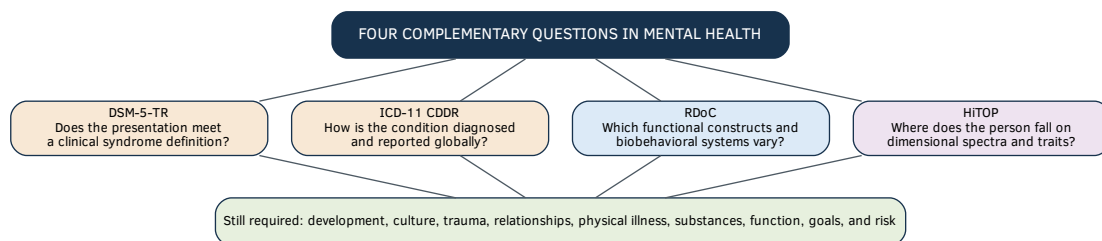


Figure 11.: Four complementary mental-health models. *DSM and ICD prioritize clinical classification; RDoC prioritizes functional systems and units of analysis; HiTOP prioritizes empirical dimensions and hierarchy.*

32. How medical schools organize health today

32.1. The dominant curricular pattern

Many contemporary medical schools use integrated preclerkship curricula organized around organ systems, clinical presentations, or longitudinal themes rather than separate traditional science courses. Foundational concepts recur in increasing complexity, followed by clerkships and advanced clinical experiences. Recent literature describes broad adoption of organ-system integration, though structures vary. (17,102)

Typical preclerkship blocks include:

- molecules, cells, genetics, and metabolism;
- host defense, microbiology, and inflammation;
- cardiovascular;
- respiratory;
- renal;
- gastrointestinal and nutrition;
- endocrine and reproduction;
- musculoskeletal and skin;
- neuroscience and behavior;
- hematology and oncology;
- growth, development, aging, and sometimes multisystem integration.

Longitudinal strands often include clinical skills, communication, ethics, health systems, evidence, quality and safety, population health, professionalism, and social determinants.

32.2. Accreditation expectations

LCME standards require content spanning biomedical, behavioral, and socioeconomic sciences; each organ system and life phase; prevention; acute, chronic, rehabilitative, and end-of-life care; scientific method; communication; structural/cultural factors and disparities; and interprofessional collaboration. (14)

This means contemporary accreditation is broader than a purely biomedical curriculum. The implementation problem is **weight and integration**: psychosocial, environmental, disability, and systems content may be present as short threads while organ pathology and testable facts dominate time and assessment.

32.3. Competency-based education

The AAMC's Core Entrustable Professional Activities for entering residency specify integrated professional tasks such as gathering histories, prioritizing differential diagnoses, recommending tests, documenting encounters, presenting patients, recognizing urgent conditions, obtaining consent, performing procedures, collaborating, identifying system failures, and practicing evidence-based medicine. (15)

The ACGME's six core competencies are:

1. patient care and procedural skills;
2. medical knowledge;

3. practice-based learning and improvement;
4. interpersonal and communication skills;
5. professionalism;
6. systems-based practice. (16)

CanMEDS organizes physician competence through the roles of Medical Expert, Communicator, Collaborator, Leader, Health Advocate, Scholar, and Professional. The UK GMC specifies outcomes in professional values, skills, and knowledge. WFME provides global standards for medical education. (103–105)

32.4. The hidden curriculum

Formal curricula may teach holistic care while assessment, workflow, and hierarchy teach different lessons. Learners notice what receives time, marks, prestige, procedures, and reimbursement. Common hidden messages include:

- acute disease is more “medical” than chronic disability;
- objective tests outrank patient report regardless of validity;
- specialist knowledge outranks coordination;
- uncertainty is failure rather than an explicit state;
- social problems belong to someone else;
- professionalism means endurance even when systems are unsafe;
- documentation and billing categories define reality.

32.5. A better educational architecture

The atlas recommends teaching every clinical problem through six lenses:

1. **mechanism and organ physiology;**
2. **mind, behavior, and lived experience;**
3. **function and participation;**
4. **determinants and environment;**
5. **measurement and evidence;**
6. **care system, ethics, and intervention.**

This preserves biomedical depth while preventing it from becoming the only map.

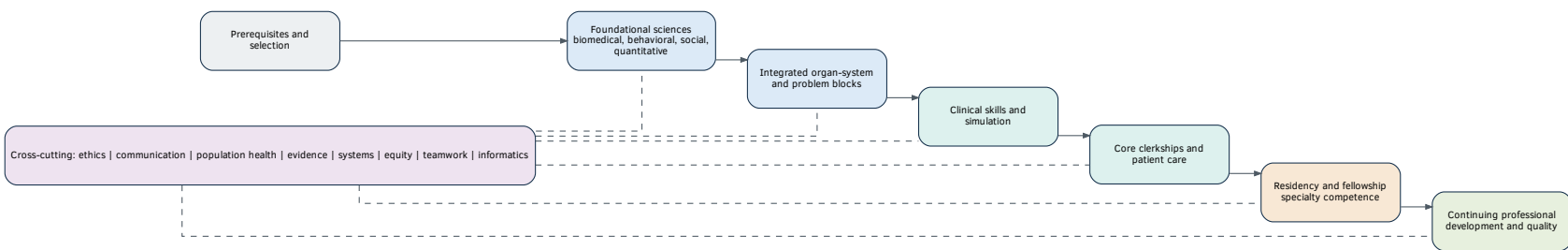


Figure 12.: A typical medical-education pathway and the integrative strands that should cross every stage. Competency frameworks broaden the curriculum, but institutional departments and assessments still strongly shape what learners perceive as central.

33. Professions, specialties, and the division of labor

33.1. Why professional maps differ from health maps

Professional categories define legal scope, training, accountability, labor markets, and service delivery. They should not be mistaken for a taxonomy of nature. Cardiology exists because cardiovascular expertise is deep and technically demanding, not because cardiovascular health is independent of kidneys, lungs, behavior, or environment.

33.2. Major professional families

33.2.1. Medicine and surgery

Primary care, internal medicine, pediatrics, obstetrics and gynecology, psychiatry, neurology, anesthesiology, radiology, pathology, emergency medicine, preventive medicine, physical medicine and rehabilitation, dermatology, ophthalmology, otolaryngology, and surgical specialties divide clinical expertise by population, organ, setting, method, and procedure. Subspecialties multiply along technical and disease-specific lines. Professional board structures enumerate dozens of specialties and many more subspecialties. (106)

33.2.2. Nursing

Nursing spans assessment, monitoring, care coordination, education, prevention, treatment delivery, symptom management, advocacy, and continuity across settings. Advanced practice roles may diagnose and prescribe depending on jurisdiction. Nursing frameworks often preserve whole-person and family context more explicitly than procedure-centered structures. (107)

33.2.3. Pharmacy

Pharmacists specialize in medication selection, formulation, dispensing, interactions, dosing, monitoring, adherence, stewardship, and medication-system safety. Their domain crosses every organ and setting.

33.2.4. Psychology and behavioral health

Psychology includes clinical, counseling, school, neuropsychological, developmental, social, cognitive, health, rehabilitation, industrial-organizational, and other fields. The American Psychological Association's divisions illustrate a professional map based on population, method, theory, and application rather than one unified hierarchy. (108,109)

33.2.5. Rehabilitation professions

Physical therapy, occupational therapy, speech-language pathology, audiology, prosthetics and orthotics, rehabilitation counseling, recreation therapy, and related professions organize around function, activity, adaptation, and participation.

33.2.6. Social work and community professions

Social workers, community health workers, case managers, peer specialists, and legal/benefits navigators address relationships, resources, systems, safety, and access. Their expertise is often essential but structurally underpowered in medical organizations.

33.2.7. Dentistry and oral health

Dentistry, dental hygiene, oral surgery, orthodontics, periodontics, endodontics, prosthodontics, and oral medicine are often financed and recorded separately from general health, creating preventable discontinuity.

33.2.8. Public health

Epidemiology, biostatistics, environmental health, occupational health, health policy, health education, infectious-disease control, global health, and emergency preparedness organize around populations and systems.

33.2.9. Laboratory, engineering, and informatics

Laboratory medicine, medical physics, biomedical engineering, clinical engineering, health informatics, data science, human factors, and device regulation mediate what can be observed and safely acted upon.

33.3. Interprofessional education and team design

WHO and professional frameworks support interprofessional education because collaborative practice can improve coordination and safety. (110,111)

Team design should specify:

- shared goals and a common problem representation;
- role clarity and scope;
- information flow;
- escalation and accountability;
- patient and caregiver participation;
- conflict resolution;
- longitudinal ownership;
- measurement of coordination burden.

Adding professions without redesigning workflow can increase fragmentation. The unit of integration must be the **care process and person-defined goal**, not the meeting itself.

34. Pharmacology: several overlapping ways to classify medicines

34.1. Anatomy and therapeutic use: ATC

The Anatomical Therapeutic Chemical system classifies active substances in a five-level hierarchy. The first level contains fourteen main anatomical or pharmacological groups; lower levels refine therapeutic, pharmacological, chemical, and substance identity. Defined daily doses support drug-utilization research but are technical units, not necessarily recommended patient doses. (93)

ATC is valuable for population analysis. It can be awkward when one molecule has several indications, mechanisms, routes, or combination products.

34.2. Clinical drug identity: RxNorm

RxNorm provides normalized names and relationships for clinical drugs in the United States, linking ingredients, strengths, dose forms, and branded products. It supports medication reconciliation and interoperability. It is not a complete pharmacodynamic or prescribing ontology. (19)

34.3. Pharmacologic class

FDA pharmacologic-class concepts may incorporate mechanism of action, physiological effect, and chemical structure. This is closer to mechanism but still represents curated classes rather than a complete patient-specific model. (94)

34.4. Target and bioactivity resources

ChEMBL organizes bioactive molecules, assays, targets, and activity data. The IUPHAR/BPS Guide to Pharmacology provides expert-curated ligand–target relationships and nomenclature. These resources support drug discovery and mechanism mapping. (25,26,112)

34.5. Pharmacokinetics and pharmacodynamics

A therapeutic model also needs:

- absorption, distribution, metabolism, and excretion;
- concentration over time;
- exposure at target tissue;
- receptor occupancy and downstream effect;
- delay and hysteresis;
- variability from genes, age, pregnancy, organ function, interactions, adherence, and formulation;
- therapeutic and toxic windows. (113–115)

A basic one-compartment model may be written:

$$\frac{dC}{dt} = \frac{\text{input}(t)}{V} - \frac{CL}{V}C,$$

with pharmacodynamic effect represented by an Emax model,

$$E(C) = E_0 + \frac{E_{\max}C}{EC_{50} + C}.$$

Real systems may require multiple compartments, active metabolites, target-mediated disposition, tolerance, feedback, disease progression, and time-varying covariates.

34.6. The medication profile needed for whole health

Each medication should be linked to ingredient and product identity, target/mechanism, intended indication, dose and schedule, actual use, exposure modifiers, expected benefit, measurable response, adverse effects, interactions, treatment burden, cost/access, and deprescribing plan. A flat medication list is insufficient.

35. Research organization: institutions, indexing, and translational stages

35.1. Institutional organization

Research agencies often divide health by disease, organ, population, or scientific method. The U.S. NIH consists of 27 Institutes and Centers with missions spanning cancer, heart/lung/blood, neurology, mental health, aging, allergy/infection, child health, environmental health, nursing, imaging, minority health, complementary health, genomics, translational science, and more. (116)

This structure mobilizes expertise and advocacy but can create boundaries around funding, datasets, and review panels. Cross-cutting programs are therefore required for pain, multimorbidity, women's health, data science, implementation, and climate-related health.

35.2. Literature indexing

Medical Subject Headings (MeSH) provide a controlled vocabulary for indexing and searching biomedical literature. Hierarchical placement and multiple headings improve retrieval, but novel and cross-disciplinary concepts may lag or be divided across trees. (117)

35.3. Translational spectrum

NCATS and related scholarship describe translational stages from basic discovery through preclinical research, clinical research, implementation, and population impact. “Bench to bedside” is too linear: translation is iterative, with feedback from clinical and community settings to foundational science. (118,119)

35.4. Research registries and reporting

ClinicalTrials.gov supports registration and results reporting for clinical studies. FAIR principles encourage data that are findable, accessible, interoperable, and reusable. Reporting guidelines improve transparency but do not ensure unbiased design or complete data access. (48,49,120,121)

35.5. Evidence hierarchies and causal questions

Different questions require different evidence:

- mechanism: laboratory and translational experiments;
- prognosis: longitudinal cohorts;
- treatment efficacy: randomized trials when feasible;
- rare harm: large observational and surveillance systems;
- diagnostic accuracy: blinded comparison with reference standards;
- policy: natural experiments, quasi-experiments, modeling, and implementation evidence;
- experience and acceptability: qualitative and mixed methods;
- complex systems: triangulation across designs.

A hierarchy without a question is not evidence-based reasoning.

36. Clinical terminologies: SNOMED CT, LOINC, and medication vocabularies

36.1. SNOMED CT

SNOMED CT is a comprehensive, logic-based clinical terminology containing concepts, descriptions, and relationships. It supports polyhierarchy and postcoordination, enabling detailed expression of findings, disorders, procedures, organisms, substances, situations, and context. (18)

Advantages: rich semantics, computable subsumption, clinical detail, mapping potential.

Challenges: implementation complexity, local subsets, interface terminology, versioning, compositional consistency, and the gap between what clinicians document and what the formal model permits.

36.2. LOINC

LOINC identifies laboratory tests, clinical observations, survey instruments, and documents. A laboratory term typically specifies component, property, timing, system, scale, and method where relevant. The 2026 release continued a large, actively maintained vocabulary. (5)

LOINC identifies *what was measured*; result value, units, reference range, device, specimen quality, interpretation, and uncertainty are carried elsewhere.

36.3. RxNorm and related medication data

RxNorm standardizes clinical drug identity, while National Drug Codes and product identifiers address packaged products. ATC provides therapeutic classification; target databases provide mechanism. Medication interoperability requires mapping across these layers.

36.4. Interface terminology

Clinicians and patients use ordinary language, abbreviations, and local terms. Interface terminology should allow familiar entry while mapping to standards. A perfect backend ontology with an unusable interface produces poor data.

37. FHIR, OMOP, and UMLS: exchange, analytics, and mapping

37.1. FHIR

HL7 FHIR defines modular resources for exchange, including Patient, Observation, Condition, Procedure, Medication, DiagnosticReport, CarePlan, Goal, QuestionnaireResponse, Device, Consent, and many others. Profiles and implementation guides constrain resources for specific uses. The current published major specification remains R5, while many production systems continue to use R4-based implementations. (6)

FHIR improves transport and API access. It does not guarantee that two systems mean the same thing by a code, use the same units, preserve provenance, or implement the same profile.

37.2. OMOP Common Data Model

The OMOP CDM standardizes observational health data into common tables and vocabularies for distributed analytics. OHDSI develops tools and methods for reproducible characterization, population-level estimation, and prediction. (20,122)

Its strengths are analytic consistency and multi-database research. Limitations include information loss during extraction, transformation, and mapping; variable source quality; and difficulty representing complex narratives, device waveforms, social context, and uncertain phenotypes.

37.3. UMLS

The UMLS integrates many biomedical vocabularies through concept identifiers and semantic networks. It helps bridge terms across source systems. Mapping does not imply perfect equivalence; source concepts may differ in granularity, scope, and purpose. (21,123)

37.4. The interoperability stack

A reliable stack separates:

1. **source data and provenance;**
2. **canonical clinical concepts;**
3. **observation identities, units, and methods;**
4. **exchange resources and profiles;**
5. **analytic common models;**
6. **knowledge and causal relationships;**
7. **decision logic and user interfaces.**

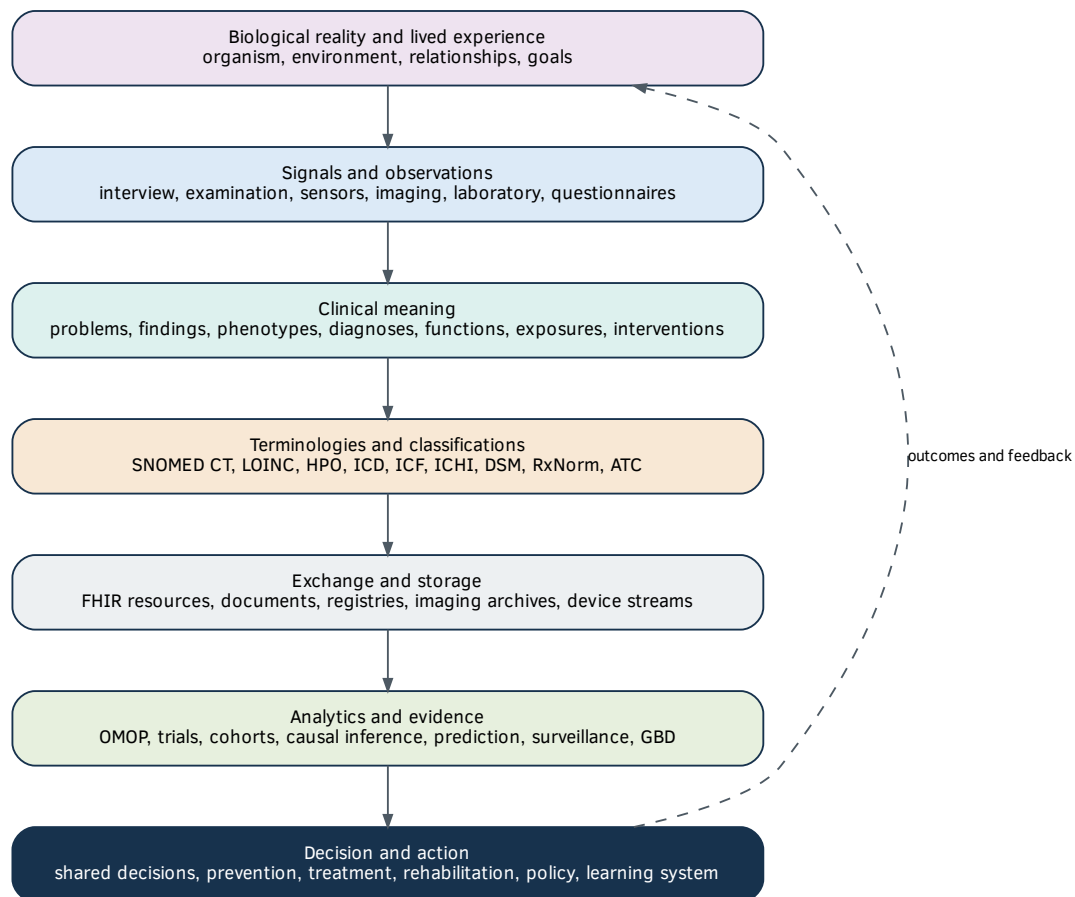


Figure 13.: A layered health-data stack. Standards at one layer cannot substitute for missing semantics, provenance, quality, or governance at another.

38. Biomedical ontologies: phenotype, disease, function, and compounds

38.1. Human Phenotype Ontology

HPO provides standardized terms and relationships for phenotypic abnormalities, widely used in rare-disease diagnosis and genomic analysis. It supports detailed, computable phenotype profiles and annotations. (22,124)

A phenotype record should include presence/absence, onset, severity, frequency, progression, laterality, measurement method, and uncertainty—not only a term list.

38.2. Mondo Disease Ontology and Disease Ontology

Mondo integrates disease concepts and mappings from multiple sources, aiming to support consistent disease representation across research. Disease Ontology provides a formal human-disease ontology. (23,125)

These resources are valuable for cross-database integration but inherit uncertainty and disagreement about disease identity, synonymy, and granularity.

38.3. Gene Ontology

Gene Ontology represents molecular function, biological process, and cellular component, with annotations supported by evidence codes. It is foundational for functional genomics. (24,126)

GO does not represent the whole person; it represents a critical molecular/cellular layer. Linking GO processes to phenotypes, exposures, and clinical outcomes requires additional ontologies and evidence.

38.4. Compound and target ontologies

ChEMBL, IUPHAR/BPS, and chemical ontologies connect compounds, targets, assays, and mechanisms. These support target discovery and pharmacology but must be linked to formulation, exposure, patient covariates, and outcomes for clinical use.

39. Burden, functioning, and patient-reported outcomes

39.1. Global Burden of Disease

GBD organizes health through causes, sequelae, impairments, risk factors, locations, ages, sexes, and years, estimating mortality and disability burden with uncertainty. It enables comparisons across conditions that otherwise use incompatible metrics. (73,89,90,127)

The DALY is:

$$DALY = YLL + YLD,$$

where years of life lost depend on deaths and a reference life expectancy, and years lived with disability depend on prevalence and disability weights. (71)

Critical limitations: data scarcity, model assumptions, attribution among correlated risks, disability-weight interpretation, and the inability of one burden metric to capture equity, caregiver impact, cost, meaning, or rights.

39.2. PROMIS

PROMIS provides calibrated patient-reported measures across physical, mental, and social health domains. Item-response theory supports short forms and computerized adaptive tests. Domains include physical function, fatigue, pain, sleep, emotional distress, social roles, and global health. (128,129)

PROMIS demonstrates how function and symptoms can be measured comparably across diseases. Challenges include language/cultural validation, accessibility, response burden, and clinical interpretation of score changes.

39.3. WHOQOL

WHOQOL instruments assess quality of life across physical, psychological, social, and environmental dimensions from the person's perspective and were developed cross-culturally. (130,131)

Quality of life should not be used to discount objective harm or justify poor conditions. It is one essential outcome among several.

39.4. OECD PaRIS

The OECD Patient-Reported Indicator Surveys program measures outcomes and experiences of adults living with chronic conditions in primary care. Its international focus illustrates the shift from service activity toward what people report about health, function, confidence, coordination, and care experience. (132,133)

39.5. Linking biology to quality of life

The Wilson–Cleary model links biological and physiological variables to symptoms, functioning, general health perceptions, and overall quality of life, modified by individual and environmental characteristics. (134)

This sequence is useful but should be treated as a network: quality of life, behavior, and environment can feed back to physiology.

40. Crosswalk: which system answers which question?

Table 4.: Crosswalk of major health-knowledge systems.

System	Primary question	Best represented	Typical user	Key output	What must be added
ICD-11	What disease, injury, or cause should be classified?	Diagnostic and statistical categories	clinicians, coders, governments	code/linearization	function, mechanism, determinants, uncertainty
ICF	How does the person function in context?	body function, activity, participation, environment	rehabilitation, policy, disability services	functioning profile	diagnosis, time, intervention detail
ICHI	What intervention occurred?	action–target–means	health systems, statisticians	intervention code	indication, dose, fidelity, outcome
DSM-5-TR	Which psychiatric syndrome criteria are met?	descriptive mental-disorder categories	mental-health clinicians/researchers	diagnosis/specifiers	dimensions, mechanism, context, function
RDoC	Which functional system and unit of analysis is studied?	transdiagnostic processes	researchers	construct matrix	clinical thresholds, service use, broader social structure
HiTOP	What dimensional psychopathology hierarchy fits covariation?	spectra and dimensions	researchers/clinicians	dimensional profile	mechanisms, decisions, implementation
SNOMED CT	How can clinical meaning be encoded?	detailed compositional concepts	EHRs, informaticians	concept expression	workflow, measurement quality, analytics
LOINC	What observation or document is this?	test/measure identity	labs, EHRs, instrument vendors	observation code	result, units, reference, uncertainty
RxNorm	Which clinical drug product is meant?	ingredient/strength/form	pharmacy and EHR systems	normalized drug concept	mechanism, indication, response, burden
FHIR	How is structured health data exchanged?	modular resources	software systems	API/resource instance	common semantics, data quality, governance
OMOP	How can observational data be analyzed consistently?	standardized analytic records	research networks	common tables/cohorts	source nuance, narrative, real-time use
UMLS	How can vocabularies be connected?	cross-terminology concepts	informaticians, NLP	mappings/CUIs	equivalence review, purpose-specific semantics

System	Primary question	Best represented	Typical user	Key output	What must be added
HPO	Which phenotypic abnormalities are present?	granular phenotypes	genetics/rare disease	phenotype profile	ordinary function, context, outcomes
GO	What molecular function/process/component is involved?	gene-product biology	basic and translational science	ontology annotation	clinical phenotype and person context
GBD	What is population burden and attributable risk?	mortality/disability trends	public-health planners	rates, DALYs, uncertainty	local goals, equity, lived experience
PROMIS/WHOQOL	What does the person report about symptoms, function, or life quality?	patient-reported outcome	clinicians/researchers	calibrated score/profile	mechanism, objective state, context

Key synthesis

The crosswalk demonstrates why a unified health atlas should not replace established standards. It should orchestrate them: each concept retains the identifiers and representations required for its purpose, while a higher layer records relationships, provenance, time, uncertainty, function, context, and goals.

41. Part IV — Where Present Health Organization Breaks Down

42. Failure mode 1 — Organ and specialty silos

42.1. The fragmentation problem

Specialization creates depth but divides the causal system. A person with diabetes, heart failure, chronic kidney disease, depression, arthritis, and housing insecurity may receive six internally reasonable plans that are collectively impossible. Each plan may assume that another service owns medication reconciliation, exercise tolerance, nutrition, transportation, cognition, finances, and follow-up.

Fragmentation appears in several forms:

- **clinical fragmentation:** many clinicians with weak longitudinal ownership;
- **informational fragmentation:** records, waveforms, images, medications, and goals in disconnected systems;
- **temporal fragmentation:** acute episodes treated without trajectory or recovery data;
- **professional fragmentation:** different vocabularies and priorities;
- **financial fragmentation:** separate payers, benefits, formularies, and authorization rules;
- **geographic fragmentation:** care distributed across institutions and regions;
- **conceptual fragmentation:** biological, psychological, social, and functional explanations treated as competing.

Evidence links fragmented ambulatory care with burdens such as duplicated testing, inconsistent treatment, avoidable utilization, and poor experience, although measurement definitions vary. (135)

42.2. Why a referral is not integration

Referral transfers or shares responsibility; it does not by itself create a common model. Integration requires:

1. one coherent problem representation;
2. explicit priorities and tradeoffs;
3. role and decision ownership;
4. shared medication and intervention logic;
5. synchronized monitoring;
6. feedback after transitions;
7. a place for patient and caregiver goals;
8. reconciliation of conflicting recommendations.

42.3. The “orphan interaction”

Health systems assign diseases to departments, but interactions often have no owner. Examples include:

- anticoagulation versus fall risk;
- diuretic treatment versus renal function and orthostasis;
- stimulant treatment versus appetite, sleep, and blood pressure;
- pain relief versus sedation, constipation, function, and dependence;
- cancer treatment versus fertility and cognition;
- psychiatric medication versus metabolic risk;

- dialysis schedule versus employment and caregiving;
- pregnancy versus chronic disease medication.

A unified profile must model interactions as first-class entities, not comments scattered across notes.

43. Failure mode 2 — Reductionism without level awareness

43.1. When reduction is productive

Reduction is a necessary scientific strategy. Breaking a system into parts can identify a pathogen, mutation, receptor, lesion, pressure gradient, or faulty circuit. Problems arise when success at one level is generalized into the claim that other levels add no causal information.

The following statements can all be true:

- depression involves brain and body processes;
- depression can be influenced by loss, violence, poverty, sleep, medication, inflammation, and learning;
- neural mechanisms mediate psychological and social effects;
- social and psychological explanations remain causally relevant even after neural mediation is described;
- different interventions can act at different points in the same pathway.

43.2. Downward and upward causation

Lower-level components constrain higher-level functions, while higher-level organization changes the conditions under which components act. A workplace policy can change schedules, sleep, autonomic activation, food access, and injury risk. The policy is not a molecule, but it produces molecular consequences. Conversely, a thyroid hormone disturbance can alter mood, cognition, energy, and social participation.

The atlas avoids metaphysical debates by using **intervention-relative causation**: a variable is causally relevant when changing it under specified conditions changes the outcome. Multiple intervention levels can be valid.

43.3. Biomarker substitution

A biomarker becomes dangerous when it substitutes for the outcome it only imperfectly predicts. Lower blood pressure usually reduces cardiovascular risk, but a specific intervention can lower a marker while causing other harms. Tumor shrinkage, viral load, glucose, bone density, and inflammatory markers can be valuable intermediate outcomes, yet they must be linked to survival, symptoms, function, and adverse effects.

Surrogate validity is intervention- and context-specific. A marker that predicts outcome observationally may not capture all causal effects of treatment.

44. Failure mode 3 — Categorical thresholds and diagnostic reification

44.1. Why thresholds exist

Decisions require boundaries: start therapy, qualify for support, report a case, enroll in a trial, or trigger an alarm. Thresholds can improve consistency and make systems accountable.

44.2. How thresholds mislead

- values immediately above and below may have nearly identical risk;
- measurement error can move a person across the boundary;
- repeated measurements may classify differently;
- cutoffs may be derived in populations unlike the current person;
- thresholds can reflect cost or capacity, not biology;
- labels can change identity and future interpretation;
- service eligibility may encourage strategic documentation;
- category-specific research can hide transdiagnostic dimensions.

44.3. The solution: dual representation

Every threshold-defined state should store:

1. the raw continuous measure;
2. measurement uncertainty and protocol;
3. the threshold rule and version;
4. the reason the rule is being applied;
5. modifiers and exceptions;
6. resulting action;
7. outcome after action.

This preserves both scientific continuity and practical decision-making.

45. Failure mode 4 — Multimorbidity and polypharmacy

45.1. Multimorbidity is the common case in complex care

Multimorbidity means coexistence of multiple long-term conditions, often with functional, psychological, and social complexity. Large population studies show that it is common and strongly patterned by age and socioeconomic disadvantage, appearing earlier in deprived populations. (27)

A person's burden is not the sum of diagnosis counts. Interactions may be:

- causal: one condition causes or accelerates another;
- treatment-mediated: therapy for one worsens another;
- measurement-related: one condition changes the meaning of a test;
- behavioral: symptoms interfere with self-management;
- functional: combined small impairments produce major disability;
- social: care burden exceeds available time and resources.

45.2. Single-disease guidelines can conflict

Applying every recommended action from multiple guidelines can generate excessive visits, monitoring, medications, diet rules, and self-management tasks. A classic analysis showed the complexity created by applying disease-specific guidance to a hypothetical older adult with several conditions. (28)

NICE and other multimorbidity guidance recommend identifying people who may benefit from an individualized approach, clarifying goals, reviewing treatment burden, considering prognosis and frailty, and stopping low-value interventions. (29,136)

45.3. Polypharmacy is a systems property

Polypharmacy is not simply “many medications.” Appropriate polypharmacy may be necessary. Problematic polypharmacy occurs when net benefit is unfavorable, indications are unclear, monitoring is missing, prescribing cascades occur, interactions accumulate, or treatment burden exceeds capacity.

A medication graph should represent:

- drug–drug interactions;
- drug–condition interactions;
- drug–measurement effects;
- duplicate mechanisms;
- cumulative anticholinergic, sedative, bleeding, QT, renal, hepatic, metabolic, and fall burden;
- adherence complexity;
- cost and access;
- withdrawal and deprescribing risk.

45.4. Goal-oriented care

For complex patients, optimization must be constrained by goals and capacity. A technically ideal regimen that cannot be followed is not ideal. The target may be reducing falls and fatigue while preserving cognition rather than maximizing every disease-specific biomarker.

46. Failure mode 5 — The mental–physical split

46.1. Administrative dualism

Mental and physical health are often separated in insurance, records, facilities, training, and stigma. This division persists even though sleep, pain, autonomic regulation, endocrine function, inflammation, substance exposure, medication, cognition, behavior, and social conditions cross it continuously.

Consequences include:

- medical symptoms dismissed after normal testing;
- psychiatric symptoms missed in medical settings;
- medical illness missed in psychiatric settings;
- fragmented medication management;
- crisis care without longitudinal support;
- unequal access and reimbursement;
- diagnostic overshadowing, especially for disabled and psychiatric patients.

46.2. Two opposite errors

Psychologization: attributing poorly explained symptoms to stress, anxiety, or behavior without sufficient biological investigation.

Biological exceptionalism: treating subjective experience, learning, trauma, and social conditions as unscientific because they are not captured by a standard laboratory test.

The correct approach is parallel rather than sequential: evaluate dangerous and plausible biological processes while also characterizing cognition, emotion, behavior, function, and context. Psychological treatment can be useful for symptom regulation even when a biological cause is present; its benefit does not prove a symptom was “all psychological.”

46.3. Diagnostic overshadowing

Once a psychiatric, developmental, obesity-related, or disability label exists, clinicians may interpret new symptoms through it. The same problem occurs in reverse when medical diagnoses obscure depression, trauma, substance use, or cognitive decline. Structured differential diagnosis and baseline physiological data can reduce this bias.

47. Failure mode 6 — Function and lived experience are secondary

47.1. The clinic measures what it can

Health systems more reliably capture diagnoses, procedures, laboratory values, and claims than fatigue, pain interference, communication, parenting, work, autonomy, sexual function, or meaning. What is easy to code becomes visible; what is difficult to quantify becomes optional.

47.2. Why function is not downstream noise

Function can be:

- an early signal before diagnostic thresholds;
- an independent treatment target;
- a mediator between disease and quality of life;
- a predictor of hospitalization and mortality;
- a determinant of adherence and safety;
- the outcome most valued by the person.

The ICF, PROMIS, WHOQOL, and patient-reported outcome initiatives provide methods, but routine integration remains incomplete. (8,128,130,132)

47.3. Response shift and adaptation

People may recalibrate expectations, reprioritize values, or reconceptualize quality of life after illness. This “response shift” can make stable self-reported quality of life coexist with objective decline. That does not invalidate the report. It means subjective and objective outcomes answer different questions.

47.4. Caregiver and family outcomes

Illness redistributes work, time, sleep, income, and emotional burden across families. Pediatric, dementia, disability, and end-of-life care cannot be fully evaluated without caregiver capacity and well-being. The person receiving care remains central, but the care system includes others.

48. Failure mode 7 — Social and environmental determinants are underweighted

48.1. The downstream clinical bias

Clinical systems encounter people after upstream exposures have accumulated. Housing, work, education, food systems, discrimination, pollution, violence, and transportation may be documented but not treated as part of the causal model because clinicians lack authority or resources to change them.

WHO's 2025 social-determinants report reiterates that avoidable health gaps are driven by unequal conditions and broader economic and political forces. (86)

48.2. The social-risk checklist problem

Screening for food insecurity or housing instability can be valuable, but it can also become performative if no response pathway exists. Asking without assistance may increase frustration or risk. A determinant record should include:

- the condition and severity;
- person-perceived priority;
- mechanism of health impact;
- immediate safety implications;
- available intervention and eligibility;
- referral completion;
- outcome and unresolved barrier.

48.3. Individualization of structural problems

Telling a person to eat better, exercise, sleep, or reduce stress can obscure shift work, unsafe neighborhoods, caregiving, food prices, disability, advertising, trauma, or lack of transport. Behavior matters, but opportunity structures behavior. Interventions should separate knowledge, motivation, capacity, opportunity, and constraint.

48.4. Environmental invisibility

Exposure data are rarely linked to clinical timelines. Air pollution, heat, noise, water contamination, occupation, mold, pesticides, and neighborhood conditions may be absent from the record. The exposome concept highlights this gap, but comprehensive measurement is difficult. (66,67)

49. Failure mode 8 — Coding, billing, and incentive distortion

49.1. Codes become proxies for reality

Administrative systems require standardized codes. Once payment, quality reporting, risk adjustment, and authorization depend on them, coding behavior changes. Problems include:

- upcoding or defensive coding;
- undercoding stigmatized or poorly reimbursed conditions;
- copying stale diagnoses forward;
- coding rule complexity;
- pressure to choose one principal condition;
- omission of uncertainty;
- conflation of “not documented” with “not present.”

49.2. Problem-list pollution

Problem lists often contain duplicates, resolved conditions, rule-outs, billing artifacts, and vague labels. They may omit function, social context, and goals. A useful problem representation needs status, onset, evidence, certainty, relationships, responsible team, monitoring plan, and resolution criteria.

49.3. Procedure incentives

Fee structures may reward discrete procedures more than communication, prevention, deprescribing, coordination, or environmental intervention. This can shape specialty growth and research priorities. The system’s taxonomy is therefore partly an economic map.

49.4. Risk adjustment and algorithmic feedback

Risk adjustment can promote fair comparison but can also reward documentation intensity, normalize lower expectations for disadvantaged groups, or hide inequitable care. Algorithms trained on cost may predict prior access and spending rather than health need, reproducing bias. (76)

50. Failure mode 9 — Semantic and technical fragmentation

50.1. Syntactic interoperability is not semantic interoperability

Two systems can exchange a FHIR Observation yet disagree about:

- specimen and collection conditions;
- device and calibration;
- units and reference interval;
- fasting or posture state;
- data cleaning;
- patient identity;
- time zone and event time;
- whether a value is measured or inferred;
- whether absence means negative or missing.

50.2. Mapping loss

Mapping local codes to standards can collapse distinctions. One-to-many and partial mappings are common. UMLS concept links help but cannot guarantee equivalence. OMOP harmonization may enable analysis while discarding source nuance. (20,21)

50.3. Version drift

Ontologies, clinical guidance, software, and algorithms change. A code without version and effective date is ambiguous. Longitudinal data may span several coding systems. Reproducibility requires preserving original terms, mapped terms, mapping method, and version.

50.4. Narrative versus structure

Narrative captures uncertainty, causal reasoning, priorities, and context. Structured data support retrieval and computation. Forcing all narrative into codes loses meaning; leaving everything in prose blocks computation. The solution is linked structure plus narrative, not one replacing the other.

50.5. Waveforms, images, and high-frequency data

Clinical information models are strongest for discrete observations. Device waveforms, images, raw sensor streams, calibration records, and preprocessing pipelines require additional standards and storage. Derived features should link back to raw data and algorithm version.

51. Failure mode 10 — Bias, inequity, and cultural invalidity

51.1. Data are generated by systems

Clinical datasets are not neutral samples of biology. They reflect who had access, who was tested, what clinicians suspected, which services were covered, and how institutions coded. Missingness is often informative.

51.2. Reference populations

Laboratory ranges, growth charts, risk equations, device optical performance, imaging thresholds, and psychometric instruments can perform differently across populations. The appropriate response is not to encode broad social categories as biological constants. It is to measure causal variables, validate performance, and report uncertainty.

51.3. Race correction

Race-based adjustments have been used in kidney, pulmonary, obstetric, and other algorithms. Critiques show that unexamined correction can delay treatment, misstate biology, or reproduce inequity. (75)

An ancestry marker may be useful for a specific genetic question; self-identified race may be useful for studying discrimination and inequity; neither should be substituted for the other.

51.4. Cultural validity

Symptoms, distress, function, family decision-making, and quality-of-life responses depend partly on language and culture. Translation alone does not establish measurement equivalence. Instruments require cognitive testing, psychometric validation, and community participation.

51.5. Algorithmic bias

Bias can enter through target selection, sampling, labels, measurement, feature choice, optimization, deployment, and feedback. Fairness metrics can conflict. Evaluation should include subgroup calibration, error distribution, clinical consequences, access, and the possibility that the target itself is unjust.

52. Failure mode 11 — Prevention and public health are structurally peripheral

52.1. Acute care dominates visibility

Hospitals, procedures, and specialist encounters generate detailed records and concentrated spending. Prevention often operates across sectors with diffuse benefits and long time horizons. Successful prevention may be invisible because the event never occurs.

52.2. The prevention paradox

A small risk reduction across a large population can prevent more events than a large reduction among a small high-risk group. Yet individual clinical systems tend to prioritize identifiable high-risk patients. Both strategies are necessary.

52.3. Upstream–downstream separation

Clinical and public-health agencies may use different data, funding, legal authority, and timelines. Environmental, housing, education, and labor policies may have major health effects while being governed outside health departments. “Health in All Policies” approaches attempt to make these effects explicit. (137)

52.4. Screening enthusiasm

Early detection sounds universally beneficial, but screening can cause overdiagnosis and treatment harm. Programs should be evaluated as end-to-end systems, including confirmatory testing, treatment effectiveness, access, and inequity—not just test sensitivity. (81,82)

53. Failure mode 12 — Evidence mismatch, reproducibility, and uncertainty

53.1. Average evidence, individual decisions

Randomized trials estimate effects in enrolled populations under specific conditions. Individual decisions require translation across baseline risk, comorbidity, adherence, competing events, preferences, and feasibility. Absolute benefit can differ greatly even when relative effect is similar.

53.2. External validity

Trial participants may be younger, healthier, less diverse, or more supported than routine-care populations. Exclusion of pregnancy, disability, severe mental illness, multimorbidity, and older adults creates evidence gaps exactly where decisions are complex.

53.3. Outcome selection

Studies may prioritize short-term biomarkers or composite endpoints. Composite outcomes can combine events of very different importance. Patient-reported and functional outcomes may be secondary or absent.

53.4. Publication and analysis bias

Selective publication, flexible analysis, small samples, weak replication, and incentive structures can inflate evidence. Ioannidis's critique of false findings and subsequent reproducibility reforms underscore the need for preregistration, transparent methods, data/code sharing where ethical, and replication. (53,121)

53.5. Evidence hierarchy misuse

A randomized trial can be biased; an observational study can provide strong evidence for harm or long-term outcome; qualitative research can identify implementation failure; mechanistic evidence can explain heterogeneity. GRADE appropriately evaluates certainty across domains rather than assuming design alone decides truth. (51)

53.6. Uncertainty suppression

Clinical interfaces often force a diagnosis or normal/abnormal flag. A better system distinguishes:

- aleatory variability;
- measurement error;
- model uncertainty;
- causal uncertainty;
- missing data;

- disagreement among experts;
- value uncertainty.

Uncertainty should be actionable: what observation or time will reduce it, and what decision is safe meanwhile?

54. Failure mode 13 — Medicalization, overdiagnosis, and underdiagnosis

54.1. Medicalization

Medicalization describes the process by which human problems become defined and treated as medical. It can bring recognition, research, treatment, and protection. It can also narrow social problems into individual pathology, expand markets, and reduce variation to disorder. (138)

54.2. Overdiagnosis

Overdiagnosis identifies a real abnormality that would not have caused symptoms or harm during the person's lifetime. It differs from false-positive error. Sensitive imaging, screening, biomarkers, and expanded thresholds can detect indolent states. (139,140)

54.3. Overtreatment and low-value care

Overtreatment follows when intervention burden and harm exceed likely benefit. Waste also arises from research questions of low relevance, poor methods, inaccessible results, and failure to implement known effective care. (141)

54.4. Underdiagnosis and undertreatment

The opposite occurs when symptoms are dismissed, access is poor, reference data are biased, or stigmatized groups receive inadequate investigation. A system can overdiagnose low-risk abnormalities in well-served populations while underdiagnosing serious disease elsewhere.

54.5. The classification expansion dilemma

Broader definitions can catch early treatable states but increase false labeling and treatment burden. Narrow definitions improve specificity but can deny support. The solution is not one universal cutoff; it is explicit risk, uncertainty, preferences, staged intervention, and monitoring.

55. The fragmentation loop

The failure modes reinforce one another:

1. categories define departments;
2. departments define workflows and data;
3. data define research cohorts and outcomes;
4. research defines guidelines and payment;
5. guidelines produce more siloed care;
6. missing outcomes remain unmeasured;
7. the apparent evidence base confirms the original categories.

Breaking the loop requires integration at several points, not a new vocabulary alone.

Critical appraisal

The health system is not “failing” because specialization, diagnosis, coding, trials, or devices are inherently wrong. It fails when these useful reductions are not reassembled into a longitudinal person-in-context model and when incentives make the missing dimensions systematically invisible.

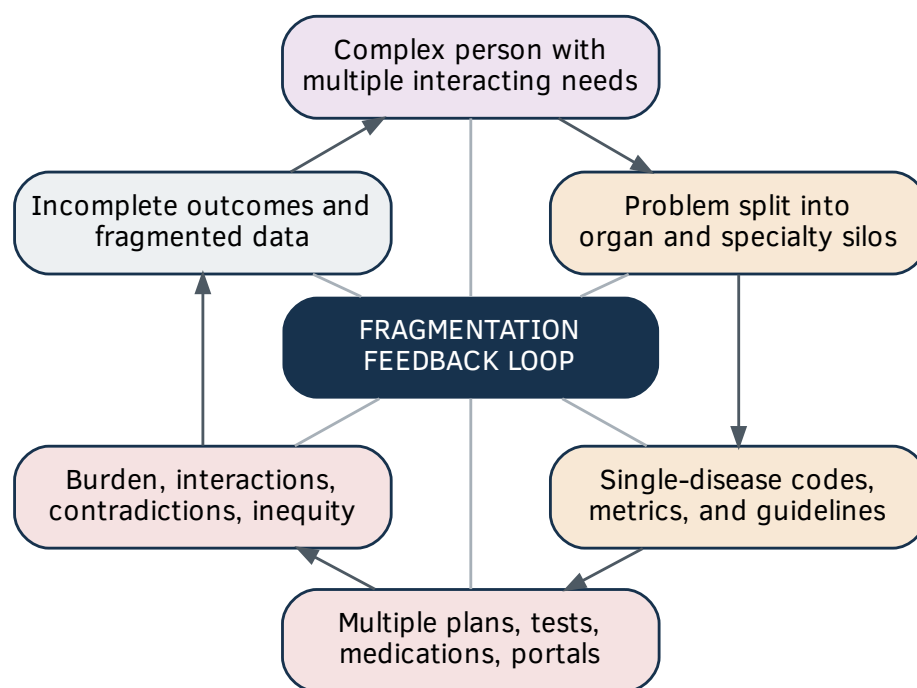


Figure 14.: *The fragmentation loop. Narrow categories shape institutions and data; fragmented data reinforce narrow evidence and care; outcomes outside the silo remain invisible, perpetuating the original categories.*

56. Part V — A Practical Multi-Axial Health Knowledge Architecture

57. Design requirements

A master health architecture should satisfy ten requirements.

57.1. 1. Preserve multiple legitimate views

A condition may have diagnostic, mechanistic, functional, experiential, social, and administrative representations. The architecture must link them without forcing one to replace the others.

57.2. 2. Be longitudinal by default

Health is a trajectory. Every observation, inference, diagnosis, exposure, intervention, and outcome requires valid time, recording time, and version. The system should represent onset, episode, recurrence, progression, remission, and uncertainty.

57.3. 3. Separate observation from interpretation

A raw waveform, a derived heart-rate value, a clinician's rhythm interpretation, and an ICD diagnosis are different objects. They may be linked, but provenance must remain visible.

57.4. 4. Represent uncertainty explicitly

Possible, probable, confirmed, ruled out, historical, and resolved states require different semantics. Confidence should be attached to assertions and relationships, with the evidence that supports them.

57.5. 5. Model relationships, not only lists

The architecture should represent:

- causes and contributors;
- manifestations and complications;
- treatment targets;
- contraindications and interactions;
- shared mechanisms;
- temporal precedence;
- conflicting goals;
- environmental barriers and facilitators.

57.6. 6. Link biological outcomes to lived outcomes

Every technical measurement program should identify the symptoms, functions, participation, risk, or decisions it is intended to improve. A sensor feature with no validated relation to a meaningful use is an engineering output, not yet a health outcome.

57.7. 7. Remain compatible with standards

The system should attach ICD, ICF, ICHI, SNOMED CT, LOINC, RxNorm, ATC, HPO, Mondo, FHIR, OMOP, and other identifiers where appropriate. It should not invent local synonyms when a maintained standard exists.

57.8. 8. Support person and population views

An individual longitudinal profile and a population surveillance system need different aggregation, privacy, uncertainty, and causal methods. The underlying concepts should remain linkable.

57.9. 9. Make values and governance visible

Goals, consent, access, equity, rights, and decision authority should be represented alongside technical facts. A recommendation is incomplete without who decides and whose outcome is optimized.

57.10. 10. Permit modular expansion

Each branch must be expandable into a dedicated packet—such as the cardiovascular packet—without redesigning the top-level architecture.

58. The health-coordinate record

58.1. Core record object

A **HealthConceptInstance** represents an assertion about a person, group, environment, intervention, or process at a time. It is not limited to diagnoses.

Field group	Required content
Identity	unique ID, human-readable label, standard terminology identifiers, version
Subject	person, cohort, family, organization, place, device, or ecosystem
Domain	one or more of the fifteen atlas branches and subbranches
Scale	molecule through planetary level
Mechanism	asserted or hypothesized process, direction, mediator, confidence
Time	valid interval, onset, recurrence, duration, trajectory, recording time
Determinants	exposures, resources, behaviors, contexts, susceptibilities
Function/outcome	symptom, capacity, activity, participation, quality of life, risk, survival
Evidence	observation IDs, method, source, uncertainty, reference standard, citations
Intervention	action, target, dose/intensity, actor, setting, adherence/fidelity
Governance	consent, purpose, access, legal/ethical constraints, equity flags
Relationships	causes, manifests-as, contraindicates, modifies, responds-to, conflicts-with
Status	observed, inferred, suspected, confirmed, inactive, resolved, disproven

58.2. Observation object

A **HealthObservation** should include:

- observed property and standard code;
- value, unit, scale type, and allowable range;
- acquisition timestamp and duration;
- specimen or body site;
- device, model, serial number, firmware, calibration state;
- protocol, posture, activity, fasting state, environment, and operator;
- raw-data location and checksum where applicable;
- preprocessing and feature algorithm version;
- quality flags and artifact segments;
- reference interval or decision threshold with source/version;

- uncertainty distribution or interval;
- whether the result was measured, reported, estimated, or inferred.

This model is essential for device development. “Heart rate 78” is incomplete without whether it came from ECG, PPG, a manually palpated pulse, an exercise interval, or an algorithm during motion.

58.3. Relationship object

A typed relationship should state:

- source and target concepts;
- relation type;
- direction;
- temporal scope;
- population/context;
- evidence level;
- effect estimate if available;
- uncertainty;
- whether the relation is associative, causal, mechanistic, diagnostic, prognostic, or normative.

Examples:

- smoking **causally increases risk of** atherosclerotic disease;
- atrial fibrillation **increases risk of** embolic stroke;
- beta blocker **may worsen** symptomatic bradycardia;
- inaccessible transport **restricts participation in** outpatient rehabilitation;
- fatigue **interferes with** medication management;
- pregnancy **modifies pharmacokinetics of** a medication.

58.4. Goal and preference object

Goals need structure:

- outcome desired or harm avoided;
- priority and time horizon;
- minimum meaningful change;
- unacceptable tradeoffs;
- decision-maker and consent;
- confidence/stability of preference;
- review date.

A preference is not a fixed trait. It may change with prognosis, experience, family context, and new options.

59. A computable integrated health profile

59.1. Profile layers

The recommended longitudinal profile has eight visible layers.

1. **Identity and context** — demographics relevant to care, language, accessibility, relationships, occupation, geography, and consent.
2. **Current whole-person state** — symptoms, function, participation, well-being, safety, vital risk, and goals.
3. **Physiological and biological state** — organ systems, laboratory values, imaging, waveforms, molecular and genetic data.
4. **Mechanisms and causal graph** — known and hypothesized pathways with confidence.
5. **Life-course and exposure timeline** — development, reproductive events, infection, injury, behavior, work, environment, treatment.
6. **Intervention and medication graph** — what is being done, why, adherence, response, burden, and interaction.
7. **Care-system map** — responsible clinicians, transitions, appointments, resources, barriers, and escalation.
8. **Uncertainty and next-information plan** — unresolved questions and measurements most likely to change decisions.

59.2. State vector and trajectory

For technical systems, a person's latent health state can be represented as a partially observed dynamic system:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t, \mathbf{e}_t, \mathbf{c}_t) + \mathbf{w}_t$$

$$\mathbf{y}_t = g(\mathbf{x}_t, \mathbf{m}_t) + \mathbf{v}_t$$

where:

- $\mathbf{x}_t$ is the latent biological, cognitive, functional, and psychosocial state;
- $\mathbf{u}_t$ represents interventions and behaviors;
- $\mathbf{e}_t$ represents exposures and external events;
- $\mathbf{c}_t$ represents context;
- $\mathbf{y}_t$ is the set of observations;
- $\mathbf{m}_t$ captures measurement method and conditions;
- $\mathbf{w}_t$ and $\mathbf{v}_t$ represent process and measurement uncertainty.

The point is not to assume one solvable mathematical model. It is to enforce a distinction between health state and noisy observations, and to make interventions and context part of the dynamics.

59.3. Change detection versus diagnosis

Longitudinal devices may detect change before they can diagnose cause. The system should distinguish:

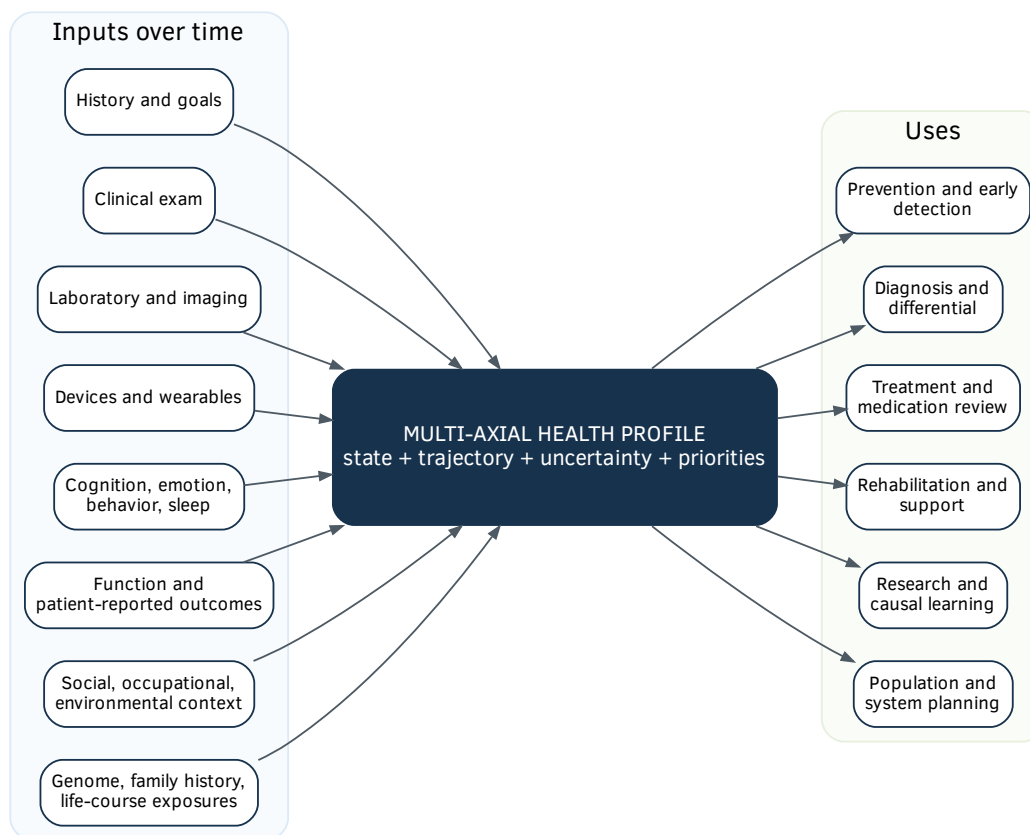


Figure 15.: Integrated health profile. Heterogeneous inputs are preserved with provenance, assembled into a longitudinal multi-axial model, and used for care, research, device validation, prevention, and governance.

- **within-person deviation:** change from personal baseline;
- **population abnormality:** deviation from reference distribution;
- **clinically important change:** magnitude linked to outcome or action;
- **diagnostic pattern:** evidence meeting criteria for a condition;
- **causal interpretation:** explanation supported by further evidence.

A wearable that detects reduced activity and higher resting heart rate may identify deterioration but cannot determine whether the cause is infection, anemia, arrhythmia, sleep loss, medication, heat, stress, or device artifact without additional context.

60. Worked examples using all nine axes

61. Example 1 — Cardiovascular health

Domain: cardiovascular/circulatory, whole-person outcome, measurement, prevention.

Scale: molecular lipoproteins and coagulation; endothelial and myocardial cells; heart, vessels, kidneys, lungs; person; food, work, and care systems.

Mechanisms: atherosclerosis, hypertension, thrombosis, electrical instability, pump dysfunction, inflammation, metabolic dysregulation, congenital structure, autonomic control.

Time: developmental risk, long asymptomatic accumulation, acute events, chronic remodeling, recovery, recurrence, aging.

Determinants: genetics, blood pressure, lipids, diabetes, smoking, nutrition, activity, sleep, stress, infection, pollution, socioeconomic conditions, care access.

Function: exercise tolerance, dyspnea, chest symptoms, cognition after stroke, mobility, work, sexual function, independence.

Context: age, pregnancy, kidney disease, occupation, geography, medication access, family support.

Measurement: ECG, PPG, blood pressure, laboratory lipids and biomarkers, echocardiography, CT/MR, exercise testing, hemodynamics, symptoms, function.

Intervention/governance: prevention, medication, procedures, surgery, rehabilitation, emergency systems, product regulation, air-quality policy, shared decisions.

The cardiovascular packet is therefore one branch packet attached to the master atlas, not a standalone definition of health.

62. Example 2 — Long COVID or post-acute sequelae after infection

Domain: infection/immunity; respiratory, cardiovascular, neurologic, metabolic, cognitive, mental-health, functional, occupational, and care-system branches.

Mechanisms: potentially heterogeneous combinations of persistent antigen or viral reservoirs, immune dysregulation, endothelial or microvascular effects, autonomic dysfunction, organ injury, altered metabolism, sleep disturbance, deconditioning, and symptom amplification. Mechanistic certainty varies by phenotype and study.

Time: begins after acute infection, may fluctuate, relapse after exertion, improve, persist, or evolve.

Determinants: acute severity, prior health, reinfection, vaccination, sex-related biology, age, occupational exposure, care access, social legitimacy, and rehabilitation fit.

Function: fatigue, post-exertional symptom exacerbation, cognition, breathlessness, work and school participation, self-care.

Measurement: infection history, symptom and function profiles, orthostatic testing, cardiopulmonary exercise where safe and indicated, pulmonary/cardiac evaluation, cognition, sleep, laboratory and imaging guided by differential diagnosis. No single device defines the condition.

Intervention: symptom-directed care, pacing or individualized activity management, treatment of identified organ disease, rehabilitation tailored to post-exertional worsening, workplace accommodations, prevention of reinfection, research enrollment. The atlas keeps mechanism hypotheses labeled rather than converting uncertainty into one asserted cause.

63. Example 3 — Depression

Domain: emotion, cognition, behavior, sleep, social and existential health; psychiatric diagnosis; physical health and care access.

Scale: neurotransmission and endocrine/immune pathways; brain networks; whole-person learning and behavior; relationships; work and social structure.

Mechanisms: heterogeneous—genetic liability, developmental adversity, loss, chronic stress, sleep/circadian disturbance, medical illness, medication/substance effects, cognitive and reinforcement processes, social isolation.

Time: single or recurrent episodes, persistent symptoms, seasonal patterns, perinatal timing, bipolar-spectrum differential, treatment response and relapse.

Function: self-care, concentration, parenting, work, social participation, motivation, safety.

Measurement: diagnostic interview, dimensional symptom scales, suicide-risk assessment, sleep and activity, medical differential, function and goals. Passive sensing may supplement but not replace clinical assessment.

Intervention: psychotherapy, pharmacotherapy, social and sleep interventions, exercise where feasible, neuromodulation, crisis support, treatment of medical contributors, workplace and family support.

64. Example 4 — Hypertension

Domain: cardiovascular, renal, endocrine, measurement, prevention, population health.

Mechanisms: vascular resistance, arterial stiffness, volume regulation, sympathetic activity, renin–angiotensin–aldosterone signaling, kidney disease, endocrine causes, medication/substance effects, sleep apnea, genetics, environment.

Time: variable within minutes and days, persistent over years, pregnancy-specific states, acute severe presentations, cumulative organ damage.

Measurement: cuff size, posture, rest, repeated visits, home/ambulatory monitoring, device validation, white-coat and masked patterns. The diagnosis is partly a measurement protocol.

Function/outcomes: often asymptomatic until complication; future stroke, coronary disease, heart failure, kidney disease, and cognitive decline.

Intervention: environment and diet, activity, sleep-apnea treatment, medications, adherence and side-effect management, population sodium and tobacco policy. A threshold code should never replace the time series and protocol.

65. Example 5 — Obesity and body composition

Domain: endocrine/metabolic, nutrition, behavior, psychosocial, musculoskeletal, cardiovascular, respiratory, reproductive, public health, and ethics.

Mechanisms: energy balance over time, appetite and reward regulation, genetics, medications, sleep, endocrine disorders, adipose distribution and signaling, environment, food systems, stress, and life course.

Measurement: body mass index is a population-screening proxy, not a direct measure of adiposity, distribution, fitness, metabolism, or health. Waist measures, body composition, metabolic markers, function, sleep, and comorbidity may add relevant information.

Function and experience: mobility, pain, stigma, eating-disorder risk, self-image, fertility, work, and care access.

Intervention: nutrition and environment, behavioral support, activity, medication, metabolic/bariatric surgery, treatment of complications, anti-stigma care, and policy. Weight change is one outcome; cardiometabolic health, function, adverse effects, sustainability, and quality of life matter.

66. Example 6 — Childhood asthma

Domain: respiratory, immune/allergic, developmental, family, school, environment, devices, public health.

Mechanisms: airway inflammation, hyperresponsiveness, bronchoconstriction, mucus, infection, allergen and irritant exposure.

Time: episodic symptoms, seasonality, developmental change, acute exacerbation, long-term control.

Determinants: housing, mold, pests, smoke, air pollution, viral exposure, medication access, school policy, caregiver understanding.

Measurement: history, examination, spirometry when age-appropriate, peak flow in selected plans, symptom/control instruments, inhaler-use data, environmental assessment.

Function: sleep, play, school attendance, exercise, family burden, emergency risk.

Intervention: controller and reliever therapy, inhaler technique, action plan, trigger reduction, vaccination, school access, housing remediation, emergency recognition.

67. Example 7 — Pregnancy

Domain: maternal, fetal, cardiovascular, renal, hematologic, endocrine, immune, mental, social, occupational, ethical.

Mechanisms/time: rapidly changing physiology by trimester and postpartum; placental function; altered volume, coagulation, glucose, renal filtration, and pharmacokinetics.

Measurement: gestational-age-specific reference, fetal and maternal observations, blood pressure, laboratory tests, ultrasound, symptoms, mental health, social safety.

Intervention: prenatal care, nutrition, medication review, infection prevention, management of complications, delivery planning, postpartum monitoring, lactation support, mental-health care, and workplace accommodation.

Pregnancy demonstrates why a person cannot be represented by ordinary adult reference ranges with one additional diagnosis code.

68. Example 8 — Loneliness

Domain: psychosocial, mental health, behavior, cardiovascular/immune risk pathways, function, community.

Mechanisms: perceived social threat, stress regulation, sleep, behavior, material support, health-care engagement, meaning.

Measurement: subjective loneliness, objective isolation, relationship quality, network structure, cultural norms, digital connection.

Intervention: not simply “more social contact.” Options include relationship repair, group activity, psychotherapy, peer support, transport, hearing care, accessible community design, bereavement support, and policy addressing exclusion.

69. Example 9 — Sepsis

Domain: infection, immunity, cardiovascular, respiratory, renal, neurologic, coagulation, critical care, safety.

Mechanisms: dysregulated host response, tissue hypoperfusion, endothelial injury, organ dysfunction, pathogen and source.

Time: rapid progression, time-critical treatment, recovery with possible long-term cognitive and functional sequelae.

Measurement: vital signs, laboratory trends, cultures, imaging, organ-function scores, hemodynamics, source evaluation; every measure is time-dependent.

Intervention: rapid recognition, antimicrobials, source control, fluids and vasoactive support where indicated, organ support, stewardship, rehabilitation, and follow-up. The acute event and post-intensive-care trajectory belong in one record.

70. Example 10 — Chronic pain

Domain: sensory, nervous, musculoskeletal, immune, psychological, sleep, function, social, occupational, pharmacology.

Mechanisms: nociceptive, neuropathic, nociplastic, inflammatory, ischemic, visceral, mixed; central and peripheral sensitization; learning and threat; sleep and mood feedback.

Measurement: intensity, quality, distribution, temporal pattern, triggers, interference, physical examination, neurological findings, imaging or laboratory testing when indicated, medication effect, function and goals.

Intervention: mechanism-matched medical treatment, physical and occupational rehabilitation, psychological skills, sleep care, social/occupational adaptation, interventional procedures for selected conditions, and careful opioid/nonopioid risk management.

The model avoids both “pain equals tissue damage” and “normal imaging means psychological.”

71. Example 11 — Dementia

Domain: cognition, nervous system, vascular/metabolic factors, function, caregiver/family, safety, legal and ethical governance.

Measurement: cognitive domains, daily function, trajectory, sensory status, mood, medication, laboratory and imaging differential, biomarkers where validated, driving and home safety.

Intervention: cause-specific therapy where available, risk-factor management, cognitive and functional support, environmental design, caregiver services, advance planning, and supported decision-making.

72. Example 12 — Workplace heat exposure

Domain: environmental/occupational, thermoregulation, cardiovascular, renal, public health, labor governance.

Measurement: ambient temperature, humidity, radiant heat, work intensity, clothing/PPE, hydration, core or skin temperature where appropriate, heart rate, symptoms, kidney markers, shift timing.

Intervention: engineering controls, shade/cooling, acclimatization, work–rest cycles, hydration, scheduling, training, emergency plans, labor protection, and climate adaptation. Individual fitness data cannot substitute for safe workplace design.

73. Designing a research dataset around the atlas

73.1. Minimum tables or entities

A relational or graph implementation should include:

- Person/Population;
- Encounter/Event;
- Observation;
- Specimen;
- Device;
- Condition/Phenotype;
- Function/Participation;
- Symptom/Experience;
- Exposure/Determinant;
- Intervention/Procedure;
- Medication/Administration;
- Goal/Preference;
- Relationship;
- CareTeam/Organization;
- Location/Environment;
- Evidence/Source;
- Consent/Governance;
- Algorithm/Model;
- Outcome.

73.2. Temporal semantics

At minimum store:

- event time;
- start/end interval;
- assertion time;
- source-system time;
- timezone;
- uncertainty in time;
- recurrence or episode identifier;
- version and correction history.

73.3. Missingness

Differentiate:

- not measured;
- measured but unavailable;
- below detection;

- person declined;
- not applicable;
- device failure;
- clinically not indicated;
- lost to follow-up;
- value withheld for privacy.

Treating all as null destroys causal and operational information.

73.4. Provenance and lineage

Derived features must link to:

- source data;
- transformation code and version;
- parameters;
- training data and model version for algorithms;
- quality-control decisions;
- operator or automated process;
- validation status.

73.5. FAIR but not careless

Findable, accessible, interoperable, and reusable data principles improve science, but accessibility must be governed by consent, privacy, community interests, and risk. (121)

73.6. Common analytic views

The system should generate purpose-specific views:

- clinical summary;
- device-validation dataset;
- longitudinal trajectory;
- causal cohort;
- safety surveillance;
- population dashboard;
- intervention response;
- multimorbidity interaction graph;
- patient-facing goals and trends.

74. Implications for health-device and sensor development

74.1. A device measures a signal, not “health” directly

Every device chain is:

physiological or behavioral process → transduction → analog/digital acquisition → signal processing → feature → i

Validation must address each arrow. High agreement with another consumer device is not enough if both share the same error. A feature can be technically reproducible yet clinically useless.

74.2. Device taxonomy

74.2.1. Electrical

- ECG, EEG, EMG, EOG, electrodermal activity, bioimpedance;
- neural and muscular stimulation;
- impedance spectroscopy.

74.2.2. Optical

- PPG and pulse oximetry;
- spectroscopy;
- optical imaging;
- retinal imaging;
- fluorescence and biosensing.

74.2.3. Pressure, force, and strain

- blood pressure;
- tonometry;
- respiratory pressure;
- plantar pressure;
- force plates;
- strain gauges;
- intracavitary and compartment pressure.

74.2.4. Acoustic and vibration

- phonocardiography;
- lung sounds;
- voice and speech;
- cough analysis;
- seismocardiography and ballistocardiography;
- ultrasound and Doppler.

74.2.5. Motion and position

- accelerometry, gyroscopy, magnetometry;
- gait, posture, falls, tremor, sleep movement;
- GPS and ecological mobility where ethically justified.

74.2.6. Chemical and biochemical

- glucose and metabolite sensors;
- sweat, saliva, interstitial fluid, breath, and exhaled gases;
- electrochemical biosensors;
- point-of-care immunoassays;
- environmental exposure sensors.

74.2.7. Thermal and environmental

- body and skin temperature;
- heat flux;
- ambient temperature, humidity, light, sound, particulate matter, gases, and allergens.

74.2.8. Imaging

- radiography, CT, MRI, nuclear medicine, ultrasound, optical and endoscopic imaging;
- portable and wearable imaging where feasible.

74.2.9. Self-report and interaction

A questionnaire, ecological momentary assessment, voice diary, or structured task is also a measurement device. Human factors and psychometrics are as important as electronic accuracy.

74.3. Measurement-context packet

Every device study should collect a synchronized context packet:

- posture and activity;
- location and ambient conditions;
- time of day and sleep history;
- meals, caffeine, nicotine, alcohol, and relevant medications;
- acute symptoms and stressors;
- device placement, fit, skin contact, and operator;
- demographic and physiological factors relevant to performance;
- reference measurement;
- raw signal and signal-quality index.

74.4. Reference standards

A reference should be selected for the intended measurand, not convenience. Examples:

- ECG reference for rhythm timing;

- arterial catheter for invasive pressure validation under appropriate settings;
- validated cuff protocol for noninvasive blood pressure;
- laboratory co-oximetry for oxygen saturation validation;
- polysomnography for sleep-disordered breathing or sleep staging studies;
- metabolic cart for oxygen consumption;
- clinical laboratory methods for chemistry;
- expert-adjudicated imaging or event diagnosis where no perfect gold standard exists.

The reference itself has uncertainty and may not be synchronized perfectly.

74.5. Population and subgroup validation

Validation should cover intended-use ranges and challenging conditions:

- age and body size;
- skin pigmentation and optical properties;
- sex-related biology and pregnancy where relevant;
- arrhythmia, low perfusion, motion, edema, tremor;
- chronic disease and medication;
- temperature and altitude;
- device placement and user variability;
- language, cognition, and accessibility for interface-based devices.

Subgroup analysis should be prespecified and powered when safety or performance may differ.

74.6. Clinical validation ladder

1. **bench verification** — electronics, mechanics, calibration, environmental limits;
2. **analytical validation** — accuracy, precision, repeatability, artifact rejection;
3. **physiological validation** — signal relation under controlled perturbation;
4. **clinical validity** — association/prediction of intended condition or outcome;
5. **clinical utility** — decisions and outcomes improve;
6. **human factors** — users can operate and interpret safely;
7. **implementation** — workflow, equity, cost, maintenance, and adoption;
8. **post-deployment surveillance** — drift, rare harm, cybersecurity, and real-world performance.

74.7. Multi-modal profiling

No single sensor captures cardiovascular, metabolic, cognitive, immune, and psychosocial health. A modular suite may combine:

- ECG/PPG and pressure;
- respiration and oxygenation;
- movement and sleep;
- temperature and environment;
- intermittent laboratory or chemical measurements;
- symptoms, function, and context;
- imaging or formal testing at lower frequency.

Fusion should occur only after per-modality quality control. A model should be allowed to abstain when data are insufficient.

74.8. Avoiding the “complete health score” trap

A single score is attractive for dashboards and marketing. It can hide uncertainty, incompatible tradeoffs, and value judgments. Prefer a profile with domain scores, trajectories, confidence, and reason-for-alert. Composite indices may be useful for a specific decision if their derivation and weights are validated.

75. Measurement map by health branch

Table 6.: Representative measurement families by atlas branch.

Branch	Direct or proximal measures	Functional/context measures	Typical gaps
Whole-person outcomes	mortality, events, global ratings, symptom burden	participation, goals, quality of life, caregiver burden	tradeoffs and long-term trajectory
Biological foundations	omics, chemistry, cellular assays, physiology	reserve/challenge response	causal interpretation across scales
Organ systems	electrical, pressure, flow, imaging, lab, examination	organ-specific function and daily impact	cross-organ interactions
Pathological processes	pathogen, inflammation, pathology, biomarkers	symptom and progression pattern	mechanism heterogeneity
Mind/cognition/behaviors	assess, interviews, scales, EEG/imaging, passive sensing	school/work, relationships, safety	ecological validity and privacy
Psychosocial	surveys, interviews, network and material indicators	belonging, role, meaning, security	structural causation and response pathways
Life course	growth, milestones, reproductive and aging measures	developmental tasks and transitions	intergenerational linkage
Infection/immunity	microbiology, serology, immune phenotyping, surveillance	isolation burden, recovery, participation	post-acute outcomes
Environment/occupation	exposure monitors, GIS, job and housing assessment	accessibility, displacement, work capacity	personal exposure and mixtures
Function/disability	performance tests, assistive-device data	activity, participation, environmental barriers	routine longitudinal capture
Population/public health	rates, surveillance, DALYs, coverage	inequity, community priorities	local data and causal attribution
Care systems	process, utilization, safety, wait time, continuity	experience, burden, trust	coordination quality
Measurement/data	calibration, validity, provenance, completeness	usability, interpretability	semantic interoperability
Intervention	dose, fidelity, adherence, exposure	benefit, harm, burden, goal attainment	real-world comparative effectiveness
Ethics/governance	consent, access, audit, disparity metrics	dignity, contestability, trust	values hidden in technical choices

76. Medical education redesigned around the atlas

76.1. A matrix rather than a replacement curriculum

The proposed atlas does not eliminate organ blocks or clerkships. It overlays a matrix so every topic is taught across relevant axes.

For each condition or presentation, learners should answer:

1. What normal physiology and development are involved?
2. What mechanisms could produce the presentation?
3. Which organ systems and cross-system interactions matter?
4. What does the person experience, and how does it affect function?
5. Which psychosocial, cultural, and environmental determinants shape it?
6. What is the time course and life-stage context?
7. Which measurements are valid, and what uncertainty remains?
8. Which interventions act at which level, with what benefit, harm, and burden?
9. Which profession or system owns each task, and how is care integrated?
10. What ethical, legal, equity, and governance issues are present?

76.2. Spiral progression

76.2.1. Stage 1 — Foundational maps

Teach scale, control systems, causation, measurement, bias, function, and social/environmental determinants alongside anatomy and physiology.

76.2.2. Stage 2 — Organ and mechanism integration

Each organ block includes development, behavior, function, environment, pharmacology, devices, epidemiology, and patient experience.

76.2.3. Stage 3 — Presentations and multimorbidity

Use symptoms and cases that cross specialties: fatigue, dizziness, pain, weight change, dyspnea, cognitive change, fever, falls, sleep disturbance, and functional decline.

76.2.4. Stage 4 — Clinical systems

Learners follow patients across primary care, hospital, rehabilitation, home, pharmacy, public health, and community resources.

76.2.5. Stage 5 — Entrustable integration

Assessment requires constructing and updating a multi-axial problem representation, reconciling plans, communicating uncertainty, and measuring outcomes that matter.

76.3. Assessment redesign

What is tested becomes what is learned. Assessment should include:

- mechanistic reasoning;
- data and measurement critique;
- diagnostic calibration;
- function and goals;
- social and environmental intervention;
- interprofessional coordination;
- medication and treatment burden;
- ethics and equity;
- longitudinal change;
- reflection on uncertainty and error.

77. Organizational redesign

77.1. Teams organized around trajectories and goals

Institutions can preserve specialty departments while creating longitudinal integrator roles and condition-agnostic pathways for multimorbidity, complex symptoms, frailty, and disability.

77.2. Shared problem representation

A concise team summary should include:

- active threats;
- person priorities;
- causal/mechanistic model;
- functional constraints;
- key determinants/barriers;
- medication/intervention conflicts;
- uncertainties and next information;
- responsible owner and follow-up.

77.3. Integration metrics

Systems should measure:

- continuity and handoff completion;
- duplicated or conflicting actions;
- medication burden and reconciliation quality;
- time spent coordinating by patient/caregiver;
- goal attainment;
- function and symptom trajectory;
- preventable emergency use;
- access and equity;
- clinician/team workload;
- data completeness and semantic consistency.

77.4. Learning health system

A learning system links routine care to evidence generation while protecting consent and privacy. It uses standardized data, rapid-cycle evaluation, safety monitoring, and feedback. The main danger is turning every encounter into opaque experimentation. Governance must distinguish quality improvement, research, surveillance, and commercial use.

78. Implementation roadmap

78.1. Phase 0 — Governance and scope

- define use cases and excluded uses;
- establish patient, caregiver, clinical, research, engineering, ethics, disability, and community governance;
- choose standard terminologies and version policy;
- define privacy, consent, and access;
- create a claims/evidence ledger.

78.2. Phase 1 — Core ontology and crosswalk

- implement the fifteen branches and nine axes;
- map to ICD-11, ICF, ICHI, SNOMED CT, LOINC, RxNorm, ATC, HPO, Mondo, and local systems;
- represent uncertainty, time, provenance, and relationships;
- build human-readable and machine-readable definitions.

78.3. Phase 2 — Cardiovascular pilot

Use the existing cardiovascular packet as the first deep domain. Build:

- measurement registry;
- device and protocol schema;
- risk/condition/function relationships;
- intervention and medication graph;
- patient-facing longitudinal profile;
- validation dataset.

78.4. Phase 3 — Cross-domain pilots

Add domains selected for interaction value:

1. sleep and circadian health;
2. metabolic/endocrine health;
3. mental, cognitive, and psychosocial health;
4. respiratory health;
5. infection/immunity;
6. renal health;
7. function, pain, and rehabilitation;
8. environmental and occupational exposure.

78.5. Phase 4 — Multi-domain fusion

Test worked profiles for multimorbidity, pregnancy, aging, post-infectious syndromes, and chronic pain. Evaluate whether the architecture reduces missing relationships and improves decisions.

78.6. Phase 5 — Operational integration

- FHIR APIs and profiles;
- OMOP research export;
- device ingestion and waveform linkage;
- terminology service;
- consent and audit service;
- dashboards for clinicians, researchers, and users;
- quality and equity monitoring.

78.7. Phase 6 — Evaluation and revision

Evaluate semantic accuracy, usability, time burden, clinical utility, subgroup performance, patient understanding, coordination, research reproducibility, and unintended consequences. Version the ontology and preserve backward compatibility.

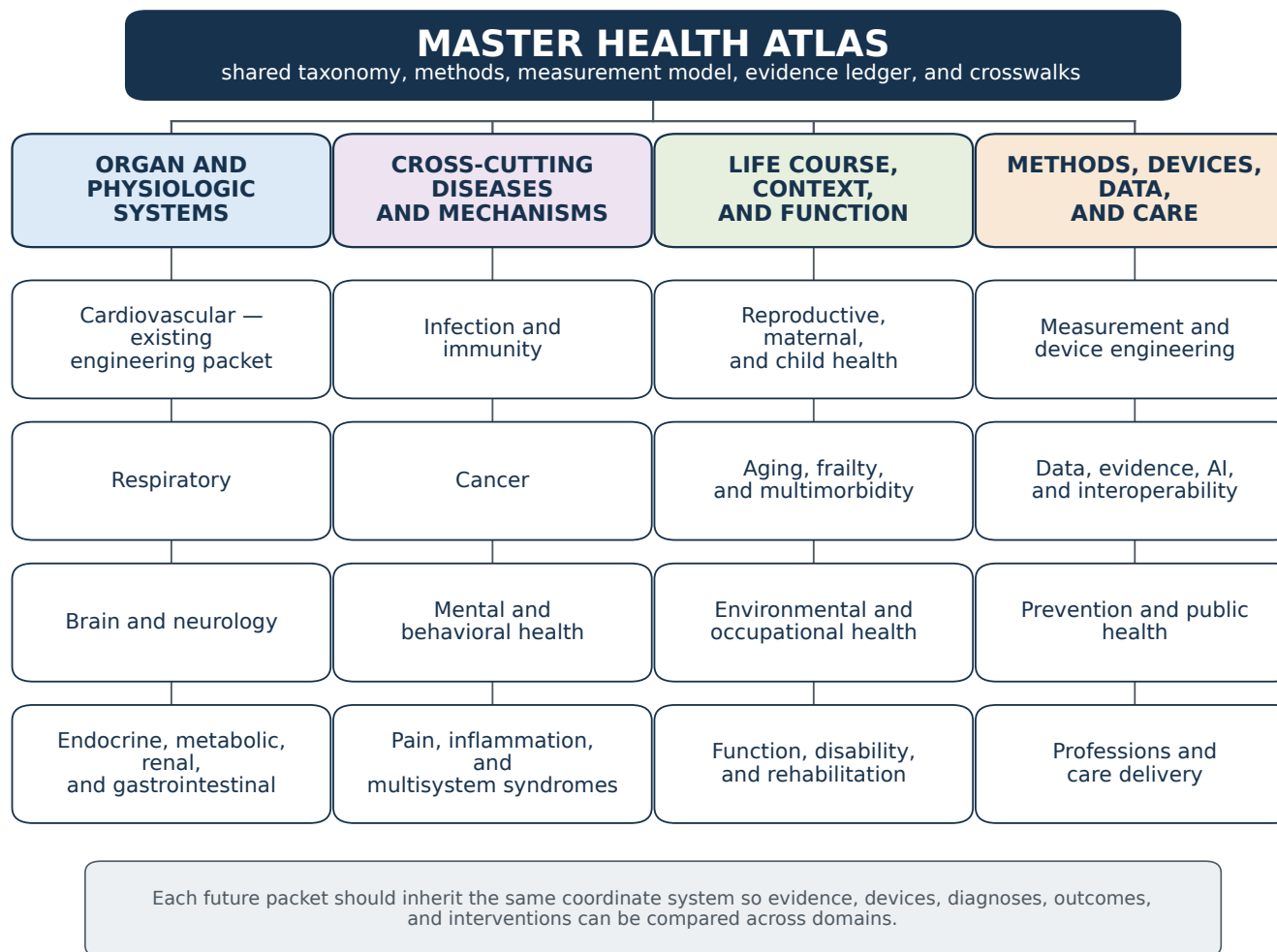


Figure 16.: Expansion roadmap. The cardiovascular packet serves as the first deeply engineered branch; subsequent packets should share a common taxonomy, measurement model, evidence ledger, and interoperability layer.

79. Recommended future domain packets

The master atlas should become a library. Each packet follows a common template:

1. scope and taxonomy;
2. history and landmark evidence;
3. normal anatomy, physiology, development, and regulation;
4. pathophysiology and mechanisms;
5. genetics, exposures, behavior, psychosocial and environmental determinants;
6. epidemiology and burden;
7. symptoms, function, and lived experience;
8. measurement modalities and devices;
9. clinical interpretation and differential diagnosis;
10. prevention and intervention;
11. validation, evidence gaps, and controversies;
12. engineering requirements and safety;
13. data model and interoperability;
14. implementation and equity;
15. research roadmap.

79.1. Priority sequence

79.1.1. Tier 1 — Core physiological and measurement systems

- cardiovascular;
- respiratory;
- sleep/circadian;
- endocrine/metabolic;
- renal/fluid/electrolyte;
- neurologic and autonomic;
- hematologic and oxygen transport.

79.1.2. Tier 2 — Cross-system and high-burden domains

- immunity/infection/microbiome;
- cancer;
- pain and fatigue;
- mental health and substance use;
- cognition and neurodevelopment;
- musculoskeletal/function/rehabilitation;
- nutrition and digestive/liver health.

79.1.3. Tier 3 — Life-course and context

- maternal/fetal/reproductive;
- child and adolescent health;
- aging/frailty/end of life;
- occupational and environmental health;

- social, relational, and community health;
- disability, accessibility, and assistive technology.

79.1.4. Tier 4 — Enabling systems

- health measurement and sensor engineering;
- pharmacology and therapeutic systems;
- imaging;
- laboratory and molecular diagnostics;
- health data, AI, and informatics;
- public-health surveillance and One Health;
- ethics, governance, and regulation.

80. Final synthesis

The history of medicine can be read as a sequence of increasingly powerful decompositions: organ, tissue, cell, microbe, molecule, gene, circuit, behavior, population, and data element. Each decomposition revealed causes and enabled interventions. None is the whole person.

A complete organization of health must therefore do two things at once:

- preserve the precision of specialized science; and
- continuously reassemble observations into a dynamic model of people living in bodies, relationships, institutions, environments, and time.

The fifteen-branch tree makes the field navigable. The nine axes prevent the tree from becoming another silo system. Existing classifications provide maintained standards for particular purposes. The proposed architecture connects them through time, provenance, mechanism, function, context, goals, uncertainty, and governance.

Key synthesis

The best health taxonomy is not the one with the most categories. It is the one that helps a team ask the right questions, preserve what is known and unknown, connect measurements to meaningful outcomes, recognize interactions, and choose interventions at the level where beneficial change can actually occur.

A. Full master taxonomy tree

This appendix is the compact reference hierarchy. It is intentionally more granular than the narrative chapters. Terms may appear under more than one parent when they serve different navigational purposes. Each leaf should also receive tags from the nine-axis codebook in the next appendix.

0

Whole-person health states and outcomes

- **0.1 Vital status and longevity**
 - alive/deceased status; cause and manner of death;
 - all-cause and cause-specific mortality;
 - survival time, event-free survival, remission, recurrence;
 - life expectancy, healthy life expectancy, longevity;
 - years of life lost and premature mortality.
- **0.2 Morbidity and condition burden**
 - incidence, prevalence, severity, duration, recurrence;
 - multimorbidity and complexity;
 - frailty and vulnerability;
 - symptom count and burden;
 - treatment burden and self-management workload;
 - years lived with disability and DALYs.
- **0.3 Risk, susceptibility, and reserve**
 - absolute and relative risk;
 - inherited and acquired susceptibility;
 - physiological reserve and stress tolerance;
 - cognitive reserve;
 - immune competence;
 - resilience and recovery speed.
- **0.4 Subjective health and well-being**
 - self-rated health;
 - life satisfaction;
 - positive and negative affect;
 - meaning, purpose, coherence, hope;
 - suffering and distress;
 - flourishing and eudaimonic well-being.
- **0.5 Safety**
 - risk of acute deterioration;
 - self-harm and suicide risk;
 - violence and abuse exposure;
 - falls, injury, poisoning, overdose;
 - medical and medication harm;
 - infection and environmental hazard;
 - safeguarding and emergency readiness.
- **0.6 Goal attainment and preference concordance**
 - survival and disease-control goals;
 - symptom relief;

- function, independence, work, parenting, and education;
 - fertility and sexual goals;
 - cognitive clarity and alertness;
 - place-of-care and end-of-life preferences;
 - unacceptable tradeoffs and treatment burden.
- **0.7 Equity and distribution**
 - within-population gradients;
 - avoidable and unfair differences;
 - access, quality, experience, and outcome disparities;
 - intergenerational distribution;
 - geographic and environmental distribution.

1**Biological foundations and regulation**

- **1.1 Physical and chemical foundations**
 - water, electrolytes, pH, gases, osmolarity, temperature;
 - diffusion, convection, filtration, osmosis;
 - electrical potentials and current;
 - mechanics, pressure, flow, resistance, compliance;
 - thermodynamics and bioenergetics;
 - redox chemistry and reactive species.
- **1.2 Macromolecules and metabolites**
 - nucleic acids and nucleotides;
 - proteins, peptides, and enzymes;
 - lipids and membranes;
 - carbohydrates and glycoconjugates;
 - vitamins, minerals, cofactors, and trace elements;
 - metabolic intermediates and waste products.
- **1.3 Genetics and genomics**
 - genome structure and variation;
 - Mendelian and non-Mendelian inheritance;
 - chromosomal and structural variation;
 - mitochondrial genetics;
 - mosaicism and somatic mutation;
 - polygenic architecture;
 - penetrance, expressivity, pleiotropy;
 - population genetics and ancestry;
 - pharmacogenomics.
- **1.4 Epigenetics and gene regulation**
 - chromatin state and DNA methylation;
 - histone modification;
 - noncoding RNA;
 - transcriptional and post-transcriptional regulation;
 - genomic imprinting;
 - developmental and exposure-linked regulation.
- **1.5 Cell biology**
 - membranes and transport;

- organelles and intracellular trafficking;
- cytoskeleton, motility, and polarity;
- cell cycle, proliferation, differentiation;
- apoptosis, necrosis, autophagy, senescence;
- stem cells and lineage;
- cell–cell and cell–matrix signaling.
- **1.6 Tissue organization**
 - epithelia and barriers;
 - connective tissue and extracellular matrix;
 - muscle and contractile tissue;
 - nervous tissue;
 - vascular and lymphatic tissue;
 - tissue niches and microenvironments;
 - fibrosis, remodeling, and repair.
- **1.7 Signaling and communication**
 - receptors and ligands;
 - second-messenger pathways;
 - endocrine, paracrine, autocrine, and juxtacrine signaling;
 - neurotransmission and synaptic plasticity;
 - electrical conduction and excitability;
 - mechanotransduction;
 - immune signaling.
- **1.8 Control and regulation**
 - feedback and feedforward control;
 - set points and defended ranges;
 - homeostasis and allostasis;
 - reserve, redundancy, adaptation, and compensation;
 - oscillation and biological rhythms;
 - autonomic and neuroendocrine integration.
- **1.9 Metabolism and energy**
 - digestion and substrate availability;
 - glycolysis, oxidative phosphorylation, fatty-acid metabolism;
 - amino-acid and nitrogen metabolism;
 - nutrient sensing;
 - anabolic/catabolic balance;
 - metabolic flexibility;
 - thermogenesis and storage.
- **1.10 Growth, repair, regeneration, and aging biology**
 - growth signaling and developmental patterning;
 - wound healing and regeneration;
 - DNA repair and genomic stability;
 - proteostasis;
 - mitochondrial maintenance;
 - stem-cell exhaustion;
 - cellular senescence;
 - aging clocks and biomarkers;
 - biological reserve and frailty mechanisms.

- **2.1 Cardiovascular and circulatory**

- cardiac structure and mechanics;
- cardiac electrophysiology and rhythm;
- coronary circulation and myocardial perfusion;
- valves and congenital structure;
- arterial and venous systems;
- microcirculation and endothelial biology;
- lymphatic return and edema;
- pressure-flow regulation;
- exercise and circulatory reserve;
- atherosclerosis, hypertension, ischemia, infarction;
- arrhythmia, cardiomyopathy, heart failure;
- valve, aortic, peripheral arterial, venous thromboembolic disease;
- shock and circulatory failure.

- **2.2 Respiratory**

- upper and lower airways;
- ventilation and respiratory mechanics;
- alveolar diffusion and gas exchange;
- pulmonary circulation;
- respiratory control and neuromuscular drive;
- mucociliary clearance and airway defense;
- asthma, COPD, bronchiectasis;
- interstitial and restrictive disease;
- infection and occupational lung disease;
- pleural disease;
- pulmonary hypertension and embolic disease;
- sleep-related breathing disorder;
- respiratory failure and ventilatory support;
- lung cancer.

- **2.3 Nervous system**

- brain, spinal cord, peripheral nerves;
- autonomic and enteric nervous systems;
- motor, sensory, and integrative pathways;
- cerebrovascular supply and barriers;
- neural development and plasticity;
- stroke and hemorrhage;
- epilepsy and paroxysmal disorders;
- headache and pain disorders;
- infection and inflammation;
- trauma;
- demyelination;
- movement disorders;
- neuropathy and neuromuscular disease;
- neurodegeneration and dementia;
- tumors;
- functional neurological disorders.

- **2.4 Endocrine and metabolic**

- hypothalamic-pituitary axes;
- thyroid and parathyroid;
- adrenal function;
- pancreatic endocrine function;
- gonadal hormones;
- growth hormone and growth;
- glucose and insulin regulation;
- lipoprotein metabolism;
- adipose biology and energy balance;
- calcium, phosphate, and bone metabolism;
- diabetes and hypoglycemia;
- thyroid, adrenal, pituitary, gonadal disorders;
- obesity and metabolic syndrome;
- inborn metabolic disease.

- **2.5 Gastrointestinal, hepatobiliary, and pancreatic**

- oral–pharyngeal ingestion interfaces;
- esophagus and reflux;
- gastric function;
- small and large intestine;
- motility and enteric nervous control;
- digestion, absorption, and barrier function;
- liver metabolism, synthesis, storage, and detoxification;
- biliary system;
- pancreatic exocrine function;
- gut microbial ecology;
- inflammatory, immune, infectious, functional, structural, and neoplastic disease.

- **2.6 Renal and urinary**

- renal blood flow and filtration;
- tubular transport;
- fluid, electrolyte, and acid-base regulation;
- renin–angiotensin–aldosterone system;
- erythropoietin and vitamin-D activation;
- urinary tract, bladder, and continence;
- acute kidney injury;
- chronic kidney disease;
- glomerular, tubulointerstitial, vascular, hereditary, and cystic disease;
- stones, infection, obstruction, incontinence, and cancer.

- **2.7 Hematologic**

- hematopoiesis and marrow;
- erythrocytes and oxygen carrying;
- leukocytes;
- platelets, coagulation, and fibrinolysis;
- blood groups and transfusion;
- anemia and hemoglobinopathy;
- cytopenia and marrow failure;
- bleeding and thrombophilia;

- leukemia, lymphoma, and myeloproliferative disease.
- **2.8 Immune and lymphatic**
 - innate and adaptive immunity;
 - tolerance and immune memory;
 - lymphoid organs and lymphatic transport;
 - mucosal and barrier immunity;
 - immunodeficiency;
 - allergy and hypersensitivity;
 - autoimmunity and autoinflammation;
 - transplant immunity;
 - tumor immunity;
 - lymphedema and lymphatic dysfunction.
- **2.9 Musculoskeletal and connective tissue**
 - bone, cartilage, joint, skeletal muscle, tendon, ligament, fascia;
 - posture, biomechanics, gait, strength, endurance;
 - fracture and injury;
 - osteoporosis and metabolic bone disease;
 - osteoarthritis and inflammatory arthritis;
 - spine disorders;
 - myopathy and neuromuscular overlap;
 - connective-tissue disorders;
 - repetitive strain and occupational disease.
- **2.10 Integumentary**
 - skin barrier, thermoregulation, sensation, pigmentation;
 - hair, nails, sweat and sebaceous glands;
 - wounds, burns, ulcers, pressure injury;
 - dermatitis and inflammatory disease;
 - infection and infestation;
 - autoimmune and blistering disease;
 - pigment disorders;
 - benign and malignant neoplasia.
- **2.11 Sensory systems**
 - vision and ocular health;
 - hearing and auditory processing;
 - vestibular function and balance;
 - smell and taste;
 - touch, proprioception, temperature, and nociception;
 - sensory integration;
 - assistive technology and communication access.
- **2.12 Oral, dental, and craniofacial**
 - teeth, periodontium, oral mucosa, saliva;
 - mastication and jaw function;
 - craniofacial development;
 - caries, periodontal disease, infection, ulceration;
 - oral pain, temporomandibular disorders;
 - oral and oropharyngeal cancer;
 - speech/swallow interfaces.

- **2.13 Reproductive and sexual**
 - reproductive anatomy and gametogenesis;
 - fertility and infertility;
 - menstrual and ovulatory health;
 - contraception and reproductive planning;
 - sexually transmitted infection;
 - sexual function, desire, arousal, orgasm, and pain;
 - gender-affirming health needs;
 - reproductive cancers;
 - consent, safety, pleasure, and rights.
- **2.14 Maternal, fetal, and perinatal**
 - conception, implantation, placenta, fetal development;
 - normal pregnancy physiology;
 - prenatal screening and diagnosis;
 - hypertensive, metabolic, thrombotic, infectious, and placental complications;
 - labor and delivery;
 - postpartum and lactation;
 - newborn transition and neonatal health;
 - maternal mental health;
 - maternal mortality and severe morbidity.
- **2.15 Integrated physiological systems**
 - autonomic regulation;
 - sleep and circadian rhythms;
 - thermoregulation;
 - fluid/electrolyte and acid-base balance;
 - exercise and cardiorespiratory fitness;
 - pain and interoception;
 - stress-response systems;
 - energy balance;
 - immune–neural–endocrine integration.

3

Cross-system pathological processes

- **3.1 Genetic and congenital** — monogenic, polygenic, chromosomal, mitochondrial, epigenetic, developmental-malformation, teratogenic, mosaic, and gene–environment processes.
- **3.2 Infectious** — bacterial, viral, fungal, parasitic, prion, polymicrobial; acute, latent, persistent, congenital, healthcare-associated, vector-borne, food/water/air/contact transmitted.
- **3.3 Inflammatory** — acute/chronic inflammation, resolution failure, fibrosis, granulomatous response, cytokine dysregulation.
- **3.4 Immune dysregulation** — allergy, autoimmunity, autoinflammation, immunodeficiency, immune exhaustion, transplant reaction.
- **3.5 Neoplastic** — benign, premalignant, malignant, hematologic, solid, local invasion, metastasis, treatment resistance.
- **3.6 Vascular and perfusion** — endothelial dysfunction, atherosclerosis, arteriosclerosis, thrombosis, embolism, ischemia, reperfusion, hemorrhage, aneurysm, dissection, microvascular dysfunction.
- **3.7 Endocrine and metabolic** — hormone excess/deficiency/resistance, substrate excess/deficiency, storage disease, toxic metabolite, malnutrition, dyslipidemia.

- **3.8 Degenerative** — protein aggregation, cellular senescence, mitochondrial dysfunction, loss of repair, fibrosis, atrophy, network failure.
- **3.9 Trauma and injury** — mechanical, blast, thermal, cold, electrical, radiological, chemical, psychological; acute and chronic sequelae.
- **3.10 Toxicologic** — medication, substance, environmental, occupational, food, plant/animal toxin, radiation, mixture effects.
- **3.11 Mechanical/structural** — obstruction, stenosis, insufficiency, compression, instability, rupture, overload, altered compliance, malalignment.
- **3.12 Pain and interoceptive dysregulation** — nociceptive, neuropathic, nociplastic, inflammatory, visceral, ischemic, mixed; sensitization and symptom persistence.
- **3.13 Autonomic and regulatory failure** — orthostatic intolerance, dysautonomia, baroreflex failure, circadian dysregulation, thermoregulatory failure.
- **3.14 Nutritional deficiency/excess** — energy/protein malnutrition, micronutrient deficiency/toxicity, refeeding, disordered intake, absorption failure.
- **3.15 Iatrogenic and care-related** — adverse drug, procedure, device, infection, radiation, diagnostic error, delayed care, overtreatment, withdrawal, fragmentation.
- **3.16 Aging and frailty** — reserve loss, multimorbidity, sarcopenia, immunosenescence, sensory decline, vulnerability to stressors.
- **3.17 Repair and regeneration** — wound healing, scar, fibrosis, remodeling, compensation, regeneration, rehabilitation response.
- **3.18 Functional/somatic syndromes** — persistent symptom patterns requiring positive characterization across neural, physiological, psychological, behavioral, and social mechanisms.

4

Brain, cognition, emotion, and behavior

- **4.1 Consciousness and arousal** — wakefulness, alertness, sleep–wake transition, delirium, coma, anesthesia, awareness.
- **4.2 Perception and interoception** — sensory perception, pain, dyspnea, visceral sensation, body ownership, hallucination, illusion, neglect.
- **4.3 Attention** — selective, sustained, divided, orienting, vigilance.
- **4.4 Learning and memory** — working, episodic, semantic, procedural, emotional, autobiographical, reinforcement, extinction.
- **4.5 Language and communication** — production, comprehension, literacy, pragmatics, speech motor control, augmentative communication.
- **4.6 Executive function** — inhibition, switching, planning, decision-making, error monitoring, cognitive flexibility.
- **4.7 Emotion and regulation** — affect, fear, anger, sadness, joy, shame, guilt, regulation, emotional awareness.
- **4.8 Motivation and reward** — incentive, effort, anhedonia, apathy, impulsivity, compulsivity, habit.
- **4.9 Social cognition and attachment** — empathy, perspective, trust, attachment, cooperation, moral cognition, belonging.
- **4.10 Personality and identity** — temperament, traits, self-concept, narrative identity, values, gender identity, agency.
- **4.11 Behavior and self-regulation** — eating, movement, sleep, substance use, sexual behavior, adherence, risk, help-seeking.
- **4.12 Sleep and circadian** — duration, quality, timing, architecture, insomnia, hypersomnolence, parasomnia, circadian misalignment.
- **4.13 Stress and trauma** — appraisal, acute and chronic stress, trauma-related symptoms, moral injury, recovery.

- **4.14 Neurodevelopment** — intellectual, learning, language, autism-spectrum, attention/hyperactivity, motor, tic, sensory-development conditions.
- **4.15 Psychiatric syndrome families** — psychosis, mood, anxiety, obsessive-compulsive, trauma, dissociative, somatic, eating, sleep, impulse/conduct, substance/addictive, personality, neurocognitive.
- **4.16 Cognitive aging and dementia** — reserve, mild impairment, dementia syndromes, behavior, capacity, safety, caregiver context.

5

Psychosocial, relational, cultural, and existential health

- **5.1 Attachment and caregiving** — caregiver availability, sensitivity, attachment security, dependency, reciprocal care.
- **5.2 Family systems** — structure, communication, conflict, parenting, co-parenting, sibling relationships, intergenerational patterns.
- **5.3 Intimate and sexual relationships** — connection, consent, support, conflict, coercion, violence, sexual satisfaction.
- **5.4 Social networks and support** — network size, quality, reciprocity, practical support, belonging, loneliness, isolation.
- **5.5 Violence, abuse, and safeguarding** — child, partner, sexual, elder, peer, community, institutional, trafficking, war.
- **5.6 Work and occupational psychosocial health** — demand, control, reward, justice, security, harassment, schedule, meaning.
- **5.7 School and educational health** — learning environment, belonging, bullying, accessibility, support, academic stress.
- **5.8 Economic and material security** — income, wealth, debt, food, housing, utilities, transport, digital access.
- **5.9 Community and neighborhood** — cohesion, safety, services, recreation, green space, civic participation, collective efficacy.
- **5.10 Culture and language** — explanatory models, identity, norms, communication, interpretation, cultural safety.
- **5.11 Religion and spirituality** — belief, practice, community, meaning, spiritual struggle, chaplaincy, ritual.
- **5.12 Discrimination and structural inequity** — racism, sexism, ableism, ageism, classism, homophobia, transphobia, xenophobia, coloniality.
- **5.13 Migration and legal context** — migration, asylum, displacement, detention, family separation, documentation, language.
- **5.14 Digital and media environment** — social media, gaming, online community, misinformation, harassment, surveillance, telehealth.
- **5.15 Caregiving burden** — time, sleep, financial strain, emotional burden, role conflict, respite, bereavement.
- **5.16 Meaning and existential health** — purpose, identity, dignity, hope, grief, mortality, moral injury, legacy.

6

Development, reproduction, aging, and life course

- **6.1 Preconception and intergenerational** — parental health, genetics, fertility, exposure, reproductive autonomy, resources.
- **6.2 Prenatal and fetal** — implantation, placentation, organogenesis, growth, maturation, congenital conditions.
- **6.3 Birth and neonatal transition** — labor, delivery, prematurity, adaptation, feeding, bonding, neonatal intensive care.

- **6.4 Infancy** — growth, attachment, feeding, sleep, sensory/motor development, immunization, safety.
- **6.5 Early childhood** — language, play, executive development, school readiness, family and childcare context.
- **6.6 Middle childhood** — learning, peer relationships, physical development, chronic illness, school participation.
- **6.7 Adolescence** — puberty, identity, autonomy, sleep shift, risk, sexuality, mental health, transition planning.
- **6.8 Emerging adulthood** — education/work transition, independence, relationships, substance exposure, care transition.
- **6.9 Midlife** — work/family roles, reproduction, caregiving, cumulative risk, chronic-disease emergence.
- **6.10 Reproductive transitions** — menstruation, fertility, pregnancy, postpartum, lactation, menopause, gonadal aging.
- **6.11 Later life** — sensory/cognitive change, multimorbidity, frailty, falls, retirement, bereavement, social support.
- **6.12 End of life** — prognosis, goals, palliative care, dying, death, bereavement, legacy.
- **6.13 Life-course processes** — critical periods, accumulation, chains of risk, mobility, turning points, reserve, intergenerational transmission.

7

Infection, immunity, host-microbe biology, and biosecurity

- **7.1 Host barriers and innate defense** — skin/mucosa, microbiota, complement, phagocytes, pattern recognition, inflammation.
- **7.2 Adaptive immunity** — B/T cells, antibodies, memory, tolerance, vaccination, immunologic aging.
- **7.3 Microbial classes** — bacteria, viruses, fungi, parasites, prions, polymicrobial communities.
- **7.4 Microbiome ecology** — composition, function, metabolites, colonization resistance, dysbiosis, site-specific communities.
- **7.5 Infection stages** — exposure, entry, colonization, invasion, dissemination, latency, persistence, clearance, shedding.
- **7.6 Transmission** — airborne, droplet/contact, fecal-oral, bloodborne, sexual, vertical, vector, food/water, zoonotic.
- **7.7 Clinical infection syndromes** — respiratory, gastrointestinal, urinary, skin, bloodstream, neurologic, congenital, opportunistic.
- **7.8 Post-infectious sequelae** — immune-mediated, organ injury, chronic symptoms, functional and cognitive outcomes.
- **7.9 Vaccines and immune prevention** — platform, schedule, efficacy, effectiveness, safety, durability, coverage, trust.
- **7.10 Antimicrobials and stewardship** — antibacterial, antiviral, antifungal, antiparasitic; resistance and appropriate use.
- **7.11 Antimicrobial resistance** — selection, transmission, agriculture, environment, manufacturing, surveillance.
- **7.12 Outbreak science** — detection, case definitions, tracing, genomic surveillance, modeling, communication.
- **7.13 Infection prevention and control** — hand hygiene, PPE, ventilation, isolation, sterilization, healthcare design.
- **7.14 Biosecurity and biosafety** — laboratory containment, dual use, accidental release, deliberate threats, governance.

- **7.15 One Health infectious ecology** — animal reservoirs, vectors, food systems, wildlife, land use, climate.

8

Environmental, occupational, One Health, and planetary health

- **8.1 Air** — particulate matter, ozone, nitrogen oxides, smoke, indoor combustion, ventilation, allergens.
- **8.2 Water and sanitation** — drinking water, wastewater, pathogens, metals, industrial contaminants, drought/flood.
- **8.3 Soil and food environment** — pesticides, metals, nutrients, food safety, contamination, agricultural systems.
- **8.4 Chemical exposure** — solvents, persistent pollutants, endocrine disruptors, consumer products, mixtures.
- **8.5 Physical exposure** — heat, cold, noise, vibration, radiation, altitude, light, weather extremes.
- **8.6 Built environment** — housing, crowding, accessibility, transport, walkability, green space, urban design.
- **8.7 Occupational hazards** — injury, ergonomic, chemical, biological, physical, psychosocial, shift work, precarity.
- **8.8 Climate health** — heat, extreme events, smoke, infectious shifts, food/water, displacement, mental health.
- **8.9 Biodiversity and ecosystem health** — habitat, pollinators, zoonotic ecology, ecosystem services, toxic blooms.
- **8.10 Food systems** — production, labor, nutrition, affordability, marketing, sustainability, animal health.
- **8.11 One Health** — human–animal–plant–ecosystem interfaces, AMR, zoonosis, food safety.
- **8.12 Planetary health** — climate, biodiversity, land, freshwater, oceans, pollution, energy, civilization systems.
- **8.13 Environmental justice** — unequal exposure, political power, segregation, labor, indigenous rights, remediation.
- **8.14 Disaster and humanitarian health** — hazard, vulnerability, response, displacement, continuity, recovery.

9

Function, disability, rehabilitation, and participation

- **9.1 Body functions** — mental, sensory, voice/speech, cardiovascular/respiratory, digestive/metabolic/endocrine, genitourinary/reproductive, neuromusculoskeletal, skin.
- **9.2 Body structures** — nervous, sensory, voice/speech, cardiovascular/immune/respiratory, digestive/metabolic/endocrine, genitourinary/reproductive, movement, skin.
- **9.3 Mobility** — changing position, transferring, walking, wheeled mobility, transport, driving.
- **9.4 Self-care** — washing, dressing, toileting, eating, drinking, health management.
- **9.5 Domestic life** — meals, household tasks, shopping, caring for possessions and others.
- **9.6 Communication** — receiving/producing messages, conversation, devices, language access.
- **9.7 Learning and applying knowledge** — attention, reading, writing, calculation, decision-making, problem solving.
- **9.8 Interpersonal interaction** — family, intimate, informal, formal, social boundaries.
- **9.9 Major life areas** — education, employment, economic life, parenting, civic roles.
- **9.10 Community/social/civic participation** — recreation, religion, politics, culture, community life.
- **9.11 Environmental facilitators/barriers** — products/technology, natural and built environment, support, attitudes, services, policy.

- **9.12 Rehabilitation** — restoration, compensation, adaptation, skill learning, conditioning, symptom management.
- **9.13 Assistive technology** — mobility, prosthetic/orthotic, hearing/vision, communication, cognition, environmental control.
- **9.14 Accessibility and universal design** — physical, sensory, cognitive, digital, communication, policy.
- **9.15 Caregiver and support needs** — assistance level, training, respite, burden, sustainability.

10**Population health, prevention, and public health**

- **10.1 Demography and population structure** — age, sex, migration, fertility, mortality, household, geography.
- **10.2 Epidemiology** — incidence, prevalence, association, causation, transmission, prognosis, surveillance.
- **10.3 Burden of disease** — mortality, YLL, YLD, DALYs, healthy life expectancy, attributable risk.
- **10.4 Primordial prevention** — economic, commercial, environmental, educational, and policy conditions preventing risk emergence.
- **10.5 Primary prevention** — vaccination, risk reduction, protective environment, prophylaxis.
- **10.6 Secondary prevention** — screening, early detection, case finding, preventive treatment.
- **10.7 Tertiary prevention** — complication prevention, chronic-disease control, rehabilitation.
- **10.8 Quaternary prevention** — avoiding unnecessary diagnosis, treatment, and harm.
- **10.9 Health promotion** — policy, supportive settings, community action, skills, service reorientation.
- **10.10 Public-health services** — assessment, investigation, communication, partnership, policy, enforcement, access, workforce, evaluation, infrastructure.
- **10.11 Social determinants** — economic stability, education, care access, neighborhood, social/community context, political forces.
- **10.12 Commercial determinants** — product, marketing, pricing, lobbying, labor, research influence, digital platforms.
- **10.13 Global health** — transnational disease, inequity, systems, aid, trade, conflict, migration.
- **10.14 Health security** — preparedness, surveillance, emergency response, supply chains, critical infrastructure.
- **10.15 Community health** — local assets, needs, participation, trusted institutions, place-based intervention.
- **10.16 Health equity and justice** — distribution, avoidability, fairness, rights, reparative action.

11**Health care, professions, and delivery systems**

- **11.1 Self-care and informal care** — health literacy, family care, peer support, traditional/community care.
- **11.2 Primary care** — first contact, continuity, comprehensiveness, coordination, prevention.
- **11.3 Emergency and acute care** — triage, stabilization, urgent diagnosis, trauma, intensive care.
- **11.4 Specialty and subspecialty care** — organ, disease, population, procedure, setting, technology.
- **11.5 Surgical and procedural care** — preoperative, anesthesia, intervention, recovery, complications.
- **11.6 Mental-health and substance-use services** — outpatient, inpatient, crisis, community, peer, rehabilitation.
- **11.7 Rehabilitation and long-term support** — inpatient/outpatient/home rehabilitation, skilled nursing, personal assistance.
- **11.8 Palliative and hospice care** — symptom, communication, goals, caregiver, bereavement.

- **11.9 Diagnostic services** — laboratory, pathology, imaging, physiology, genetics, point-of-care testing.
- **11.10 Pharmacy and medication systems** — prescribing, dispensing, monitoring, reconciliation, stewardship.
- **11.11 Public-health and community services** — surveillance, vaccination, maternal/child, school, occupational, environmental.
- **11.12 Digital and home care** — telehealth, remote monitoring, hospital-at-home, digital therapeutics.
- **11.13 Professional functions** — assess, diagnose, monitor, treat, rehabilitate, educate, coordinate, advocate, research, govern.
- **11.14 Quality and safety** — effectiveness, person-centeredness, timeliness, efficiency, equity, diagnostic and medication safety.
- **11.15 Financing and access** — insurance, benefit design, payment, authorization, affordability, universal coverage.
- **11.16 Operations and resilience** — workforce, supply chains, facilities, procurement, emergency continuity.

12**Measurement, evidence, data, and informatics**

- **12.1 History and narrative** — symptoms, timeline, context, goals, explanatory model.
- **12.2 Physical and behavioral observation** — examination, structured observation, task performance.
- **12.3 Patient/proxy-reported measurement** — symptoms, function, quality of life, experience, burden.
- **12.4 Laboratory measurement** — chemistry, hematology, microbiology, immunology, genomics, pathology.
- **12.5 Electrical measurement** — ECG, EEG, EMG, EOG, electrodermal, bioimpedance.
- **12.6 Optical measurement** — PPG, oximetry, spectroscopy, retinal/optical imaging.
- **12.7 Pressure/flow/mechanical** — blood pressure, spirometry, tonometry, force, gait, vibration.
- **12.8 Acoustic and ultrasound** — heart/lung/voice sounds, Doppler, diagnostic ultrasound.
- **12.9 Imaging** — radiography, CT, MRI, nuclear, endoscopy, microscopy.
- **12.10 Physiological challenge** — exercise, orthostatic, glucose tolerance, pulmonary challenge, stress tasks.
- **12.11 Environmental and ecological monitoring** — air, water, noise, heat, location, workplace, household.
- **12.12 Digital phenotyping** — passive sensing, interaction, activity, sleep, voice, mobility, communication proxies.
- **12.13 Measurement science** — validity, reliability, calibration, traceability, uncertainty, invariance, usability.
- **12.14 Study design** — experimental, observational, diagnostic, qualitative, implementation, synthesis.
- **12.15 Statistics and causal inference** — estimation, prediction, missingness, bias, mediation, transportability.
- **12.16 Terminology and ontology** — ICD, ICF, SNOMED CT, LOINC, RxNorm, HPO, Mondo, GO, UMLS.
- **12.17 Interoperability and data models** — FHIR, OMOP, imaging/waveform standards, APIs, provenance.
- **12.18 AI and decision support** — model development, validation, calibration, explainability, abstention, monitoring.

- **12.19 Privacy and security** — consent, access, minimization, cybersecurity, audit, federated methods.
- **12.20 Evidence synthesis and guidance** — systematic review, meta-analysis, GRADE, guideline, decision aid.

13**Interventions and therapeutics**

- **13.1 Primordial and structural prevention** — policy, taxation, regulation, infrastructure, income, education, labor, environment.
- **13.2 Primary prevention** — vaccine, prophylaxis, nutrition, activity, sleep, substance prevention, protective equipment.
- **13.3 Screening and early detection** — tests, surveillance, risk assessment, follow-up pathways.
- **13.4 Pharmaceutical** — small molecule, biologic, vaccine, cell/gene therapy, replacement, antimicrobial, psychopharmacology.
- **13.5 Nutrition and metabolic support** — oral/enteral/parenteral nutrition, supplementation, dietary pattern, metabolic therapy.
- **13.6 Surgical and procedural** — resection, repair, reconstruction, transplant, endoscopic, catheter, ablative, image-guided.
- **13.7 Device-based therapy** — pacing, defibrillation, stimulation, ventilation, dialysis, infusion, prosthesis, closed loop.
- **13.8 Psychological and behavioral** — cognitive, behavioral, exposure, interpersonal, family, motivational, contingency, trauma-focused.
- **13.9 Rehabilitation** — physical, occupational, speech, cognitive, cardiac, pulmonary, vocational, pain rehabilitation.
- **13.10 Assistive and accommodative** — devices, environmental control, communication, workplace/school modification.
- **13.11 Social intervention** — housing, income, food, transport, legal aid, family support, violence prevention.
- **13.12 Environmental and occupational** — engineering control, exposure reduction, ventilation, heat plans, ergonomic redesign.
- **13.13 Public-health intervention** — sanitation, vaccination program, surveillance, communication, vector control, product regulation.
- **13.14 Palliative/supportive** — symptom control, communication, spiritual care, caregiver support, hospice.
- **13.15 Emergency and critical care** — resuscitation, stabilization, antidote, source control, organ support.
- **13.16 Implementation and adherence** — acceptability, feasibility, fidelity, access, cost, maintenance, deimplementation.
- **13.17 Deprescribing and harm reduction** — stop/reduce low-value treatment, tapering, safer-use strategies, overdose prevention.

14**Ethics, law, rights, equity, and governance**

- **14.1 Autonomy and supported decision-making** — consent, capacity, assent, surrogate, advance directive.
- **14.2 Benefit and harm** — proportionality, safety, precaution, treatment burden, opportunity cost.
- **14.3 Justice and equity** — fair distribution, access, need, benefit, priority, anti-discrimination.
- **14.4 Dignity and identity** — respect, stigma, personhood, cultural and gender identity, end-of-life dignity.

- **14.5 Privacy and confidentiality** — clinical, genetic, reproductive, mental-health, location, family data.
- **14.6 Data and algorithm governance** — purpose, minimization, access, bias, transparency, contestability, monitoring.
- **14.7 Public-health law** — surveillance, isolation/quarantine, vaccination policy, emergency powers, safeguards.
- **14.8 Professional regulation** — licensure, scope, standards, duty, negligence, conflicts of interest.
- **14.9 Research ethics** — scientific validity, fair selection, consent, risk–benefit, review, monitoring, dissemination.
- **14.10 Reproductive and genetic ethics** — contraception, abortion, fertility, prenatal testing, editing, family implications.
- **14.11 Disability rights and accessibility** — accommodation, universal design, participation, independent living.
- **14.12 Child and elder protection** — best interests, assent, guardianship, safeguarding, neglect/abuse.
- **14.13 Resource allocation** — triage, coverage, cost-effectiveness, priority setting, global allocation.
- **14.14 Commercial and conflict governance** — marketing, sponsorship, lobbying, intellectual property, procurement.
- **14.15 Environmental and intergenerational justice** — climate, pollution, indigenous rights, future generations.
- **14.16 Biosecurity and dual use** — biosafety, cybersecurity, misuse, proportional oversight.

Design principle

The full tree is a navigation index. It should be implemented as a polyhierarchical ontology or knowledge graph, not as a database table that requires one parent. Cross-links are expected and desirable.

B. Nine-axis codebook

This codebook turns the atlas from a reading structure into a reusable annotation system.

B.1. Axis D — Domain or target

Use one or more branch codes and the deepest applicable subbranch. A concept should have one **primary navigation domain** and any number of secondary domains.

Code family	Meaning	Examples
D0	whole-person state/outcome	survival, burden, flourishing, safety
D1	biological foundation/regulation	gene expression, control system, metabolism
D2	organ system	cardiovascular, respiratory, renal
D3	cross-system mechanism	inflammation, ischemia, injury, toxicity
D4	cognition/emotion/behavior	attention, mood, sleep, addiction
D5	psychosocial/cultural/existential	family, work, loneliness, meaning
D6	life course	prenatal, childhood, pregnancy, aging
D7	infection/immunity/biosecurity	pathogen, vaccine, outbreak, AMR
D8	environment/occupation/planetary	air, heat, work, climate
D9	function/disability/participation	mobility, communication, work participation
D10	population/prevention/public health	screening, surveillance, equity
D11	care/professions/delivery	primary care, emergency, coordination
D12	measurement/evidence/data	ECG, assay, trial, ontology
D13	intervention/therapeutics	medicine, surgery, rehabilitation, policy
D14	ethics/law/rights/governance	consent, privacy, allocation, rights

B.2. Axis L — Scale or level

Code	Level	Examples of entities
L1	atom/ion/chemical	sodium, oxygen, toxin
L2	macromolecule/gene	DNA variant, receptor, antibody
L3	organelle/cell	mitochondrion, neuron, lymphocyte
L4	tissue/microenvironment	endothelium, tumor niche, epithelium
L5	organ	heart, kidney, brain, placenta
L6	physiological system	circulation, immunity, sleep regulation
L7	whole person	symptoms, function, risk, goals

Code	Level	Examples of entities
L8	dyad/family/household	caregiver relationship, household exposure
L9	organization/institution	school, workplace, hospital, insurer
L10	community/place	neighborhood, city, watershed
L11	population/political economy	national policy, market, social gradient
L12	ecosystem/planetary	animal–human ecology, climate system

Record all levels required for the causal explanation. Do not force the “lowest” level to be primary.

B.3. Axis M — Mechanism or process

Mechanism codes should distinguish **established**, **probable**, **plausible**, and **speculative** relationships.

- M-GEN: genetic/genomic/epigenetic;
- M-DEV: developmental and congenital;
- M-INF: infectious and microbial;
- M-IMM: immune and inflammatory;
- M-NEO: neoplastic and clonal;
- M-VAS: vascular, ischemic, thrombotic, hemorrhagic;
- M-MET: endocrine, metabolic, nutritional;
- M-DEG: degenerative, senescent, repair failure;
- M-INJ: trauma and injury;
- M-TOX: toxicologic and exposure-mediated;
- M-MEC: mechanical, structural, pressure/flow;
- M-NEU: neural, cognitive, learning, plasticity;
- M-AUT: autonomic and regulatory;
- M-BEH: behavior, habit, reinforcement, adherence;
- M-REL: relationship, attachment, social support;
- M-STR: structural/political/economic;
- M-IAT: iatrogenic and care-system;
- M-UNK: unknown or under investigation.

B.4. Axis T — Time and life course

Field	Values
onset	congenital, sudden, gradual, uncertain, detected incidentally
duration	transient, acute, subacute, chronic, lifelong
pattern	continuous, episodic, relapsing, seasonal, cyclical, progressive
phase	exposure, latency, prodrome, active, recovery, remission, sequela

Field	Values
developmental stage	preconception, prenatal, neonatal, childhood, adolescence, adulthood, later life
reproductive stage	puberty, menstrual, preconception, pregnancy trimester, postpartum, lactation, menopause
trajectory	improving, stable, worsening, fluctuating, unknown
temporal relation	precedes, follows, co-occurs, delayed, cumulative, recurrent

B.5. Axis X — Determinants and exposures

- X-GEN: inherited susceptibility and ancestry-relevant variation;
- X-PREN: prenatal and early-life conditions;
- X-NUT: nutrition and food environment;
- X-ACT: physical activity, sedentary behavior, fitness;
- X-SLP: sleep and circadian conditions;
- X-SUB: tobacco, nicotine, alcohol, drugs, medication exposure;
- X-REL: relationship, caregiving, isolation, violence;
- X-PSY: stress, coping, trauma, cognition, emotion;
- X-MAT: income, housing, food, transport, utilities;
- X-EDU: education and health literacy;
- X-WRK: occupation, schedule, exposure, job control;
- X-ENV: air, water, heat, noise, chemical, built environment;
- X-INF: pathogen and animal/vector exposure;
- X-CARE: access, quality, delay, continuity, iatrogenic exposure;
- X-POL: policy, law, discrimination, commercial determinants;
- X-DIG: digital/media environment;
- X-PROT: protective resources and resilience factors.

B.6. Axis F — Function and lived outcome

- F-SYM: symptom and distress;
- F-BODY: body function/structure;
- F-CAP: capacity under standardized conditions;
- F-ACT: actual activity in daily life;
- F-PART: participation and role;
- F-IND: autonomy and independence;
- F-QOL: quality of life and well-being;
- F-SAFE: safety and risk;
- F-CARE: caregiver/family burden;
- F-GOAL: goal attainment;
- F-SURV: survival and event outcome.

B.7. Axis C — Context and population

Context should be used only where relevant to mechanism, measurement, access, experience, or equity.

- age and developmental stage;
- sex-related biology, pregnancy, and reproductive stage;
- gender identity and expression;
- ancestry and genetic population where scientifically relevant;
- race/ethnicity as social classification and discrimination context;
- disability and accessibility needs;
- language, literacy, and culture;
- socioeconomic position and occupation;
- family/household and caregiving;
- geography, climate, and urban/rural context;
- care setting and institution;
- legal, migration, and insurance context;
- cohort selection and inclusion/exclusion criteria.

B.8. Axis E — Evidence, measurement, and uncertainty

B.8.1. Evidence-source code

- E-OBS: direct observation;
- E-SELF: self-report;
- E-PROXY: proxy report;
- E-CLIN: clinician interpretation;
- E-DEV: device measurement;
- E-LAB: laboratory/pathology;
- E-IMG: imaging;
- E-ALG: algorithmic inference;
- E-REC: record/administrative data;
- E-EXP: experimental evidence;
- E-OBSERV: observational-study evidence;
- E-SYN: systematic synthesis/guideline;
- E-CONS: expert or community consensus.

B.8.2. Assertion confidence

- confirmed/high confidence;
- probable/moderate confidence;
- possible/low confidence;
- disputed;
- ruled out;
- unknown.

B.8.3. Measurement quality

Record calibration, validity, reliability, uncertainty, artifact, missingness, reference method, protocol, and intended use.

B.9. Axis I — Intervention and governance

- I-PRIM: primordial prevention;
- I-PREV: primary prevention/prophylaxis;
- I-SCR: screening/early detection;
- I-DX: diagnostic clarification/monitoring;
- I-PHARM: medication/biologic/cell/gene therapy;
- I-PROC: procedure/surgery;
- I-DEV: therapeutic or assistive device;
- I-PSY: psychological/behavioral;
- I-REHAB: rehabilitation/adaptation;
- I-SOC: social/resource intervention;
- I-ENV: environmental/occupational modification;
- I-PH: public-health program;
- I-PAL: palliative/supportive;
- I-EMERG: emergency/critical care;
- I-DEIMP: deprescribing/deimplementation;
- I-POL: law, regulation, financing, rights;
- I-RESEARCH: evidence generation;
- I-NONE: monitoring/watchful waiting/no intervention.

Governance fields include actor, authority, consent, payer, accountability, access, equity, monitoring, and stop/review criteria.

C. Classification-system comparison matrix

Table 10.: Detailed comparison of representative systems.

System	Governance	Primary unit	Structure	Main use	Update model	Atlas role
ICD-11	WHO	disease/disorder/injury/disease	foundational plus linearizations; postcoordination	mortality/morbidity statistics and clinical coding	maintained digital releases	diagnosis/statistical identifiers
ICF	WHO	function, structure, activity, participation, environment	hierarchical categories with qualifiers	functioning/disability	maintained classification	functional/context profile
ICHI	WHO	intervention	target–action–means axes	intervention statistics	maintained classification	intervention identity
DSM-5-TR	American Psychiatric Association	mental-disorder syndrome	categorical criteria, specifiers, dimensions	clinical/research communication	text revisions and updates	psychiatric diagnosis layer
ICD-11 CDDR	WHO	mental/behavioral/disorder	clinical descriptions and requirements	global clinical diagnosis	linked to ICD maintenance	international mental-health layer
RDoC	NIMH	functional construct	domain × unit-of-analysis matrix	transdiagnostic research	evolving research framework	mechanism/research tags
HiTOP	research consortium	psychopathology dimension	empirical hierarchy	research and dimensional clinical description	evidence-driven revisions	dimensional symptom profile
SNOMED CT	SNOMED International	clinical concept	description logic/polyhierarchy	EHR clinical meaning	frequent releases	canonical terminology
LOINC	Regenstrief Institute	observation/document identity	multi-axial term model	lab/clinical observation exchange	regular releases	measurement identity
RxNorm	U.S. NLM	clinical drug concept	normalized ingredients/forms/strengths	medication interoperability	regular releases	drug-product identity
ATC/DDD	WHO Collaborating Centre	active substance/use class	five-level hierarchy	drug-utilization research	annual maintenance	therapeutic/anatomic class
MeSH	U.S. NLM	indexing descriptor	polyhierarchical trees	literature indexing/search	annual maintenance	evidence retrieval
UMLS	U.S. NLM	cross-vocabulary concept	metathesaurus/sentiment network	terminology integration	regular releases	mapping bridge

System	Governance	Primary unit	Structure	Main use	Update model	Atlas role
FHIR	HL7 International	exchange resource	modular resources/profiles	APIs/interoperability	standard versions and implementation guides	transport layer
OMOP CDM	OHDSI community	standardized analytic record	relational tables/common vocabularies	observational analytics	community releases	research export/view
HPO	international consortium	phenotypic abnormality	ontology/polyhierarchy	any disease/genomics	continuous maintenance	phenotype layer
Mondo	Monarch Initiative/community	disease concept	ontology with mappings	research integration	regular releases	disease ontology crosswalk
Disease Ontology	community	disease concept	formal ontology	biomedical research	regular releases	disease relations
Gene Ontology	GO Consortium	molecular function/process/-component	formal ontology/evidence annotations	functional genomics	continuous maintenance	biological mechanism layer
ChEMBL	EMBL-EBI	compound/assay/target/activity	publicly curated database	drug discovery	numbered releases	pharmacology evidence
IUPHAR/BPS GtoPdb	IUPHAR/BPS curators	ligand–target relation	expert-curated database	target nomenclature/pharmacology	scheduled releases	target/mechanism layer
GBD	IHME collaborators	cause, sequela, risk, location/-time	statistical estimation framework	comparative burden	periodic studies/updates	population burden layer
PROMIS	HealthMeasures/NIA origins	self-reported domain/item	item banks and IRT	patient-reported outcomes	calibrated instrument maintenance	symptoms/function profile
WHOQOL	WHO	quality-of-life domain/item	standardized questionnaires	cross-cultural quality of life	instrument versions	subjective outcome layer

D. Profession and discipline map

The following map classifies professions by their dominant contribution. Most professions span several rows.

Table 11.: *Health-profession function map.*

Function family	Representative professions/disciplines	Characteristic contribution
Generalist longitudinal integration	family medicine, general internal medicine, pediatrics, geriatrics, primary-care nursing, community health	first contact, continuity, broad differential, prevention, coordination, goals
Organ/disease expertise	medical and surgical specialties, dental specialties, psychiatric subspecialties	deep domain knowledge, complex diagnosis, advanced therapy
Acute stabilization	emergency medicine, critical care, trauma surgery, emergency nursing, paramedicine	triage, resuscitation, time-critical diagnosis and support
Procedural intervention	surgery, interventional cardiology/radiology, endoscopy, anesthesia, dentistry	invasive diagnosis/treatment, periprocedural risk
Medication systems	pharmacy, clinical pharmacology, toxicology, pharmacometrics	selection, dosing, interactions, monitoring, stewardship, formulation
Laboratory and pathology	pathology, clinical laboratory science, microbiology, genetics, cytology	specimen-based measurement, diagnosis, quality systems
Imaging and physics	radiology, nuclear medicine, sonography, medical physics, radiation therapy	image acquisition/interpretation, radiation safety, image-guided treatment
Nursing and care continuity	registered and advanced-practice nursing, nurse specialists	monitoring, treatment delivery, education, coordination, symptom and family care
Psychological assessment/treatment	clinical/counseling psychology, neuropsychology, psychotherapy professions	cognition, emotion, behavior, diagnosis, psychotherapy, testing
Social and resource intervention	social work, case management, community health workers, legal aid, benefits navigation	safety, family, housing, resources, systems, advocacy
Function and rehabilitation	physical therapy, occupational therapy, speech-language pathology, PM&R, rehabilitation psychology, recreation therapy	capacity, activity, participation, adaptation, recovery
Sensory and communication	audiology, optometry/ophthalmology, vision rehab, speech-language pathology, interpreters	sensory function, communication access, assistive systems
Assistive technology	prosthetics/orthotics, rehabilitation engineering, biomedical engineering, accessibility specialists	device fit, mobility, communication, environmental adaptation

Function family	Representative professions/disciplines	Characteristic contribution
Nutrition	registered dietitians, clinical nutrition, food-service and public-health nutrition	assessment, therapeutic nutrition, food-system implementation
Maternal and reproductive	obstetrics, midwifery, maternal-fetal medicine, reproductive endocrinology, lactation services, doulas	pregnancy, birth, fertility, postpartum, reproductive support
Oral health	dentistry, dental hygiene, oral surgery, oral medicine	prevention, diagnosis, treatment of oral/craniofacial conditions
Population and prevention	epidemiology, biostatistics, public-health nursing, health education, preventive medicine	surveillance, prevention, program and policy evaluation
Environmental/occupational	occupational medicine, industrial hygiene, environmental epidemiology, toxicology, ergonomics	exposure assessment, control, workplace and environmental prevention
Infection and One Health	infectious disease, microbiology, infection prevention, veterinary medicine, ecology	host-agent-environment, outbreaks, AMR, zoonotic systems
Data and informatics	clinical informatics, health information management, data science, terminology, software engineering	structured data, interoperability, analytics, decision support
Human factors and safety	human-factors engineering, quality improvement, safety science, systems engineering	workflow, error prevention, usability, resilient systems
Ethics and governance	bioethics, law, compliance, institutional review, disability advocacy, community governance	rights, consent, allocation, oversight, accountability
Research and evidence	basic/translational scientists, clinical investigators, trialists, qualitative researchers, implementation scientists	mechanism, efficacy, causation, experience, translation
Peer and lived-experience expertise	peer specialists, patient partners, disability advocates, caregivers, community researchers	relevance, feasibility, trust, lived outcome, co-design
Spiritual/existential care	chaplaincy, spiritual care, cultural healers where appropriate	meaning, grief, ritual, moral and existential support
Operations and stewardship	administrators, finance, supply chain, facilities, infection control, cybersecurity	capacity, reliability, resource allocation, continuity

E. Measurement modality atlas

E.1. Electrical and electromagnetic

Modality	Primary measurand	Health domains	Principal limitations
ECG	body-surface cardiac potentials	rhythm, conduction, ischemic patterns, rate variability	electrode/contact artifact, limited structural specificity
EEG	scalp electrical activity	seizure, sleep, consciousness, cognition research	spatial mixing, artifact, interpretation dependence
EMG/nerve conduction	muscle and peripheral nerve electrical activity	neuromuscular, neuropathy, movement	invasive needles in some tests, operator dependence
EOG	eye movement potentials	sleep, oculomotor function	cross-talk and drift
electrodermal activity	skin conductance/sympathetic sweat activity	arousal, stress research, autonomic	nonspecific, temperature/motion effects
bioimpedance	tissue electrical impedance	body composition, fluid, respiration, hemodynamics	model assumptions, placement and hydration effects
MEG	magnetic fields from neural currents	neuroscience, epilepsy mapping	expensive shielded environment

E.2. Optical

Modality	Primary measurand	Health domains	Principal limitations
PPG	pulsatile optical absorption/scattering	pulse, rhythm screening, vascular timing	motion, perfusion, pigmentation, placement
pulse oximetry	arterial oxygen saturation estimate	respiratory/critical care	dyshemoglobins, low perfusion, motion, bias across conditions
NIRS	tissue oxygenation proxy	cerebral/muscle oxygenation	pathlength and extracranial contamination
retinal photography/OCT	retinal structure/vasculature	eye, diabetes, vascular/neural disease	image quality and referral pathway
spectroscopy	wavelength-dependent absorption/emission	chemistry, tissue composition	calibration, specificity, penetration
fluorescence microscopy	labeled molecular/cellular structures	pathology/research	labeling, phototoxicity, limited in vivo reach

E.3. Pressure, flow, mechanical, and motion

Modality	Primary measurand	Health domains	Principal limitations
cuff BP	arterial pressure estimate	cardiovascular/renal/pregnancy	cuff/protocol/arrhythmia dependence
arterial catheter	invasive arterial waveform	critical care, validation	invasive harm, leveling/damping errors
spirometry	airflow and volume	respiratory	effort and technique dependence
plethysmography	volume change	lung volume, limb flow, body composition	chamber/device complexity
tonometry	surface pressure waveform	arterial stiffness, ocular pressure	operator and site dependence
Doppler flow	velocity from frequency shift	vascular, cardiac, fetal	angle and acoustic-window dependence
accelerometry/gyroscopy	motion and orientation	activity, gait, tremor, sleep	inference ambiguity and placement
force plate/pressure mat	ground force/pressure distribution	balance, gait, foot health	laboratory versus real-world validity
seismocardiography/BCG	chest/body vibration from cardiac mechanics	hemodynamic research	motion and posture sensitivity

E.4. Acoustic and ultrasound

- stethoscope and digital phonocardiography;
- lung-sound analysis;
- voice, speech, cough, swallowing, and snoring analysis;
- diagnostic ultrasound and echocardiography;
- vascular, obstetric, musculoskeletal, abdominal, and point-of-care ultrasound;
- elastography and tissue stiffness;
- therapeutic ultrasound.

Acoustic AI should be validated across microphones, rooms, languages, diseases, and recording protocols. A classifier can learn device or site differences instead of pathology.

E.5. Chemical, laboratory, and molecular

- serum/plasma chemistry and metabolites;
- hematology and coagulation;
- microbiology, culture, microscopy, antigen, nucleic-acid testing;
- immunology and serology;
- pathology, cytology, histology, immunohistochemistry;
- genomics, transcriptomics, proteomics, metabolomics, epigenomics;
- therapeutic drug monitoring;
- point-of-care and home tests;
- continuous/intermittent glucose and emerging interstitial-fluid sensors;
- breath gases and volatile compounds;
- sweat, saliva, urine, stool, and environmental specimens.

Preamalytic conditions—collection, fasting, posture, time, transport, storage, and contamination—can dominate error.

E.6. Imaging

- radiography and fluoroscopy;
- CT and CT angiography;
- MRI and MR angiography/spectroscopy;
- nuclear imaging, SPECT, and PET;
- ultrasound and Doppler;
- endoscopy and capsule imaging;
- optical coherence tomography;
- microscopy and digital pathology;
- thermography in selected applications.

Imaging findings require indication, protocol, contrast/radiation details, comparison, uncertainty, and clinical correlation. Incidental findings create downstream burden.

E.7. Psychometric, cognitive, and experiential measurement

- structured and semi-structured interviews;
- symptom scales;
- patient-reported outcomes;
- performance-based cognitive tasks;
- adaptive testing;
- ecological momentary assessment;
- observer and caregiver report;
- qualitative interviews and narrative methods;
- experience and trust surveys.

Psychometric quality includes item meaning, factor structure, reliability, validity, sensitivity to change, response bias, language, culture, accessibility, and measurement invariance.

E.8. Environmental and social measurement

- personal and fixed-site pollution sensors;
- temperature, humidity, heat stress, noise, light, radiation;
- water and soil sampling;
- occupational exposure and task assessment;
- geospatial and remote-sensing data;
- housing and neighborhood audits;
- income, food, transport, safety, discrimination, and social-network measures;
- policy and commercial-environment indicators.

These data should be temporally and spatially linked without exposing sensitive location unnecessarily.

F. Intervention map by level

Level	Representative interventions	Typical evidence questions
molecule/target	drug, biologic, gene editing, antidote	target engagement, dose, efficacy, toxicity
cell/tissue	cell therapy, radiation, local ablation, regenerative scaffold	delivery, persistence, off-target effect
organ	surgery, transplant, ventilation, dialysis, pacing	technical success, complications, function
physiological system	exercise training, closed-loop control, endocrine replacement	adaptation, reserve, long-term regulation
person/behavior	psychotherapy, nutrition, sleep, adherence, self-management	acceptability, heterogeneity, maintenance
function/environment	rehabilitation, assistive device, accommodation, home redesign	participation, usability, caregiver burden
relationship/family	family therapy, parenting support, respite, safety planning	dyadic outcomes, conflict, sustainability
organization	care pathway, staffing, checklist, decision support	implementation, workflow, safety, equity
community	vaccination program, transport, violence prevention, green space	reach, distribution, collective outcome
policy/market	taxation, product regulation, income, housing, labor law	natural experiment, equity, unintended effects
ecosystem/planetary	emissions reduction, land/water protection, One Health surveillance	long horizon, co-benefits, global distribution

G. Reference implementation sketch

The following pseudo-schema is illustrative. It is not a final interoperability specification.

```
HealthConceptInstance:
  id: uuid
  subject_ref: Person|Population|Organization|Location|Device
  labels:
    preferred: string
    synonyms: [string]
  codes:
    - system: uri
      version: string
      code: string
      display: string
      mapping_relation: exact|narrower|broader|related
  atlas_domains:
    primary: D-code
    secondary: [D-code]
  scales: [L-code]
  mechanisms:
    - code: M-code
      target_ref: HealthConceptInstance
      relation: causes|contributes_to|mediates|manifests_as|complicates
      confidence: high|moderate|low|disputed
      evidence_refs: [Evidence]
  valid_time:
    onset: datetime_or_interval
    end: datetime_or_interval|null
    pattern: acute|chronic|episodic|progressive|other
    trajectory: improving|stable|worsening|fluctuating|unknown
  determinants: [ExposureRef]
  functions: [FunctionRef]
  context: [ContextRef]
  observations: [ObservationRef]
  interventions: [InterventionRef]
  goals: [GoalRef]
  status: observed|suspected|probable|confirmed|inactive|resolved|ruled_out
  provenance:
    asserted_by: actor_ref
    asserted_at: datetime
    source_refs: [Source]
  governance:
    purpose: [care|research|public_health|device_validation]
    consent_ref: Consent
    access_policy_ref: Policy

Observation:
  id: uuid
```

```
subject_ref: Person|Population|Environment|Device
code: LOINC_or_SNOMED_or_local
value: quantity|code|string|time_series|image_ref|document_ref
unit: UCUM|null
acquisition:
  start: datetime
  end: datetime
  body_site: code|null
  specimen_ref: Specimen|null
  device_ref: Device|null
  protocol_ref: Protocol
  posture: code|null
  activity: code|null
  environment_ref: Environment|null
quality:
  valid: boolean|unknown
  signal_quality: number|null
  uncertainty: distribution_or_interval|null
  flags: [artifact|out_of_range|estimated|incomplete]
derivation:
  raw_data_ref: uri|null
  algorithm_ref: Algorithm|null
  parameters_hash: string|null
interpretation:
  reference_rule_ref: VersionedRule|null
  category: normal|abnormal|indeterminate|not_applicable|null
provenance: Provenance
```


H. Minimum checklist for a new domain packet

H.1. Scope and ontology

- Define domain boundaries and explicit cross-links.
- Identify standard terms and codes.
- Separate anatomy, mechanism, syndrome, function, and outcome.
- State evidence cutoff and update plan.

H.2. Science

- Describe normal development, anatomy, physiology, regulation, and reserve.
- Map causal mechanisms and feedback.
- Address genetics, environment, behavior, social and life-course factors.
- Separate established knowledge, uncertainty, and controversy.

H.3. Epidemiology

- Incidence, prevalence, burden, natural history, and heterogeneity.
- Distribution by age, geography, social conditions, and relevant biology.
- Data quality and model dependence.

H.4. Measurement

- Measurands and reference methods.
- Noninvasive through invasive modalities.
- Device physics, signal chain, calibration, artifact, uncertainty.
- Clinical validity and utility.
- Function and patient-reported outcomes.

H.5. Intervention

- Prevention through palliation.
- Mechanism, indication, dose/intensity, benefit, harm, burden.
- Implementation, access, equity, and cost.
- Monitoring, stop rules, and deimplementation.

H.6. Engineering and data

- System architecture and safety.
- Data schema, standards, provenance, synchronization.
- Human factors and cybersecurity.
- Validation ladder and subgroup testing.

H.7. Ethics and governance

- Consent, privacy, rights, allocation, stigma, dual use.
- Patient/community involvement.
- Conflict-of-interest and claims ledger.

I. Glossary and acronyms

Absolute risk: probability of an event in a defined population and time window.

Activity: execution of a task or action in ICF terminology.

Allostasis: regulation through adaptive change in response to demand.

Allostatic load: cumulative burden associated with repeated or dysregulated adaptive responses.

AMR: antimicrobial resistance.

Analytical validity: how accurately and reliably a test measures the intended analyte or signal.

ATC: Anatomical Therapeutic Chemical classification.

Bioavailability: fraction and rate at which an administered substance reaches systemic circulation or target exposure.

Biomarker: measured characteristic indicating a biological process, disease process, exposure, or response; not automatically a patient-important outcome.

Biopsychosocial model: framework linking biological, psychological, and social processes in health and illness.

Calibration: agreement between predicted probabilities or measured outputs and reference outcomes/-values; in instruments, relation of output to a traceable standard.

Capacity: what a person can do under standardized conditions.

Causal inference: reasoning and methods for estimating effects of interventions or exposures under assumptions.

Clinical utility: evidence that using a test or intervention improves decisions or outcomes relative to alternatives.

Construct validity: degree to which a measure represents the intended theoretical construct.

DALY: disability-adjusted life year, the sum of years of life lost and years lived with disability.

Determinant: condition or process shaping exposure, susceptibility, health state, access, or outcome.

Disability: interaction involving impairment, activity limitation, participation restriction, and contextual barriers/facilitators; definitions vary by framework and law.

Disease: recognized pathological process or condition; not synonymous with illness experience.

DSM-5-TR: fifth edition, text revision, of the Diagnostic and Statistical Manual of Mental Disorders.

EBM: evidence-based medicine.

Ecological validity: extent to which measurement or findings reflect real-world conditions.

Effect modification: variation in an exposure or treatment effect across levels of another variable.

EHR: electronic health record.

Etiology: cause or origin of a disease or condition.

Exposome: totality of environmental exposures across life and their biological consequences.

FAIR: findable, accessible, interoperable, and reusable data principles.

FHIR: Fast Healthcare Interoperability Resources.

Frailty: state of reduced reserve and increased vulnerability to stressors.

Functioning: umbrella term for body functions/structures, activities, and participation.

GBD: Global Burden of Disease study framework.

Gene–environment interaction: variation in genetic effect by environment or environmental effect by genotype.

GO: Gene Ontology.

GRADE: framework for rating certainty of evidence and strength of recommendations.

Health disparity: measurable difference in health or care between groups; **health inequity** emphasizes avoidability and unfairness.

Health literacy: ability and opportunity to access, understand, evaluate, and use health information and services.

HiTOP: Hierarchical Taxonomy of Psychopathology.

Homeostasis: maintenance of regulated variables within viable ranges.

HPO: Human Phenotype Ontology.

ICD-11: eleventh revision of the International Classification of Diseases.

ICF: International Classification of Functioning, Disability and Health.

ICHI: International Classification of Health Interventions.

Interoperability: ability of systems to exchange and meaningfully use information; includes technical, semantic, organizational, and governance layers.

Intervention: intentional action intended to alter health, function, exposure, care, or context.

Iatrogenic: caused by health care, including treatment, diagnosis, systems, or omission.

Life course: health development over time, including timing, accumulation, transitions, and intergenerational pathways.

LOINC: Logical Observation Identifiers Names and Codes.

Measurement invariance: evidence that a measure has comparable meaning across groups or conditions.

Mechanism: process through which a cause produces an effect; may be represented at several levels.

Medicalization: expansion of medical framing into conditions or problems previously understood differently.

Mondo: integrative disease ontology developed by the Monarch Initiative and collaborators.

Multimorbidity: coexistence of multiple long-term health conditions in one person.

Nosology: classification of diseases or disorders.

Nociplastic pain: pain associated with altered nociception without clear evidence that ongoing tissue damage or a somatosensory lesion alone explains it.

OMOP CDM: Observational Medical Outcomes Partnership Common Data Model.

Ontology: formal representation of entities and relations in a domain.

Overdiagnosis: diagnosis of a real condition that would not have caused harm during the person's lifetime.

Participation: involvement in a life situation.

Pathophysiology: disordered physiological processes associated with disease or injury.

Patient-reported outcome: report directly from a patient about health, function, or experience without interpretation by another person.

Phenotype: observable characteristic or pattern resulting from biology and environment.

Planetary health: health of human civilization and the natural systems on which it depends.

Polypharmacy: use of multiple medicines; may be appropriate or problematic depending on net benefit, interaction, and burden.

Postcoordination: combining codes or attributes to express a more detailed concept after selecting a primary code.

Precision medicine: tailoring prevention or treatment using individual variability in biology, environment, behavior, and response.

Prognosis: expected future course or outcome conditional on current information.

PROMIS: Patient-Reported Outcomes Measurement Information System.

Provenance: origin, custody, transformation, and version history of data or knowledge.

Psychometrics: science of measuring psychological and reported constructs.

Quaternary prevention: action to protect people from unnecessary medical intervention and harm.

RDoC: Research Domain Criteria.

Reference interval: distribution-based interval from a defined reference population; not automatically a diagnostic or optimal range.

Reliability: consistency or repeatability of measurement.

Resilience: capacity to maintain or recover function under disturbance; defined differently across levels.

Risk factor: characteristic associated with future outcome; may or may not be causal.

RxNorm: normalized terminology for clinical drugs in the United States.

Screening: testing an apparently asymptomatic population to identify possible disease or risk requiring further evaluation.

SDOH: social determinants of health.

Sensitivity/specificity: test performance conditional on disease status; they do not directly provide post-test probability without prevalence and context.

SNOMED CT: comprehensive clinical terminology maintained by SNOMED International.

Social determinants of health: conditions of birth, growth, work, life, and aging and the wider forces shaping them.

Surrogate endpoint: biomarker or intermediate outcome used in place of a direct patient-important outcome.

Taxonomy: classification into organized groups, often hierarchical.

Treatment burden: work and disruption imposed by healthcare and self-management.

UMLS: Unified Medical Language System.

Uncertainty: lack of complete knowledge due to variability, measurement error, missing data, model limits, causal ambiguity, disagreement, or value conflict.

Validity: degree to which a measure, inference, or conclusion supports its intended interpretation and use.

WHO-FIC: WHO Family of International Classifications.

WHOQOL: WHO quality-of-life measurement framework and instruments.

YLD: years lived with disability.

YLL: years of life lost.

J. Bibliographic method and evidence boundaries

The references combine official standards, primary or landmark scholarship, major reviews, and representative critiques. Current web-based standards were checked through the evidence cutoff. The following principles were used:

- official governing bodies for current classification and standard status;
- primary papers for foundational models where available;
- systematic or landmark empirical literature for major claims;
- explicit separation of descriptive history from causal evidence;
- no assumption that institutional endorsement proves scientific validity;
- no assumption that critique invalidates all use of a system;
- attention to purpose, implementation, and governance.

The atlas should be updated when major classification releases, medical-education standards, data standards, or new evidence materially alter the map. Versioning should preserve prior definitions and mappings.

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